A central-charge-eight CFT with Haagerup symmetry

Construction, spectrum and observables

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Abstract

We study a unitary two-dimensional conformal field theory at $c_L = c_R = 8$ with Haagerup non-invertible symmetry. Its construction uses an explicit six-current chiral algebra $U \subset V_{E_6 \oplus A_2}$, with affine levels 1 and 39. Under the stated exact construction inputs, we establish rationality and complete unitarity, identify the twelve Evans–Gannon characters, and obtain both a diagonal U theory and a chosen holomorphic E_8 extension with Haagerup defects. These full theories have different physics: their first scalar dimensions are respectively 2/3 and 2, while the first scalars invariant under the specified full categorical symmetry have dimension two. We compare these points with published modular-bootstrap bounds and compute spectra with spin through dimension 24, finite-volume thermodynamics with explicit truncation bounds, exact chiral three-point data and four-point functions, and defect and Cardy-boundary observables. The accompanying tables retain exact arithmetic and normalization conventions. We distinguish actual lattice correlators from character consequences and from nonvacuum bulk sewing data that remain uncomputed. The mathematical arguments, finite certificate inputs, and limited Lean formalizations are documented separately for review.

REVIEW — Status of the construction

This is a research draft. The realization statements depend on the explicit finite inputs in Section M; the subsequent algebraic and analytic implications are argued in the text. Exact replays check the specified finite identities, and the available Lean lemmas cover only limited parts of the argument. Neither is a machine verification of the complete CFT construction. The physics tables state which construction results each observable uses.

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1 Introduction and statement of results

Non-invertible topological lines constrain the spectrum and interactions of a two-dimensional CFT, but a consistent set of line fusion rules does not by itself produce a local theory. The Haagerup categories offer a particularly concrete test: they have small fusion rings, nonintegral quantum dimensions, and no immediate realization as an ordinary finite group symmetry. For physics, the questions are both constructive and quantitative. Does a unitary CFT realize the category, where does that theory lie in bootstrap exclusion plots, and which normalized observables can be computed beyond a torus partition function?

Evans and Gannon supplied the modular data and candidate character vectors at central charge eight [9, Sections 2.4 and 5]. Albert–Honda–Kaidi–Zheng studied a proposed Haagerup action in $(E_8)_1$ and the corresponding symmetry-resolved bootstrap [11]. Here an explicit lattice construction supplies the operator products needed to go beyond modular data. It also makes an essential distinction visible: the diagonal full CFT of the smaller chiral algebra U is different from the full CFT of its holomorphic E_8 extension. The former has many relevant scalars; the latter has none. Their symmetry-invariant scalar gaps nevertheless agree at two.

The principal spectral comparison is as follows. Degeneracies count individual Virasoro-primary components, rather than extended-algebra multiplets. In the last row the diagonal theory uses all Verlinde lines, and the extension uses the specified first chiral Haagerup factor.

Observable	Diagonal U	Selected E_8 full theory
Currents per chirality	6	248
First nonidentity Virasoro-primary dimension	2/3	1
First nonidentity scalar dimension	2/3	2
Relevant scalar components	2941	0
Dimension-two scalar components	19558	61504
Invariant scalar gap (multiplicity)	2 (36)	2 (1488)

Our interaction data include a complete three-point tensor for the 113 weight-two chiral Virasoro primaries of U, and explicit normalized four-point functions. The spectrum export reaches total dimension 24. Defect-resolved partition functions and boundary annuli give observables that distinguish the categorical action even when the ordinary E_8 partition function is already known.

We start with actual vectors in a positive-definite lattice VOA. The ambient vertex operators fix all products and their associativity from the outset. The first substantive problem is then to prove that these vectors generate a proper subalgebra. Generating additional fields in weights two and three does not settle that question: singular products of higher fields could still create new currents. We address this issue by a finite strong-generation criterion. Its input consists of exact normal-product identities and the recursive provenance of the higher generators. Its output controls every conformal weight.

The construction has three distinct stages. The first constructs the subalgebra and proves properness from explicit finite data. The second uses positive conformal vectors, branching, and fusion-category arguments to analyze its representation theory. The third compares VOA modules with conformal net sectors, proves the net realization, and establishes complete VOA unitarity through bounded annulus insertions. In particular, the existence of a positive vacuum inner product is not substituted for positivity of all fusion pairings.

Our presentation uses the normal-product conventions of De Sole–Kac [8, Section 2], separates character-theoretic information as in Evans–Gannon [9], and states analytic transport hypotheses in the operator-theoretic setting of Carpi–Kawahigashi–Longo–Weiner [10]. These works also guide the level of detail: finite input is recorded as such, and each passage to an

infinite-dimensional conclusion is made by a separate lemma. The exact arithmetic certificates are accompanying data, not substitutes for these implications.

1.1 The candidate and its finite construction

Let

$$B = V_{E_6 \oplus A_2}, \qquad H = V_{E_8}, \qquad B \subset H$$

be the standard index-three lattice extension. The notation $A_{1,k}$ means the simple affine VOA $L_k(\mathfrak{sl}_2)$. Fix the embedding $\mathcal{K} = A_{1,1} \otimes A_{1,39} \subset B$ described in Section A. Let $u \in B_2$ be the field in (83), and define the *actual generated* vertex algebra

$$U = \langle \mathcal{K}, \omega_B, u \rangle \subset B. \tag{1}$$

All products are computed in this lattice VOA. Thus associativity is inherited from the ambient algebra, rather than assumed for an abstract presentation inferred from character coefficients. Throughout the normal-space and mode formulas, $K = \mathcal{K}_1$ denotes only the six-dimensional current space, not the full affine VOA.

Proposition 1.1 (Computational properness conclusion)**.** *Accept the exact lattice identities and provenance recorded in Section M. The algebra U is strongly generated by actual fields of weights at most four. Its weight-one space is*

$$U_1 = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2, \qquad \dim U_1 = 6.$$

Consequently $U \subsetneq B$, since $\dim B_1 = 86$.

The finite certificates and the all-weight implication are presented in Sections B–E. This is the most directly checkable part of the construction.

Corollary 1.2 (Initial graded pieces)**.** *Under the same finite-certificate hypotheses, U is a simple unitary VOA of CFT type with the ambient conformal vector and central charge eight. Moreover,*

$$U_2 = S_2, \qquad U_3 = S_3, \qquad \mathrm{ch}_U(\tau) = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + O(q^4)).$$

Here S_2, S_3 are the concrete spaces in Certificate A.2. This assertion does not require a modular character calculation.

The proof is given after the properness argument. In particular the coefficient 6010 in the proposed character below has a different dependency from the first four coefficients.

We write $\mathcal{H}_3$ for the unitary Haagerup fusion category with six simple objects $\mathbf{1}, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho$ and fusion rules

$$\alpha^3 = \mathbf{1}, \qquad \alpha\rho = \rho\alpha^{-1}, \qquad \rho^2 = \mathbf{1} + \rho + \alpha\rho + \alpha^2\rho.$$

Here $\mathrm{FPdim}(\rho) = (3 + \sqrt{13})/2$. The other unitary category with this fusion ring, denoted $\mathcal{H}_2$, is Morita equivalent to $\mathcal{H}_3$. Consequently their Drinfeld centers are equivalent even though the fusion categories are not. In the statement below Z denotes this center, not merely its modular matrices.

Theorem 1.3 (Conformal-net realization)**.** *Assume the exact finite construction inputs recorded in Section M and Certificate F.1. The VOA U is strongly rational and strongly unitary, with central charge eight. Its completely rational smeared-field conformal net $\mathcal{N}_U$ satisfies*

$$\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{H}_3)$$

as unitary ribbon categories. The twelve-component character vector is the $\gamma = 0$ Evans–Gannon vector, and in particular

$$\mathrm{ch}_U(\tau) = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + 6010q^4 + \cdots). \tag{2}$$

There is a diagonal unitary full conformal net with $c_L = c_R = 8$ and partition function $\sum_{i=1}^{12} |\chi_i(\tau)|^2$.

The proof is assembled in Theorem F.8, Proposition I.4, Corollary J.3, and Proposition K.2. The character argument is given before the analytic comparison and does not presume it.

Theorem 1.4 (VOA realization and holomorphic extension)**.** *Under the same construction inputs, U is completely unitary and*

$$\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3) \tag{3}$$

as unitary ribbon categories. There is a unitary holomorphic extension $H' \cong V_{E_8}$ whose category of all algebra modules is $\mathcal{H}_3$; its algebra-bimodule category is $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$.

The additional analytic step is the pairwise annulus comparison in Proposition I.5. Together with module exhaustion, it gives Theorem I.8; the categorical identification and extension are then proved in Section J and Section K.

Here H' is obtained by transporting the canonical $\mathcal{H}_3$ Lagrangian algebra. It need not coincide with the originally embedded extension H/U; the argument identifies that original all-module category only up to the alternatives $\mathcal{H}_2, \mathcal{H}_3$. The full-CFT defects in Theorem 1.4 preserve U, and need not commute with the enlarged E_8 chiral algebra.

1.2 How the qualifications are used

Amber **REVIEW** notes identify the boundary of exact construction inputs and their replay coverage. Red **HIGH PRIORITY REVIEW** notes identify the scope of the construction and the limits of its verification. A proved lemma is not made into an additional hypothesis merely because its argument is new; its proof and external dependencies are supplied for scrutiny. Exact construction identities that are accepted as finite inputs are stated separately from these deductions.

Part of the argument	Nature of its evidence
Explicit fields and finite closure	Exact arithmetic over $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$, saved recipes and full residuals.
Properness in all weights	Recursive algebraic theorem plus finite certificates; no character upper-bound extrapolation.
Rationality and branching	Positive-stress and category arguments, embedded-field and finite-top inputs, and cited external theorems.
Haagerup modular data	Rank exclusions, indicator and character arguments printed below, with exact finite arithmetic checked separately.
Net realization and strong unitarity	Actual branching, linear reconstruction, sector counting, and unitary classification.
Complete VOA unitarity and extension	Pairwise annulus comparison, fusion positivity, unitary classification, and extension theorems.

1.3 A reading guide and the two full theories

Sections 2–7 collect the physical observables before the construction proofs. All central charges quoted in a bootstrap plot mean the common chiral value $c = c_L = c_R$, so our point is $c = 8$, not $c_L + c_R = 16$. We use $\Delta = h + \bar{h}$ and $s = h - \bar{h}$; a scalar has $s = 0$. A "primary" without further qualification is primary for the two Virasoro algebras. An extended-algebra sector can contain many such primaries, so a sector count is not an operator degeneracy.

Write $M_0 = U, M_b, M_a, M_{c_0}, \ldots, M_{d_6}$ for the twelve simple U modules, and H' for the selected extension in Theorem 1.4. The two bulk Hilbert spaces are

$$\mathcal{H}_{\mathrm{diag}} = \bigoplus_i M_i \otimes \overline{M_i}, \qquad \mathcal{H}_{E_8} = H' \otimes \overline{H'}. \tag{4}$$

The canonical Verlinde lines of the first theory form $\mathrm{Rep}(U) = Z(\mathcal{H}_3)$. In the second theory we specify the chosen first chiral factor $\mathcal{H}_3 \subset \mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$ when discussing the Haagerup bootstrap. Statements about invariance under both factors are labeled separately. These lines preserve U and the stress tensor; they need not preserve the enlarged E_8 current algebra pointwise.

2 Characters and a spin-resolved physical spectrum

The chiral algebra fixes considerably more physical data than the twelve lowest conformal weights. In this section $c_L = c_R = 8$, so the quantity often denoted simply by c in two-dimensional bootstrap plots is 8; the sum of the two central charges is 16. We distinguish two local full CFTs throughout:

$$Z_{\mathrm{diag}}(\tau, \bar{\tau}) = \sum_{i=0}^{11} \chi_i(\tau)\overline{\chi_i(\tau)}, \qquad Z_{E_8}(\tau, \bar{\tau}) = \left| \frac{E_4(\tau)}{\eta(\tau)^8} \right|^2 . \tag{5}$$

The first uses the diagonal modular invariant of U. The second is the full theory of the selected holomorphic extension carrying the Haagerup action constructed later in the manuscript. They have the same central charge and different spectra. In particular a spectral statement about the diagonal theory must not be assigned to the holomorphic extension.

2.1 An exact recurrence and its normalization

Set $q = e^{2\pi i \tau}$ and write

$$\chi_i(q) = q^{h_i - 1/3} A_i(q), \qquad A_i(q) = \sum_{n \geq 0} a_i(n) q^n .$$

The grading variable n measures distance above the actual lowest weight, including for d_3, whose lowest weight is $15/13$ rather than its twist representative $2/13$. The parameter-zero solution of Evans–Gannon [9, Sections 1.4 and 2.4] has the following initial coefficients; the two rows a, c_0 represent distinct simple modules with equal unflavoured characters.

i	h_i	$a_i(0)$	$a_i(1)$	$a_i(2)$	$a_i(3)$
0	0	1	6	120	956
b	1	80	1250	10630	65042
a	1	81	1377	11583	71037
c_0	1	81	1377	11583	71037
c_1	1/3	3	243	2916	21870
c_2	2/3	27	594	5967	39852
d_1	6/13	7	292	3204	23010
d_2	11/13	42	777	7147	45367
d_3	15/13	119	1623	12996	76429
d_4	5/13	5	229	2738	19942
d_5	7/13	13	347	3804	26390
d_6	8/13	14	441	4445	30329

Here is an explicit recurrence that extends this table without a lattice-state calculation. Write

$$\Xi = q^{\Lambda} \sum_{n \geq -1} X_n q^n, \quad X_{-1} = I, \quad X_0 = \chi, \quad C = \chi + [\Lambda, \chi],$$

where Λ, χ are the rational exponent matrix and integral characteristic matrix printed in [9, Section 2.4]. They are also recorded in full in the machine-readable dataset. With $J = q^{-1} + 196884q + \cdots$, define

$$g = \frac{\eta^{24}}{E_4 E_6} = \sum_{r \geq 0} g_r q^r, \qquad f = (J - 240)g = \frac{E_4^2}{E_6} - 984g = \sum_{r \geq 0} f_r q^r.$$

Thus $f_0 = 1$, $f_1 = 0$, and $g_0 = 0, g_1 = 1, g_2 = 240$. The differential equation $q\, d\Xi/dq = \Xi(f(\Lambda - I) + gC)$ gives

$$(n + 1 + \lambda_i - \lambda_j)(X_n)_{ij} = \sum_{r=1}^{n+1} \left[f_r(\lambda_j - 1)(X_{n-r})_{ij} + g_r(X_{n-r}C)_{ij} \right]. \tag{6}$$

All denominators are nonzero. In particular this specifies the convolution index directly, independently of typographical renderings of the g index in equation (1.24) of the cited paper. The desired characters are column zero. Vacuum grade n uses $(X_{n-1})_{00}$, whereas nonvacuum grade n uses $(X_n)_{i0}$.

The exporter uses rational arithmetic, caches $X_n C$ once per grade, and compares the first four printed coefficient columns with the published data. An independent oscillator-product calculation checks

$$\frac{E_4}{\eta^8} = \chi_0 + \chi_b + 2\chi_a = \chi_0 + \chi_b + 2\chi_{c_0} = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + \cdots) \tag{7}$$

through every exported grade. Equal scalar characters do not choose between possible extension algebra structures.

2.2 Removing Virasoro descendants

We count individual primary components, including their internal multiplicities, rather than only the twelve extended-algebra sectors. Let $P(q) = \prod_{m \geq 1}(1 - q^m)$, and let $v_i(n)$ count Virasoro highest vectors of weight $h_i + n$ in the ith chiral module. Include the identity once by setting $v_0(0) = 1$. Then

$$\sum_{n \geq 0} v_i(n) q^n = P(q) A_i(q) \quad (i \neq 0), \qquad \sum_{n \geq 0} v_0(n) q^n = P(q) A_0(q) + q. \tag{8}$$

The correction $+q$ removes the nonexistent vacuum state $L_{-1}\mathbf{1}$ from the descendant subtraction.

For completeness, the relevant absence of further null vectors can be seen directly from the Kac determinant [7, Lemmas 2–3]. Parametrize $c = 13 - 6(t + t^{-1})$; at $c = 8$, one has $|t| = 1$, $t + t^{-1} = 5/6$, and $\operatorname{Im} t \neq 0$. Its possible singular weights are

$$h_{r,s} = \tfrac{1}{4}((r^2 - 1)t + (s^2 - 1)t^{-1} - 2rs + 2).$$

For positive integers r, s, reality forces $r = s$, in which case $h_{r,r} = -7(r^2 - 1)/24 \leq 0$. A positive real highest weight therefore has an irreducible Verma module. At weight zero the only primitive null is $L_{-1}\mathbf{1}$. Unitarity permits orthogonal removal of the descendant submodules at each finite grade, giving (8). For example,

$$(v_0(0), v_0(1), v_0(2), v_0(3), v_0(4)) = (1, 6, 113, 830, 4934).$$

The corresponding first nonvacuum grades are

i	$v_i(0)$	$v_i(1)$	$v_i(2)$	$v_i(3)$	i	$v_i(0)$	$v_i(1)$	$v_i(2)$	$v_i(3)$
b	80	1170	9300	53162	d_1	7	285	2905	19514
a, c_0	81	1296	10125	58077	d_2	42	735	6328	37443
c_1	3	240	2670	18711	d_3	119	1504	11254	61810
c_2	27	567	5346	33291	d_4	5	224	2504	16975
					d_5	13	334	3444	22239
					d_6	14	427	3990	25443

Every exported chiral character is reconstructed independently by convolving these multiplicities with the partition numbers and using the vacuum correction. This verifies descendant subtraction through the stated grade; it is not a separate existence proof for U.

2.3 The diagonal theory: scalars and spinning operators

An ordered pair of grades (n, m) in sector i has

$$\begin{gathered}(h, \bar{h}) = (h_i + n, h_i + m), \quad \Delta = 2h_i + n + m, \quad s = n - m, \\ d_{\text{states}} = a_i(n)a_i(m), \quad d_{\text{prim}} = v_i(n)v_i(m).\end{gathered} \tag{9}$$

The last quantity counts primaries under both Virasoro algebras. It includes chiral currents paired with the right vacuum, and vice versa. Summing equal (Δ, s) entries produces the spin-resolved tables in the dataset. Character grades through N suffice for the complete spectrum through $\Delta = N$, because all lowest weights are nonnegative. Rectangular products at larger Δ would generally be incomplete and are not reported as complete levels.

The seven relevant scalar levels are

Δ	2/3	10/13	12/13	14/13	16/13	4/3	22/13
d_{prim}	9	25	49	169	196	729	1764

(10)

and contain 2941 primary components in total. These are the squares of the seven relevant chiral ground-space dimensions. No scalar descendant can enter below dimension two: equality of left and right weights requires $n = m$, so a positive grade raises Δ by at least two.

The complete additional relevant primary levels with nonzero spin are

Δ	s	multiplicity for each displayed spin
1	± 1	6
5/3	± 1	720
23/13	± 1	1120
25/13	± 1	1995

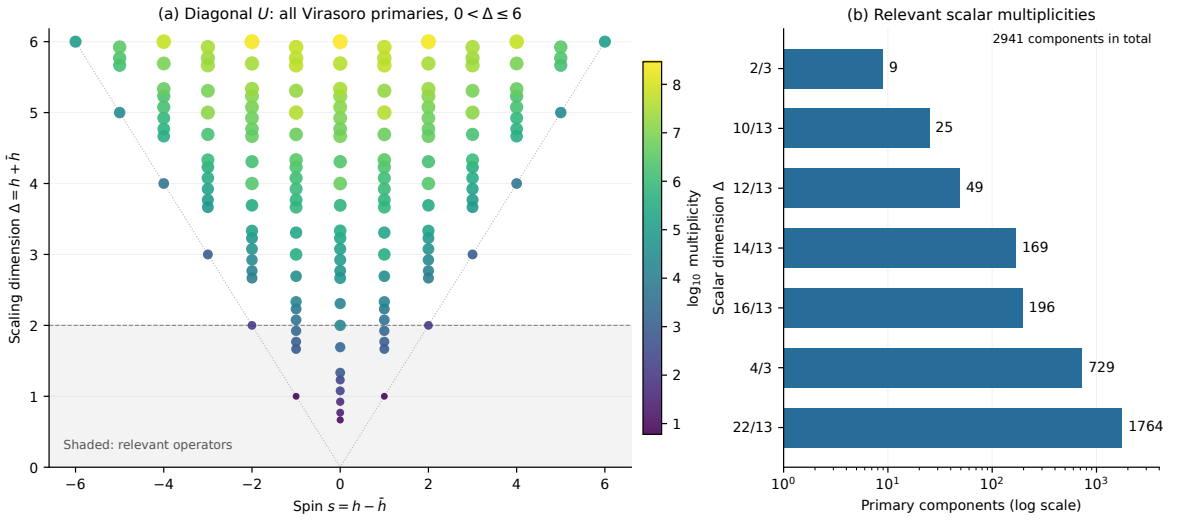


Figure 1: Exact Virasoro-primary spectrum of the diagonal U theory. Left: all nonidentity (Δ, s) levels with $0 < \Delta \leq 6$, aggregated over the twelve extended sectors; color and marker area increase with the logarithm of the component multiplicity. The shaded region has $\Delta < 2$. Right: all seven relevant scalar levels, with their exact multiplicities and a logarithmic count axis. The identity and Virasoro descendants are excluded.

For example, the second line is $3(243 - 3) = 720$: the three c_1 ground states pair with its 240 new grade-one Virasoro primaries. Including scalars, there are 10623 nonidentity relevant Virasoro-primary components. The ordinary primary gap and the scalar primary gap are both 2/3. There are six currents in each chirality, generating the affine subalgebra $\mathfrak{sl}_{2,1} \oplus \mathfrak{sl}_{2,39}$.

At dimension two the scalar-primary multiplicity is

$$80^2 + 81^2 + 81^2 + 6^2 = 19558. \tag{11}$$

The first three terms are the ground spaces of b, a, c_0; the final term comprises the current bilinears $J^A \bar{J}^B$. In addition there are 113 chiral weight-two primaries in each spin ± 2 channel. Thus dimension two contains 19784 Virasoro-primary components and 19798 states. The difference consists of the two stress tensors and twelve current derivatives. We call dimension-two scalars *marginal by dimension*; their beta functions and exact marginality require OPE data not fixed by the character vector.

2.4 The holomorphic extension and symmetry-preserving perturbations

For the full E_8 theory, the same descendant subtraction gives 248 chiral weight-one and 3875 chiral weight-two Virasoro primaries. Consequently its ordinary primary gap is one, its scalar primary gap is two, and it has no relevant scalar. The $\Delta = 2$ scalar space has dimension $248^2 = 61504$; the total dimension-two state multiplicity is $2 \cdot 4124 + 248^2 = 69752$.

For the selected canonical extension, the defect construction below specifies a first chiral $\mathcal{H}_3$ action inside the full $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$ defect category. Its invariant Hilbert space is $U \otimes \overline{H'}$: the first factor acts in its vacuum representation on U and leaves the antiholomorphic extension unrestricted. It therefore has exactly $6 \cdot 248 = 1488$ invariant scalar components at dimension two. Requiring invariance under *both independent factors* instead gives $U \otimes \overline{U}$ and $6^2 = 36$ such components. Each convention has invariant scalar gap exactly two. These invariant counts use the specified canonical-extension defect action, not only the unflavoured character

identity. Neither count establishes an exactly marginal family: that requires the corresponding OPE coefficients and beta functions.

2.5 Finite-temperature observables with controlled tails

On a circle of circumference L and velocity v, set $t = \beta v/L$, $b = 2\pi t$ and $E = \Delta - 2/3$. Then

$$H = \frac{2\pi v}{L}E, \qquad P = \frac{2\pi}{L}s, \qquad Z(b) = \sum_{\Delta,s} d_{\Delta,s} e^{-b(\Delta-2/3)}.$$

The ground energy is $-4\pi v/(3L)$. The exact finite spectral table already gives a lower bound on Z at every positive temperature. The following elementary modular estimate also controls the omitted tail and makes a finite-temperature table quantitatively meaningful.

Proposition 2.1 (A tail bound from modular invariance)**.** *Suppose $Z(b) = Z(4\pi^2/b)$ has nonnegative degeneracies, and that all states with $\Delta \le D$ are known, where $D > 2/3$. Write $Z = L_D + H_D$ for its known and omitted sums, and put $E_D = D - 2/3$. If $b > 2\pi$ and $b' = 4\pi^2/b$, then*

$$0 \le H_D(b) \le \frac{L_D(b) - L_D(b')}{e^{(b-b')E_D} - 1}. \tag{12}$$

At $b = 2\pi$ one has $0 \le H_D(b) \le L_D'(b)/E_D$. For $b < 2\pi$ the same bounds apply to $Z(b') = Z(b)$.

Proof. Every omitted energy exceeds $E_D > 0$, hence $H_D(b) \le r H_D(b')$ with $r = e^{-(b-b')E_D} < 1$. Modular invariance gives $H_D(b') = H_D(b) + L_D(b) - L_D(b')$; rearranging yields (12). At the fixed point $Z'(2\pi) = 0$, so $L_D'(2\pi) = -H_D'(2\pi) \ge E_D H_D(2\pi)$. □

The real-number rearrangement and the positive-division step at the fixed point are formalized as `modular_heavy_tail_comparison` and `selfdual_tail_first_moment_comparison` in `lean/HaagerupIzumi/CFT/PhysicsBounds.lean`. Their hypotheses explicitly include the balance, contraction, and first-moment inequalities used above. They do not formalize spectral convergence, the modular functional equation, differentiation of the trace, or the decimal interval implementation. The guarded compiler and axiom report is `c8_physics_20260927/physics_lean.log`.

To control energy and heat capacity, choose $0 < b_0 < b$ and bound $H_D(b_0)$ by the same proposition. For $k = 1, 2$,

$$\sum_{\Delta > D, s} d_{\Delta,s} E^k e^{-bE} \le H_D(b_0) \sup_{E \ge E_D} E^k e^{-(b-b_0)E}. \tag{13}$$

The supremum occurs at E_D when $(b - b_0)E_D \ge k$, and otherwise at $k/(b - b_0)$. Together with the exact low-energy moments this bounds $\langle E \rangle$, $C_V = b^2 \operatorname{Var}(E)$, and $S = \log Z + b\langle E \rangle$. At $t = 1$, modular invariance fixes $\langle E \rangle = 0$ exactly.

The supplied thermal tables use $D = 24$ and $t = 1/2, 3/4, 1, 3/2, 2$, with $b_0 = 0.9b$ for moment bounds. The implementation propagates closed decimal intervals with outward rounding and a rational enclosure of π; the tables report both endpoints. The supplied 100-digit enclosure of π is independently checked by rational alternating sums in Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, using 80 and 25 terms with explicit next-term bounds. Their certification is conditional on positivity and modular invariance of the full CFT, which are mathematical premises established elsewhere, and not consequences of finite coefficient agreement. No unbounded-temperature accuracy claim is made.

For direct comparison the following values are rounded from those intervals. Every displayed number has absolute error less than 10^{-6}, including both rounding and the omitted-state tail; the exporter checks this containment before producing the table.

theory	t	Z	$\langle\Delta - 2/3\rangle$	C_V	S
U diagonal	1/2	4367.819471	2.653128	17.089150	16.717068
U diagonal	3/4	303.785760	0.983857	16.343538	10.352640
U diagonal	1	147.219258	0.000000	19.266968	4.991923
U diagonal	3/2	565.179187	-0.621644	3.473060	0.478284
U diagonal	2	4367.819471	-0.663282	0.419054	0.046971
E_8 full	1/2	4356.000000	2.659753	16.984459	16.735170
E_8 full	3/4	297.792228	0.991916	16.622062	10.370689
E_8 full	1	144.000000	0.000000	19.157072	4.969813
E_8 full	3/2	557.169604	-0.627321	3.435582	0.410512
E_8 full	2	4356.000000	-0.664938	0.272738	0.023449

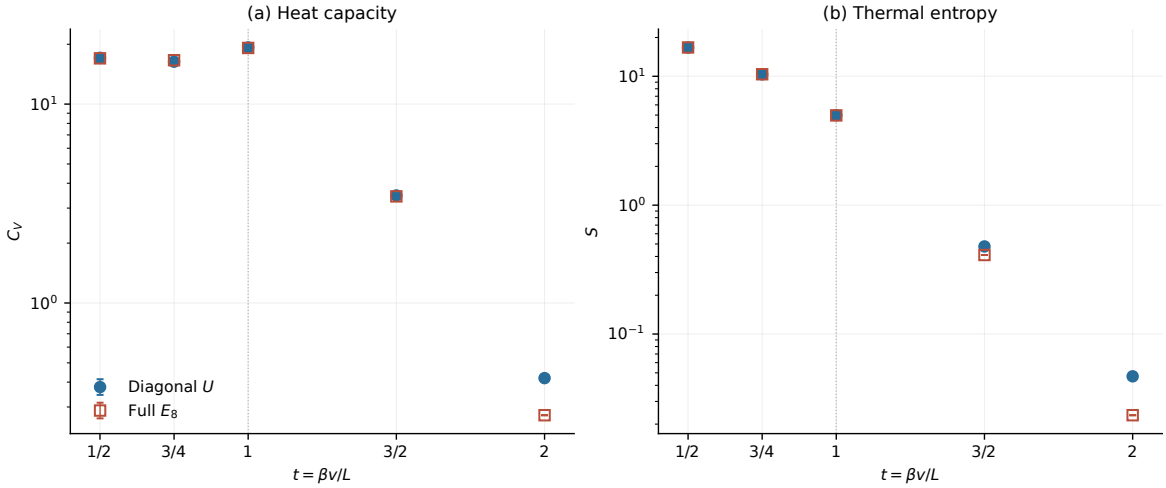


Figure 2: Heat capacity and thermal entropy of the two full theories at the five computed temperatures $t = \beta v/L$. Both vertical axes are logarithmic. No interpolation is drawn. Error bars come from the conditional certified intervals using all states through $\Delta = 24$ and the modular/moment tail bounds; they are smaller than the markers. The dotted line marks the modular fixed point $t = 1$.

In the thermodynamic limit both theories have

$$f(T) - f(0) = -\frac{4\pi}{3v}T^2, \qquad s(T) = \frac{C_V(T)}{L} = \frac{8\pi}{3v}T.$$

Their universal single-interval ground-state entanglement coefficient is $8/3$ in front of $\log(\ell/\epsilon)$, and the Rényi coefficient is $\frac{4}{3}(1+n^{-1})$. These central-charge observables do not distinguish the diagonal theory from its holomorphic extension; the low-energy spectrum and defect-resolved observables do.

2.6 Dataset, reproducibility, and limits

The standalone standard-library producer is `c8_physics_20260927/spectrum.py`. Its default run computes 24 grades of all twelve characters, including Virasoro-primary multiplicities, and complete spectra through $\Delta = 24$ for both full theories. The output directory contains `characters.json`, `characters.tsv`, `fundamental_matrix.json`, `diagonal_sector_spectrum.tsv`, `diagonal_spectrum.tsv`, `e8_spectrum.tsv`, `scalars.tsv`, `low_spectrum.json`, `thermal.json`, and `thermal.tsv`. The run summary records coefficient controls, nonnegative integrality, descendant reconstruction, spin reflection, the E_8 product check, and source/output

hashes. Paths in this paragraph are relative to `c8_physics_20260927/data/`, except the explicitly named producer.

For a replay into a fresh directory, run serially from the repository root (ordinary Python, without optimization):

```
python hi_cft_safe_run.py --limit-mib 512 -- \
  python c8_physics_20260927/spectrum.py --order 24 --delta-cut 24 \
  --outdir /tmp/c8-physics-spectrum-replay
```

Existing output files are not overwritten. The address cap in this command is lower than the permanent machine limit; the launcher also retains its host-memory reserve, process-group watchdog, and two-CPU limit.

The spectrum exporter determines unflavoured graded dimensions and the stated thermal observables. It does not compute normalized three-point coefficients, crossing matrices of conformal blocks, defect eigenvectors, or branching characters with Cartan fugacities. The separate defect and correlator datasets below provide further information of those kinds in their stated scopes. In particular an exclusion plot that resolves Haagerup charges uses the specified defect-sector decomposition; ordinary unflavoured gaps alone do not supply its inputs.

3 Exact correlators and channel data

The vacuum algebra gives several explicit interacting chiral correlators, in addition to current and stress-tensor Ward identities. We give two four-point functions of the weight-two generator, two spin-three four-point functions, and selected intertwiner blocks for the lightest nonvacuum module. All normalizations below are fixed. The last family consists of chiral blocks obtained from the containing lattice theory; it does not specify the diagonal U theory's nonvacuum bulk four-point function.

3.1 Conventions and Ward identities

Write T for the total stress tensor, with

$$T(z)T(0) \sim \frac{4}{z^4} + \frac{2T(0)}{z^2} + \frac{\partial T(0)}{z}. \tag{14}$$

The two commuting affine $\mathfrak{sl}_2$ algebras have levels $k_L = 1$ and $k_R = 39$. We use Cartan currents with $H_L(z)H_L(0) \sim 2z^{-2}$ and $H_R(z)H_R(0) \sim 78z^{-2}$; their mixed two-point function vanishes. More generally, for invariant form κ normalized by $\kappa(H, H) = 2$, the currents obey

$$J^a(z)J^b(0) \sim \frac{k\kappa^{ab}}{z^2} + \frac{f^{ab}{}_c J^c(0)}{z}. \tag{15}$$

This fixes every vacuum current correlator recursively. For example, with $f^{abc} = f^{ab}{}_d\kappa^{dc}$, $\langle J^a(z_1)J^b(z_2)J^c(z_3)\rangle = kf^{abc}/(z_{12}z_{13}z_{23})$.

Let $\widehat{u} = u/\sqrt{12}$ and let $\widehat{u}^\dagger$ denote its chiral adjoint in the transported positive lattice form. The symbol † does not introduce an antiholomorphic field. This primary has weight two, Cartan charges $(q_L, q_R) = (1, 13)$, and

$$\langle \widehat{u}(z)\widehat{u}^\dagger(0)\rangle = z^{-4}, \tag{16}$$

$$\langle H_a(z_1)\widehat{u}(z_2)\widehat{u}^\dagger(z_3)\rangle = \frac{q_a}{z_{12}z_{13}z_{23}^3}, \tag{17}$$

$$\langle T(z_1)\widehat{u}(z_2)\widehat{u}^\dagger(z_3)\rangle = \frac{2}{z_{12}^2z_{13}^2z_{23}^2}. \tag{18}$$

Consequently the squared couplings to the unit Cartan currents are $1/2$ and $13/6$. A useful mixed four-point function is

$$\langle H_a(z_1)H_b(z_2)\widehat{u}(z_3)\widehat{u}^\dagger(z_4)\rangle = \frac{1}{z_{34}^4}\left[\frac{K_{ab}}{z_{12}^2} + q_a q_b\left(\frac{1}{z_{13}}-\frac{1}{z_{14}}\right)\left(\frac{1}{z_{23}}-\frac{1}{z_{24}}\right)\right]. \tag{19}$$

Here $K = \mathrm{diag}(2, 78)$. The unit stress tensor is $\widehat{T} = T/2$; both $C_{\widehat{T}\widehat{u}\widehat{u}^\dagger}$ and $C_{\widehat{T}\widehat{T}\widehat{T}}$ equal one in these conventions. Writing $y = 1 - x$, the normalized four-point functions of any unit Cartan current $\widehat{H}$ and of $\widehat{T}$ are

$$\langle \widehat{H}(\infty)\widehat{H}(1)\widehat{H}(x)\widehat{H}(0)\rangle = 1 + x^{-2} + y^{-2}, \tag{20}$$

$$\langle \widehat{T}(\infty)\widehat{T}(1)\widehat{T}(x)\widehat{T}(0)\rangle = 1 + x^{-4} + y^{-4} + x^{-2} + y^{-2} + x^{-1} + y^{-1}. \tag{21}$$

The first follows from Heisenberg Wick contractions, the second from the Virasoro Ward identity at central charge eight.

3.2 A three-exponential four-point function

In simple-root coordinates of $E_6 \oplus A_2$, put

$$a_1 = (0, 0, 0, 0, 0, -1, 1, 1), \tag{22}$$

$$a_2 = (0, 0, 0, 1, 1, -1, 0, 0), \tag{23}$$

$$a_3 = (1, 1, 0, 0, 0, -1, 0, 0), \tag{24}$$

$$u = e^{a_1} + \frac{\sqrt{13} + 3\sqrt{-3}}{20}e^{a_2} + \frac{\sqrt{13} - 3\sqrt{-3}}{20}e^{a_3}. \tag{25}$$

The simple-root Gram matrix has diagonal entries two and entries minus one along the edges $(0,1), (1,2), (2,3), (3,4), (2,5), (6,7)$, with indices starting at zero. Thus $(a_i, a_i) = 4$ and $(a_i, a_j) = 2$ for $i \neq j$. In the torus gauge $(5, 8, 9, 8, 5, 1, 2, 2)$, the three exponential squared norms are $4, 40, 40$. Their coefficients therefore contribute $4, 4, 4$ to $\|u\|^2 = 12$. In particular this calculation uses the transported positive form, rather than treating the three displayed coefficients as coordinates in an orthonormal basis.

The lattice Gaussian formula in a specified cocycle gauge is

$$\left\langle \prod_{j=1}^{n} e^{b_j}(z_j)\right\rangle = \delta_{\sum_j b_j, 0}\,\varepsilon(b_1, \ldots, b_n)\prod_{i<j}(z_i - z_j)^{(b_i, b_j)}. \tag{26}$$

Here the integer-lattice cocycle is

$$\varepsilon(a, b) = (-1)^{\sum_i a_i b_i + \sum_{i>j} a_i b_j G_{ij}}, \tag{27}$$

and the multivariable factor is its iterated product. These formulas also give a short independent derivation: expand each of the four fields, discard nonzero total charge, and multiply their six pairwise Gaussian factors. Of the $3^4 = 81$ terms, fifteen survive. All fifteen have coefficient $1/9$ after inserting the adjoint gauge and normalizing the four external fields.

Set the insertion points to $(\infty, 1, x, 0)$, with $\phi(\infty) = \lim_{R\to\infty} R^{2h}\phi(R)$. The two useful reflection-positive channels are

$$G_0(x) = \langle \widehat{u}^\dagger(\infty)\widehat{u}(1)\widehat{u}^\dagger(x)\widehat{u}(0)\rangle = \boxed{\frac{3 - 4x + 4x^2}{3x^4(1-x)^4}}, \tag{28}$$

$$G_+(x) = \langle \widehat{u}^\dagger(\infty)\widehat{u}^\dagger(1)\widehat{u}(x)\widehat{u}(0)\rangle = \boxed{\frac{x^2(4 - 4x + 3x^2)}{3(1-x)^4}}. \tag{29}$$

For example, the fifteen neutral contributions to the first formula sum to

$$\frac{1}{3x^4(1-x)^4}+\frac{2}{3x^2(1-x)^4}+\frac{2}{3x^4(1-x)^2}. \tag{30}$$

The three inequivalent orderings follow from $G_0(1-x)=G_0(x)$, $G_+(x)=x^{-4}G_0(1/x)$, and $G_+(1-x)$; their adjoint orderings agree. These are exact rational identities, not truncated series fits.

To display positive channel data, expand in global conformal blocks:

$$x^{2h_{\mathrm{ext}}}G(x)=A_0+\sum_{n\geq 1}A_n x^n\,{}_2F_1(n,n;2n;x). \tag{31}$$

For the two channels above, A_n is the sum of squared couplings to orthonormal global quasiprimaries of weight n in the appropriate charge sector. This statement uses the positive radial pairing. It does not identify A_n with a single primary coupling or a Virasoro block coefficient. The first neutral coefficients are

$$(A_0,A_1,A_2,A_3,A_4)=\left(1,\frac{8}{3},\frac{14}{3},\frac{58}{9},\frac{107}{15}\right). \tag{32}$$

The value $A_1=1/2+13/6$ reproduces the two Cartan-current exchanges. At weight two the total stress tensor contributes one, leaving $11/3$ as the squared-coupling sum to ambient Virasoro primaries. In the equal-charge channel the first nonzero coefficient is $A_6=4/3$, followed by $A_8=19/13$; odd weights vanish. Indeed, the leading state $\widehat{u}_{(-3)}\widehat{u}$ is supported on the three momenta a_i+a_j with $i<j$, each of norm squared twelve. It is a weight-six Virasoro primary of squared norm $4/3$. The normalized leading OPE coefficient can therefore be chosen $2/\sqrt{3}$. In particular the equal-charge OPE vanishes through orders x^0 and x^1 and starts at x^2.

The four orthogonal commuting stress tensors give additional resolved weight-two data. If T_i has central charge c_i and u has T_i-weight h_i, then $C^2_{\widehat{T}_i\widehat{u}\widehat{u}^\dagger}=2h_i^2/c_i$. Here the entries are

stress	c_i	h_i	squared unit-field coupling
T_L	1	1/4	1/8
T_R	117/41	195/164	325/328
e	2400/1271	180/1271	27/1271
f	70/31	13/31	169/1085

Their charges and weights sum to eight and two, respectively.

3.3 Potts and parafermion spin-three four-point functions

The embedded Potts and parafermion algebras contain canonical spin-three generators with parameters

$$(c,\lambda)=\left(\frac{4}{5},\frac{4}{13}\right)\quad\text{and}\quad(c,\lambda)=\left(\frac{8}{5},\frac{14}{473}\right). \tag{33}$$

These are the Potts and level-thirteen parafermion cases, in the normalization of the universal $\mathcal{W}(c,\lambda)$ algebra [47, 48]. Denote either spin-three field by a, its intrinsic stress tensor by T_c, and put

$$N=\frac{c}{3},\qquad b=a_{(1)}a,\qquad A=31-16\lambda(c+2),\qquad B=NA. \tag{34}$$

The exact checked singular self-OPE is

$$a(z)a(0)\sim\frac{N}{z^6}+\frac{2T_c}{z^4}+\frac{\partial T_c}{z^3}+\frac{b}{z^2}+\frac{\frac{1}{2}\partial b-\frac{1}{12}\partial^3T_c}{z}, \tag{35}$$

$$a_{(3)}b=Aa,\qquad \|b\|^2=B. \tag{36}$$

The positive normalization is chosen so that a has two-point coefficient N; write $\widehat{a} = a/\sqrt{N}$. The exact values of (A, B) are $(1119/65, 1492/325)$ for Potts and $(69283/2365, 554264/35475)$ for the parafermion generator.

Meromorphy, the OPE and crossing determine the complete correlator:

$$\begin{aligned} N^2\langle\widehat{a}(\infty)\widehat{a}(1)\widehat{a}(x)\widehat{a}(0)\rangle = {} & N^2\big[1 + x^{-6} + (1-x)^{-6}\big] \\ & + 2c\big[x^{-4} + (1-x)^{-4} + x^{-3} + (1-x)^{-3}\big] \\ & + B\big[x^{-2} + (1-x)^{-2}\big] + (2B - 2c)\big[x^{-1} + (1-x)^{-1}\big]. \end{aligned} \tag{37}$$

For completeness, the principal part at zero follows by pairing the five terms of (35) with the outer two fields. The Ward identity gives $\langle a(\infty)a(1)T_c(0)\rangle = 3N$; its first derivative is $6N$, and its third derivative is $72N$. Likewise $\langle a(\infty)a(1)b(0)\rangle = B$, whose first derivative is $4B$. These give residues $N^2, 2c, 2c, B, 2B - 2c$ at pole orders $6, 4, 3, 2, 1$. Interchange of the equal fields fixes the principal part at one. The identity channel at infinity fixes the constant N^2. The difference between the displayed function and the correlator is therefore a holomorphic function on the sphere vanishing at infinity. Both crossing generators $x \mapsto 1 - x$ and $x \mapsto 1/x$ can also be verified directly in the displayed expression.

The self-cubic coefficient $C_{\widehat{aaa}}$ vanishes. The squared coupling to the unit intrinsic stress tensor is $18/c$. To isolate the new weight-four channel from Virasoro descendants, put

$$\Lambda =: T_cT_c : -\frac{3}{10}\partial^2 T_c, \qquad \|\Lambda\|^2 = \frac{c(5c+22)}{10}, \qquad W_4^{\mathrm{prim}} = b - \frac{32}{5c+22}\Lambda - \frac{3}{10}\partial^2 T_c. \tag{38}$$

The squared $\widehat{a}\widehat{a}\Lambda/\|\Lambda\|$ coupling is $4608/[5c(5c+22)]$. Subtracting this contribution from the global weight-four coefficient gives

$$\gamma_4^2 = \frac{48(c+2)\{8 - \lambda(5c+22)\}}{c(5c+22)} = \begin{cases} 0, & \text{Potts}, \\ 60552/2365, & \text{level-thirteen parafermion}. \end{cases} \tag{39}$$

Thus the Potts extra weight-four primary is null, whereas the parafermion channel has a strictly positive extra coupling. This separation refers to the intrinsic Virasoro algebra of central charge c, not to the total central-charge-eight Virasoro algebra. The corresponding total global weight-four coefficients are $576/13$ and $21192/473$. Equation (37) supplies all higher global coefficients without a new high-weight mode calculation.

3.4 Selected blocks for the lightest module

The two B-lattice sectors $(0, +1)$ and $(0, -1)$ both restrict to the simple U-module c_1 of lowest weight $1/3$ and top dimension three. In the A_2 simple-root basis its three fundamental weights are

$$\beta_0 = (2/3, 1/3), \quad \beta_1 = (-1/3, 1/3), \quad \beta_2 = (-1/3, -2/3), \qquad (\beta_i, \beta_j) = \delta_{ij} - \frac{1}{3}. \tag{40}$$

Their H_R charges are $q = (2, 0, -2)$; their H_L charges vanish. Choose paired lattice intertwiners with unit leading vacuum coefficient. On the branch $0 < x < 1$ with positive real powers, the nine selected four-point blocks are

$$\mathcal{F}_{ij}(x) = \langle V_{-\beta_i}(\infty)V_{\beta_i}(1)V_{-\beta_j}(x)V_{\beta_j}(0)\rangle = \boxed{x^{-2/3}(1-x)^{-(\beta_i,\beta_j)}}. \tag{41}$$

Thus the diagonal entries are $[x(1-x)]^{-2/3}$ and the off-diagonal entries are $x^{-2/3}(1-x)^{1/3}$. Analytic continuation requires a specified path; the fractional powers must not be replaced by a single-valued rational function.

In the $x \to 0$ channel these blocks factor through the B vacuum sector, whose restriction is $U \oplus X$, with X the module denoted b in the modular table. The first coefficient of the reduced block is (β_i, β_j). The decomposition of the intermediate weight-one Heisenberg vector gives the explicit split

$$(\beta_i, \beta_j) = \underbrace{\frac{q_i q_j}{78}}_{U \text{ current exchange}} + \underbrace{(\beta_i, \beta_j) - \frac{q_i q_j}{78}}_{X \text{ top exchange}} . \tag{42}$$

For $i = j = 0$ these two positive coefficients are $2/39$ and $8/13$. The three-by-three X Gram matrix has eigenvalues $0, 1, 35/39$, providing a direct positive-channel check. Off-diagonal entries need not be positive: they are overlaps, rather than squared norms. The six ordered triples with three distinct weights have nonzero three-point functions proportional to $(z_{12}z_{13}z_{23})^{-1/3}$, with coefficient of absolute square one in this normalization. Their phases depend on the lattice-intertwiner cocycle and chosen bases. Two distinct positive weights add to the negative of the remaining weight and hence exhibit a c_1 channel.

Restriction of a lattice intertwiner gives a genuine U intertwiner, so these are nonvacuum chiral data, beyond Ward identities. Their particular linear combinations and normalizations remember the extension $U \subset B$. They neither form a complete basis of all U channels nor determine the diagonal theory's local bulk correlator $\langle \Phi\Phi\Phi\Phi \rangle$ for its lightest scalar. That quantity additionally requires channel and multiplicity bases, the full relevant intertwining operators and the left-right sewing coefficients. In particular, multiplying one selected block by its complex conjugate has not been justified as that bulk correlator.

3.5 Reproducible data and verification scope

The bounded standard-library script `c8_physics_20260927/correlators.py` exports `c8_physics_20260927/correlators.json`, together with the tables `c8_physics_20260927/correlators.terms.tsv`, `c8_physics_20260927/correlators.global_blocks.tsv`, and `c8_physics_20260927/correlators.samples.tsv`. The first table retains all 486 assignments across the six charge orderings, including vanishing terms, their cocycle signs and exact coefficients in $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$. The second gives the exact global-block decomposition of the two u channels, both spin-three correlators and the two Ward correlators through the recorded cutoff, with default weight 32; the third gives exact values at five rational cross-ratios. Polynomial crossing identities, reconstruction of the truncated series, positivity in the specified channels and the lightest-module Gram decomposition are checked exactly.

The script checks the three-exponential input against `hi_cft_20260923_c8_exact_fields.json`. The spin-three parameters and norms are read from `hi_cft_20260923_proof_pf13_ope.json`; their singular OPE provenance is `hi_cft_20260923_proof_pf13_upper_poles.py` and its saved report. Source hashes are included. The export does not import the lattice mode engine or repeat its higher-grade computations. Its finite positivity tests provide reproducible consistency checks; all-order positivity follows here from the positive VOA realization, rather than from extrapolating those tests. The separately exported low-weight coupling tensor resolves many of the summed weight-two channels above into individual normalized primary couplings.

4 A complete weight-two chiral three-point dataset

The coefficient 120 of the vacuum character is small enough to resolve its first interacting primary space completely. Its 113 Virasoro primaries comprise the 96 fields in the eight charged current multiplets of the construction, together with 17 fields obtained from the currents and

stress tensors. Here "charged" refers to a nontrivial current multiplet; it does not exclude zero-weight components within that multiplet.

4.1 Actual fields and normalization

Label the eight original highest fields by $v_{m,0,0}$, $m = 0, \dots, 7$, in the order

$$(n_m, r_m) = (1,5), (1,9), (1,9), (1,13), (2,2), (2,6), (2,8), (2,14).$$

These are representation dimensions for the two current algebras. The two $(1,9)$ copies are distinct and have different coset-stress eigenvalues. In the last multiplet the highest field is the defining field u, with the normalization inherited from its saved lattice vector. Define all components by

$$v_{m,p,q} = (F_L)_0^p (F_R)_0^q v_{m,0,0}, \quad 0 \le p < n_m, \quad 0 \le q < r_m. \tag{43}$$

No factorials divide the lowering powers. The Cartan charges are $(n_m - 1 - 2p, r_m - 1 - 2q)$.

The remaining highest fields are

$$v_{8,0,0} = (E_R)_{-1} E_R, \qquad v_{9,0,0} = (E_L)_{-1} E_R,$$

of current types $(1,5)$ and $(3,3)$, and three current singlets. To specify the latter, use the four mutually commuting stress tensors

$$(T_L, T_R, T_e, T_f), \qquad (c_L^{\text{aff}}, c_R^{\text{aff}}, c_e, c_f) = \left(1, \frac{117}{41}, \frac{2400}{1271}, \frac{70}{31}\right).$$

In the next formula abbreviate the first two central charges by c_l, c_r, to distinguish them from the spacetime charges $c_L = c_R = 8$. The singlets are

$$v_{10,0,0} = c_r T_L - c_l T_R, \tag{44}$$
$$v_{11,0,0} = c_e (T_L + T_R) - (c_l + c_r) T_e, \tag{45}$$
$$v_{12,0,0} = c_f (T_L + T_R + T_e) - (c_l + c_r + c_e) T_f. \tag{46}$$

Because $\langle T_i, T_j \rangle = \delta_{ij} c_i / 2$, these three vectors are mutually orthogonal and orthogonal to the total stress tensor. A combination $\sum a_i T_i$ is a total-Virasoro primary exactly when $\sum a_i c_i = 0$, proving their primarity directly.

The right spin-two multiplet is the traceless symmetric current product. The cross-current multiplet is primary because the two current factors commute. At level one the left spin-two current product is null, so there is no additional $(5,1)$ multiplet. Together these add $5 + 9 + 3 = 17$ fields. The exact calculation checks the Virasoro-primary conditions, all possible off-diagonal Gram entries within equal Cartan weights, and positive norms for all 113 vectors. Distinct Cartan weights are orthogonal by the compact real form. Their independence and the character count $120 - 6 - 1 = 113$ prove completeness: the subtractions remove six current derivatives and the total stress tensor.

Put $N_i = \langle v_i, v_i \rangle_g$ and $\widehat{v}_i = v_i / \sqrt{N_i}$, choosing the positive square root. The exported coefficients use the antilinear first-slot Hermitian pairing and the transported lattice PCT of (84):

$$C_{ij}^{\ k} = \langle \widehat{v}_k, (\widehat{v}_i)_1 \widehat{v}_j \rangle_g = \frac{\langle v_k, (v_i)_1 v_j \rangle_g}{\sqrt{N_i N_j N_k}}. \tag{47}$$

Equivalently this is the three-point coefficient with the conjugate of $\widehat{v}_k$ at infinity. The singular OPE contains $z^{-2} C_{ij}^{\ k} \widehat{v}_k(0)$, in addition to current, stress-tensor and descendant contributions. The unnormalized numerator and norm product lie in $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$; the positive square-root normalization is left explicit. Thus the table retains the complex phase, not only $|C|^2$.

4.2 From directly computed seeds to the full tensor

Let P be orthogonal projection onto this primary space and write $P((v_i)_1 v_j) = \sum_k p_{ij}^k v_k$. Since current zero modes commute with the total Virasoro algebra, this projection is equivariant. For either lowering operator F the derivation identity gives

$$P((Fv_i)_1 v_j) = F\,P((v_i)_1 v_j) - P((v_i)_1 F v_j). \tag{48}$$

Starting with i highest in each of the thirteen multiplets determines the entire tensor recursively. In the raw basis (43), F simply advances one lowering exponent; its raising partner acts by $Ev_p = p(n-p)v_{p-1}$. Equation (48) therefore uses only exact additions and multiplications of known coefficients. Orthogonal projection is essential here: we do not discard descendants before justifying equivariance.

The direct seed calculation evaluates 1137 actual lattice mode products and 3512 charge-compatible projections, of which 1639 are nonzero. The completed tensor has

$$113^3 = 1442897 \quad \text{ordered entries,} \qquad 18365 \quad \text{nonzero entries.}$$

All omitted entries in the sparse table are exactly zero within this complete weight-two primary tensor. Every raising and lowering Ward identity is checked, including at the tops and bottoms of multiplets: there are 51076 pairwise checks. Primary skew-symmetry gives a further 12769 checks, and 33 additional direct lattice products test components outside the initial seed slice. These checks verify the specified finite tensor; they are not a test of arbitrary-weight associativity or full-CFT sewing.

A cross-check connects this tensor to an independent four-point calculation. The weight $(-1,-13)$ component is unique, so $\widehat{v}_{7,1,13}$ equals $\widehat{u}^\dagger$ up to a phase. Summing its nine nonzero primary projections gives exactly

$$\sum_{k=1}^{113} |C_{\widehat{u},\widehat{u}^\dagger}^{k}|^2 = \frac{11}{3}.$$

Adding the total-stress contribution one reproduces the global weight-two coefficient $14/3$ in the Gaussian four-point function (28). The three stress-difference contributions separately agree with their Ward values. The small standalone checker `check_physics_crosslinks.py` verifies these sums without importing the lattice mode engine.

Several entries are short enough to check without navigating a large file. The indices below are the explicit (m,p,q) labels above.

i	j	k	$\lvert C_{ij}^{k}\rvert^2$
$(0,0,0)$	$(0,0,2)$	$(0,0,0)$	4356/8645
$(1,0,0)$	$(1,0,6)$	$(0,0,0)$	39605/717288
$(7,0,0)$	$(7,1,13)$	$(10,0,0)$	13/316
$(8,0,0)$	$(8,0,3)$	$(8,0,1)$	1225/741
$(9,0,0)$	$(9,2,0)$	$(8,0,0)$	76/39
$(10,0,0)$	$(10,0,0)$	$(10,0,0)$	23104/9243
$(12,0,0)$	$(12,0,0)$	$(12,0,0)$	2916/3115

For example, for $A = c_r T_L - c_l T_R$ one has

$$\langle A, A\rangle = \frac{c_l c_r (c_l + c_r)}{2}, \qquad |C_{AAA}|^2 = \frac{8(c_r - c_l)^2}{c_l c_r (c_l + c_r)} = \frac{23104}{9243}.$$

This follows just from the two commuting stress OPEs and provides a human-sized independent normalization check on the lattice table.

The principal files are `c8_physics_20260927/full_weight_two/low_weight_basis.json`, which stores every actual lattice vector and norm; `low_weight_products.jsonl`, which stores each seed product; and `complete_primary_couplings.tsv`, which stores every nonzero normalized coefficient by its exact numerator and norm product. The accompanying JSON report documents the recurrence checks and source hashes. The earlier 96-field slice remains available as an intermediate audit, but the 113-field table is the complete release.

These are chiral vacuum-module observables of U, hence also observables of its selected extension. They give local spinning fields in either bulk theory. Factorized products of these fields give additional full-CFT vacuum-sector correlators. They do not supply the three-point constants of generic nonvacuum diagonal-sector primaries.

5 Modular data, topological defects, and conformal boundaries

We use the positive spherical structure and the Evans–Gannon ordering

$$(0, b, a, c_0, c_1, c_2, d_1, d_2, d_3, d_4, d_5, d_6).$$

The translation to the labels used in the Haagerup bootstrap literature is

$$(1, \pi_1, \pi_2, \sigma_1, \sigma_2, \sigma_3, \mu_5, \mu_2, \mu_1, \mu_3, \mu_6, \mu_4).$$

The final permutation is fixed by the distinct thirteenth-root twists in [11, arXiv version 2, Eq. (30)]; retaining the same numerical subscripts for the d and μ labels would be incorrect. In particular, a is a simple object of the center, whereas α below denotes the invertible generator of the six-object Haagerup category. These two objects have different dimensions and different roles.

Put $r = \sqrt{13}$, $d = (3 + r)/2$, $s = 2 + 3d$, and

$$A = \frac{1}{6} - \frac{1}{2r}, \qquad B = \frac{1}{6} + \frac{1}{2r}, \qquad u = \frac{1}{3}, \qquad v = \frac{1}{r}.$$

In blocks of sizes $1, 1, 4, 6$, the complete normalized matrix is

$$S = \begin{pmatrix} A & B & u\mathbf{1}_4^{\mathrm{t}} & v\mathbf{1}_6^{\mathrm{t}} \\ B & A & u\mathbf{1}_4^{\mathrm{t}} & -v\mathbf{1}_6^{\mathrm{t}} \\ u\mathbf{1}_4 & u\mathbf{1}_4 & C & 0 \\ v\mathbf{1}_6 & -v\mathbf{1}_6 & 0 & D \end{pmatrix}, \qquad C = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 & -1 \\ -1 & 2 & -1 & -1 \\ -1 & -1 & -1 & 2 \\ -1 & -1 & 2 & -1 \end{pmatrix}, \tag{49}$$

where

$$D_{ij} = -\frac{2}{\sqrt{13}} \cos \frac{2\pi ij}{13}, \qquad 1 \le i, j \le 6. \tag{50}$$

The topological twist matrix and the character matrix are distinct:

$$\Theta = \operatorname{diag}\left(1, 1, 1, 1, e^{2\pi i/3}, e^{4\pi i/3}, e^{2\pi i 6/13}, e^{2\pi i 11/13}, e^{2\pi i 2/13}, e^{2\pi i 5/13}, e^{2\pi i 7/13}, e^{2\pi i 8/13}\right), \tag{51}$$

$$T_{\mathrm{ch}} = e^{-2\pi i/3}\Theta. \tag{52}$$

Thus the d_j twist is $e^{2\pi i 6 j^2/13}$. The conformal weight of d_3 is $15/13$, although its twist exponent is $2/13$. These are the standard positive Haagerup modular data [9]; no numerical eigenvector choice enters these formulas.

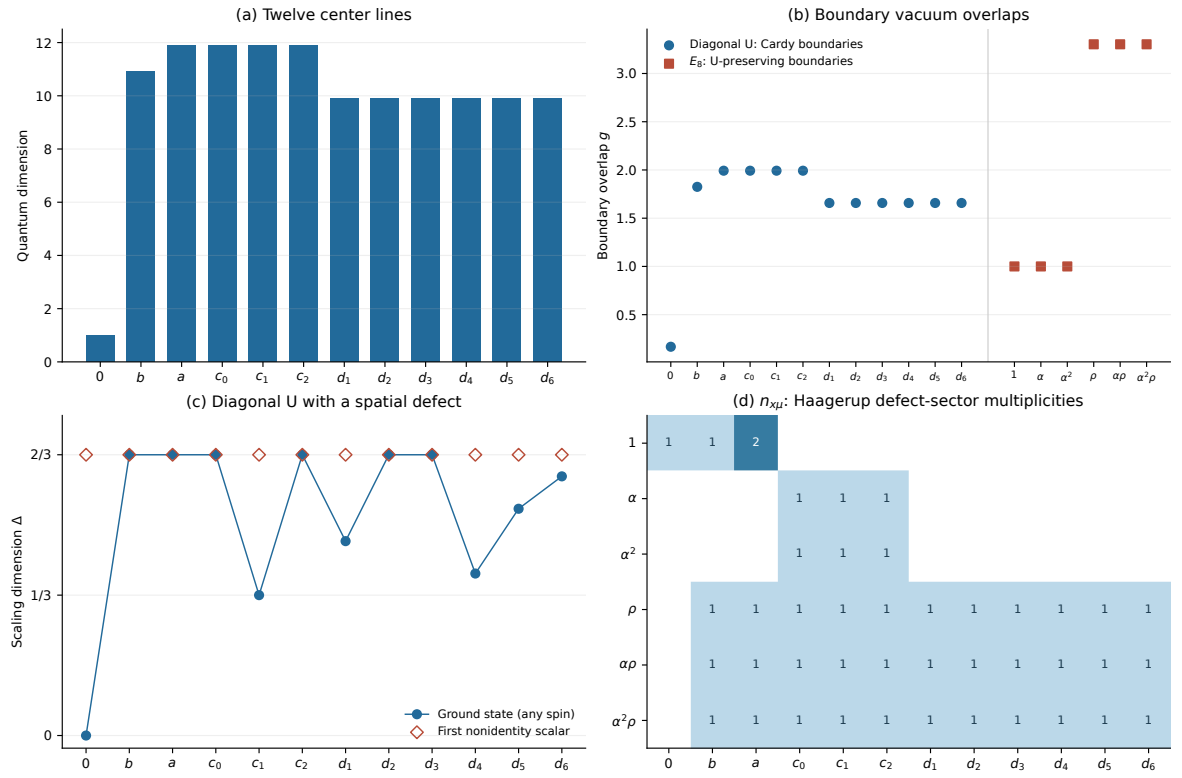


Figure 3: Exact modular and defect observables, displayed using the positive embedding of the exported algebraic data. The ground-state gap in a twisted space and its scalar gap are distinct; the latter is 2/3 for all twelve diagonal-theory lines. The final panel records the integer forgetful multiplicity matrix for the selected extension.

The categorical dimensions and total quantum dimension are

$$(\mathrm{qdim}\, M_i)_i = (1, d^2, s, s, s, s, 3d, 3d, 3d, 3d, 3d, 3d), \qquad \mathcal{D} = \frac{1}{S_{00}} = 3s. \tag{53}$$

Here $\mathcal{D}^2$, rather than $\mathcal{D}$, is the global dimension of $\mathrm{Rep}(U)$. Every simple object is self-dual because $S^2 = 1$. The six-object Haagerup category has global dimension $3 + 3d^2 = 3s = \mathcal{D}$.

5.1 Exact fusion and diagonal-theory defect spectra

Consider first the diagonal full CFT

$$\mathcal{H}_U = \bigoplus_i M_i \otimes \overline{M_i}, \qquad Z_U(\tau, \bar\tau) = \sum_i |\chi_i(\tau)|^2. \tag{54}$$

Its twelve chiral-algebra-preserving Verlinde lines form the fusion category underlying $\mathrm{Rep}(U) = Z(\mathcal{H}_3)$. Their multiplication and their action on bulk sectors are

$$N_{ab}{}^{c} = \sum_i \frac{S_{ai} S_{bi} S_{ci}}{S_{0i}}, \qquad \widehat{L}_a\big|_{M_i \otimes \overline{M_i}} = \lambda_a(i)\,\mathrm{id}, \qquad \lambda_a(i) = \frac{S_{ai}}{S_{0i}}. \tag{55}$$

Complex conjugation is absent from the displayed Verlinde sum because S is real. In particular $\lambda_a(0) = \mathrm{qdim}(M_a)$. The line insertion and its modular transform give

$$Z_U^{[a]}(\tau,\bar\tau) = \sum_i \lambda_a(i)|\chi_i(\tau)|^2, \tag{56}$$

$$Z_{U,[a]}(\tau,\bar\tau) = \sum_{i,j} N_{ai}{}^{j}\chi_i(\tau)\overline{\chi_j(\tau)}. \tag{57}$$

The first expression is a trace with a line operator and its coefficients need not be positive. The second is the partition function of the Hilbert space with an a defect intersecting the spatial circle; all its multiplicities are nonnegative integers. This is the usual distinction between a twined trace and a twisted Hilbert space [4, 1]. Indeed, applying S to the first expression makes the coefficient of $\chi_i\overline{\chi_j}$ equal to the sum in (55), proving the second formula directly.

The attached data give all 1728 integers $N_{ab}{}^{c}$, all 144 line eigenvalues, and all twelve matrices in (57). They also give the lowest scaling dimension and the lowest nonidentity scalar dimension in each twisted space. For orientation, some compact fusion products are

$$b^2 = 1 + b + a + \sum_{j=0}^{2} c_j + \sum_{j=1}^{6} d_j, \tag{58}$$

$$a^2 = 1 + 2b + 2a + \sum_{j=0}^{2} c_j + \sum_{j=1}^{6} d_j, \tag{59}$$

$$d_i^2 = 1 + a + \sum_{j=0}^{2} c_j + \sum_{\substack{1\le j\le 6\\ j\neq[2i]}} d_j. \tag{60}$$

Here $[m]$ is the representative of $\{m,-m\}$ modulo 13 in $\{1,\dots,6\}$. The gap data are particularly simple:

Line	ground Δ	multiplicity	scalar gap	multiplicity
0	0	1	2/3	9
b	2/3	18	2/3	18
a, c_0	2/3	9	2/3	9
c_1	1/3	6	2/3	18
c_2	2/3	63	2/3	9
d_1	6/13	14	2/3	9
d_2, d_3	2/3	9	2/3	9
d_4	5/13	10	2/3	9
d_5	7/13	26	2/3	9
d_6	8/13	28	2/3	9

The scalar condition is $h = \bar h$, not merely integral spin. For a summand $M_i \otimes \overline{M_j}$ it first requires $h_i - h_j \in \mathbb{Z}$. Among the supplied lowest weights the only unequal pair with this property is 0 versus 1; its scalar ground states use U_1 on the vacuum side. Consequently all the reported lowest scalar multiplicities follow from the twelve top dimensions and $\dim U_1 = 6$, without an extrapolation of character coefficients.

A useful symmetry test follows immediately from invertibility of S. If a bulk sector i is fixed by every line, its eigenvalues must obey $\lambda_a(i) = \lambda_a(0)$ for every a. The ith column of S would then be proportional to the vacuum column; hence $i = 0$. Thus the fully Verlinde-invariant space is $U \otimes \overline{U}$. Its first nonidentity scalars have weights $(1,1)$ and multiplicity $6^2 = 36$. Its scalar gap is 2, even though the unrestricted diagonal theory has scalar gap 2/3.

5.2 Cardy boundaries and their annuli

The diagonal theory has twelve elementary boundaries preserving U. With the usual normalized Ishibashi states, Cardy's solution reads

$$|a\rangle = \sum_i \frac{S_{ai}}{\sqrt{S_{0i}}} |i\rangle\!\rangle, \qquad Z_{ab}^{\text{open}}(q) = \sum_i N_{ab}{}^i \chi_i(q). \tag{61}$$

All labels are self-dual, so no conjugate label is needed in the annulus formula. Modular transformation of the closed-channel overlap gives the same Verlinde sum as above, providing a direct check of the open channel multiplicities [3].

The boundary vacuum overlaps, conventionally denoted by g_a, are

$$g_a = \frac{S_{a0}}{\sqrt{S_{00}}} = \frac{\mathrm{qdim}(M_a)}{\sqrt{3s}}, \qquad \log g_a \text{ is the boundary entropy.} \tag{62}$$

They therefore take four values, in the proportions $1 : d^2 : s : 3d$. The exact squared values lie in $\mathbb{Q}(\sqrt{13})$; the dataset supplies these as exact algebraic numbers as well as decimal values of g_a for plotting. The 144 annulus entries include the identity multiplicity, the lowest boundary weight, and the first positive boundary weight. The last quantity includes weight-one currents in the vacuum representation. Removing the entire vacuum representation would define a different gap. For the vacuum Cardy boundary the first positive boundary weight is 1, with six currents. Every other elementary Cardy boundary has boundary-preserving fields of weight $1/3$; there are six for boundary c_1 and three for each of the other ten boundaries. These are relevant boundary perturbations, without implying a particular endpoint of their renormalization-group flows.

6 The Haagerup symmetry of the holomorphic extension

The twelve diagonal-theory lines above must be distinguished from the six Haagerup lines obtained after extension. Let $H' = V_{E_8}$ be the selected holomorphic extension, with underlying U-module

$$H'|_U \simeq M_0 \oplus M_b \oplus 2M_a, \qquad \chi_{E_8} = \chi_0 + \chi_b + 2\chi_a. \tag{63}$$

Its algebra object is denoted by A, and $\mathcal{C} = \mathrm{Rep}(U)_A \simeq \mathcal{H}_3$ is the category of all right A-modules, including the nonlocal ones. The local A-module category is trivial. The six simples of $\mathcal{C}$ are $1, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho$, with

$$\alpha^3 = 1, \qquad \rho\alpha = \alpha^2\rho, \qquad \rho^2 = 1 + \rho + \alpha\rho + \alpha^2\rho. \tag{64}$$

The full E_8 theory has the corresponding $\mathcal{C} \boxtimes \mathcal{C}^{\text{op}}$ defects. In this section we select the first factor, leaving the antiholomorphic H' theory as a spectator. This convention specifies a single $\mathcal{H}_3$ action to which the twelve-sector Haagerup bootstrap applies.

6.1 Forgetful multiplicities and the physical sector assignment

Under the center equivalence $\mathrm{Rep}(U) \simeq Z(\mathcal{C})$, the free-module functor is the central forgetful functor $F : Z(\mathcal{C}) \to \mathcal{C}$. Writing $X = \rho \oplus \alpha\rho \oplus \alpha^2\rho$, its values are

$$F(0) = 1, \quad F(b) = 1 \oplus X, \quad F(a) = 2 \cdot 1 \oplus X, \quad F(c_j) = \alpha \oplus \alpha^2 \oplus X, \quad F(d_j) = X. \tag{65}$$

These are the Haagerup branching rules of [2, Eq. (4.49)]. They specify the entire 6×12 matrix $n_{x\mu} = \dim \mathrm{Hom}_{\mathcal{C}}(F(M_\mu), x)$. For example its first three rows are

$$n_1 = (1,1,2,0,0,0,0,0,0,0,0,0), \tag{66}$$
$$n_\alpha = n_{\alpha^2} = (0,0,0,1,1,1,0,0,0,0,0,0), \tag{67}$$

and each of the other three rows is $(0,1,1,1,1,1,1,1,1,1,1,1)$.

Here is an intrinsic reason for the sector identification, beyond a check of modular covariance. If R is restriction of an A-module to U, free–forgetful adjunction gives

$$\mathrm{Hom}_U(M_\mu, Rx) \simeq \mathrm{Hom}_{\mathcal{C}}(F(M_\mu), x), \qquad Rx \simeq \bigoplus_\mu n_{x\mu} M_\mu. \tag{68}$$

In the standard rational-CFT construction of the selected first-factor defects, their Hilbert spaces are $Rx \otimes \overline{H'}$. The tube-algebra decomposition is therefore

$$\mathcal{H}_x \simeq \bigoplus_\mu W_x^\mu \otimes \mathcal{V}_\mu, \qquad W_x^\mu = \mathrm{Hom}_{\mathcal{C}}(F(M_\mu), x), \qquad \mathcal{V}_\mu = M_\mu \otimes \overline{H'}. \tag{69}$$

This is the multiplicity-space convention of [1, Eq. (4.1)]. A full tube-projector image also contains the finite-dimensional factor W_x^μ; it is not in general a single copy of $\mathcal{V}_\mu$. In particular $\dim W_1^a = 2$. The physical interpretation in (69) uses the canonical extension and its rational-CFT defect construction; modular covariance alone would not establish that interpretation for an arbitrary vector-valued function.

It follows that the twelve bootstrap partition functions and the six twisted Hilbert-space partition functions are

$$Z_\mu^{3d}(\tau,\bar\tau) = \chi_\mu(\tau)\overline{\chi_{E_8}(\tau)}, \qquad Z_x(\tau,\bar\tau) = \sum_\mu n_{x\mu} Z_\mu^{3d}(\tau,\bar\tau). \tag{70}$$

For $x = 1$, (63) recovers $Z_1 = |\chi_{E_8}|^2$. Since χ_{E_8} is S-invariant and has T multiplier $e^{-2\pi i/3}$, these functions transform by (S, Θ): the left and right central-charge phases cancel. This also checks the spin convention used in the bootstrap.

6.2 Untwisted charges and scalar gaps

On the three untwisted multiplicity factors, one may choose bases in which the line actions are

	$\widehat{\alpha}$	$\widehat{\rho}$
M_0	1	d
M_b	1	$-d^{-1}$
$\mathbb{C}^2 \otimes M_a$	$\begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\omega = e^{2\pi i/3}$. (71)

The matrices obey (64); for the two-dimensional block the three noninvertible matrices sum to zero, so $\rho^2 = 1$ on that block is consistent with the categorical fusion rule. The resulting first-factor twined traces include

$$Z^{[\alpha]} = (\chi_0 + \chi_b - \chi_a)\overline{\chi_{E_8}}, \tag{72}$$
$$Z^{[\rho]} = (d\chi_0 - d^{-1}\chi_b)\overline{\chi_{E_8}}. \tag{73}$$

They act trivially on the antiholomorphic factor. Squaring their holomorphic factors in absolute value instead would correspond to particular paired left/right insertions, rather than the selected first-factor action.

Every antiholomorphic E_8 weight is an integer. Thus the sectors $c_1, c_2, d_1, \ldots, d_6$ in (70) contain no scalar states at any grade. Their nonintegral left weights cannot equal an integer right weight. The remaining sectors $0, b, a, c_0$ have their first nonidentity scalars at $(h, \bar{h}) = (1, 1)$:

μ	0	b	a	c_0
Δ_{scalar}	2	2	2	2
multiplicity in $\mathcal{V}_\mu$	$6 \cdot 248$	$80 \cdot 248$	$81 \cdot 248$	$81 \cdot 248$.

(74)

The $\mathcal{H}_3$-invariant scalar gap is therefore 2, and so is the scalar gap in $\mathcal{V}_{\pi_2} \cup \mathcal{V}_{\sigma_1}$. The physical untwisted $\mathbb{Z}_3$-charged space contains two copies of $\mathcal{V}_{\pi_2}$, with scalar multiplicity $2 \cdot 81 \cdot 248$. The union used in the bootstrap is a union of tube multiplicity spaces; it must not be identified with that entire untwisted charged space.

6.3 Six boundaries of the extended theory

The same module category supplies six elementary boundaries of the full E_8 theory which preserve U. For $x, y \in \mathcal{C}$, their open state space is $R(x^* \otimes_A y)$, hence

$$Z_{xy}^{E_8,\text{open}}(q) = \sum_{z,\mu} N_{x^*y}{}^{z} n_{z\mu} \chi_\mu(q). \tag{75}$$

The invertible duals are $\alpha^* = \alpha^2$; the three noninvertible objects are self-dual. This dual must be retained in the annulus formula, even though it could be omitted for the twelve self-dual center labels.

The normalization of the boundary entropy can be derived without an unknown proportionality constant. Restriction multiplies categorical dimensions by $\operatorname{qdim}(A) = \mathcal{D}$, so the coefficient of the vacuum in the modular transform of (75) is

$$S_{00} \operatorname{qdim}_{\text{Rep}(U)} R(x^* \otimes_A y) = \operatorname{qdim}_{\mathcal{C}}(x) \operatorname{qdim}_{\mathcal{C}}(y).$$

Reflection positivity chooses the positive overlaps. Thus

$$(g_1, g_\alpha, g_{\alpha^2}, g_\rho, g_{\alpha\rho}, g_{\alpha^2\rho}) = (1, 1, 1, d, d, d). \tag{76}$$

In particular the ordinary E_8 boundary has $g_1 = 1$ and self-annulus χ_{E_8}. Its first positive boundary weight is 1, with 248 currents. The same holds for the other two invertible boundaries. Each noninvertible boundary instead has nine fields of weight $1/3$ in its self-annulus. Between distinct noninvertible boundaries there are twelve ground fields of weight $1/3$; between two distinct invertible boundaries, or an invertible and a noninvertible boundary, there are three. All 36 annuli and their exact g values are supplied.

6.4 What the attached exact data check

The exporter `c8_physics_20260927/defects.py` works in $\mathbb{Q}(\zeta_{13})$ and represents every field element by twelve rational coefficients. Its square root is the explicit Gauss sum

$$\sqrt{13} = \sum_{j=1}^{12} \left(\frac{j}{13}\right) \zeta_{13}^{j}.$$

It checks $S^2 = 1$, $(S\Theta)^3 = 1$, exact nonnegative integrality of all Verlinde coefficients, associativity, and multiplicativity of dimensions. The 6×12 forgetful table is independently

checked as a based-ring homomorphism, and its vacuum row is checked to be S- and Θ-invariant. No coefficient is inferred by rounding. Decimal columns serve only as plot coordinates. The guarded run completed all these checks successfully; the saved JSON includes the generator hash and the precise normalization of every exported array.

The output files contain the complete fusion, line, annulus, twisted, and tube-sector recipes. These checks verify consequences of the given modular data and character tops. They neither replace the construction of U nor reconstruct associators, braidings, or operator-product coefficients from modular data alone.

7 Perturbations, bootstrap coordinates, and use of the data

7.1 Dimension-two operators and controlled deformations

The absence of a relevant symmetry-invariant scalar has a precise but limited meaning: it excludes an invariant scalar perturbation with linearized RG eigenvalue $2-\Delta>0$. It does not settle the flow of the dimension-two operators. For the diagonal theory all 36 products of vacuum currents are invariant under the full Verlinde category; their nonabelian OPEs must still be taken into account.

There is nevertheless a concrete exactly marginal subfamily. Choose the two Cartan currents and normalize them by

$$j^1 = H_L/\sqrt{2}, \qquad j^2 = H_R/\sqrt{78}, \qquad j^a(z)j^b(0) \sim \frac{\delta^{ab}}{z^2}.$$

Here L, R label the two affine factors within one chirality, not the two spacetime chiralities. The four local scalar fields

$$\mathcal{O}_{ab}(z,\bar{z}) = j^a(z)\overline{j^b}(\bar{z}), \qquad a,b \in \{1,2\}, \tag{77}$$

have the seed-point two-point functions

$$\langle \mathcal{O}_{ab}(z,\bar{z})\mathcal{O}_{cd}(0)\rangle = \frac{\delta^{ac}\delta^{bd}}{|z|^4}.$$

Thus the Zamolodchikov metric in this operator normalization is the identity. The abelian current-current deformation criterion supplies four simultaneous local exactly marginal directions [6, Section 2]. The associated current deformation space is locally $O(2,2)/(O(2)\times O(2))$, before duality identifications. We do not identify the complete conformal manifold or its global quotient. Although these operators are categorical singlets at the rational point, extending all topological lines and their junction data along the deformation requires a separate argument. Exact marginality alone does not prove that the entire deformed family retains the specified Haagerup symmetry. In particular the count 36 is not the dimension of a claimed Haagerup-preserving conformal manifold.

The ordinary full $E_{8,1}$ theory has eight Cartan currents in each chirality. Applying the same criterion gives the familiar 64 toroidal deformation directions, locally $O(8,8)/(O(8)\times O(8))$, at its enhanced-symmetry point. This statement concerns the ordinary CFT; it does not identify which deformations transport the specified Haagerup defect category.

7.2 Coordinates for published exclusion plots

For the ordinary scalar-gap bootstrap, allowing conserved currents, the diagonal theory gives $(c, \Delta_{\text{scalar}}) = (8, 2/3)$. The holomorphic extension gives $(8,2)$, the familiar $E_{8,1}$ point at the

scalar-bound kink in Collier–Lin–Yin [5, Figure 8 and Section 3.3]. Its ordinary bulk partition function is the known E_8 one; the additional information here is the specified Haagerup action and its relation to the explicit subalgebra. The diagonal theory lies well below that scalar upper bound. Chiral currents instead give the E_8 theory an all-primary gap of one, so substituting that number in a scalar-gap plot would compare different observables.

For the Haagerup bootstrap of Albert–Honda–Kaidi–Zheng, the physical sector assignment in Section 6 supplies

$$(c, \Delta^{\text{scalar}}_{\mathcal{V}_1}) = (8, 2), \qquad (c, \Delta^{\text{scalar}}_{\mathcal{V}_{\pi_2} \cup \mathcal{V}_{\sigma_1}}) = (8, 2). \tag{78}$$

These are respectively the observables in Figures 2 and 3 of [11]. The multiplicity spaces $\mathcal{V}_\mu$ exclude the finite-dimensional tube representation factors; the factor two in the untwisted π_2 block must not be counted twice when constructing the bootstrap vector. The 12-sector vector for this comparison is $\chi_\mu \overline{\chi_{E_8}}$, not the diagonal collection $|\chi_\mu|^2$. The plots are numerical exclusions at the published truncation, and our data do not rerun the associated semidefinite program. In particular, the common value two is not by itself a claim of saturation of either Haagerup bound.

Figure 4 places these two points on the original published panels. Each point lies visibly below its exclusion curve, in the shaded allowed region. We use this as a consistency comparison, without extracting a numerical curve height or claiming that the bounds single out this theory. The marker positions are calibrated from the original axis ticks; the curves themselves are reproduced without interpolation or refitting.

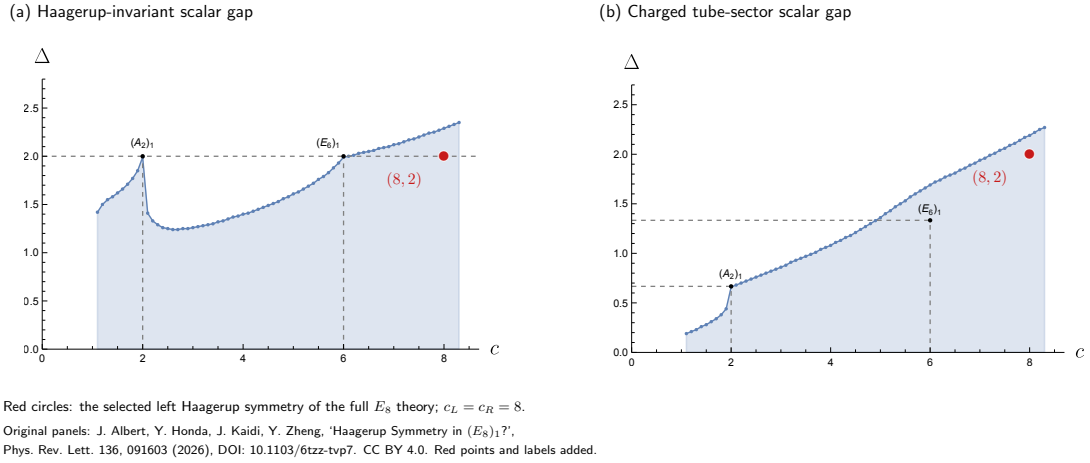


Figure 4: The $c_L = c_R = 8$ Haagerup comparison points, shown in red. Original panels from J. Albert, Y. Honda, J. Kaidi, and Y. Zheng, *Haagerup Symmetry in* $(E_8)_1$*?*, Phys. Rev. Lett. 136, 091603 (2026), doi:10.1103/6tzz-tvp7, reproduced under CC BY 4.0. Red markers and labels have been added. The left panel bounds $\mathcal{V}_1$; the right panel bounds $\mathcal{V}_{\pi_2} \cup \mathcal{V}_{\sigma_1}$.

7.3 Which observables are now supplied

The release directory `c8_physics_20260927/` contains plain TSV tables for inspection and JSON records retaining the algebraic numbers, basis conventions, and exact derivations. There are four distinct kinds of evidence.

1. The character recurrence supplies complete spin-resolved state and Virasoro-primary spectra through $\Delta = 24$, for both full theories. Thermodynamic intervals add a proved bound on the omitted tail, conditional on modular invariance and positivity.

2. The actual lattice fields supply chiral vacuum-module three-point coefficients and the displayed four-point functions. Unit normalization and PCT are explicit. The current Ward completion of the weight-two tensor is checked against direct additional mode calculations.

3. The positive modular data supply all Verlinde coefficients, diagonal-theory line eigenvalues, twisted partition functions and Cardy annuli. The selected extension and its free–forgetful adjunction supply the Haagerup-resolved E_8 sectors and boundaries.

4. Universal CFT identities then give finite-size energies, momenta, entropy coefficients and the stress-tensor and current Ward observables. They do not determine the microscopic velocity v.

The central unresolved interaction problem is the normalized bulk OPE of the nonvacuum diagonal sectors, beginning with the nine scalar components of dimension $2/3$. Neither their character coefficients nor their fusion multiplicities determine their three-point constants. An actual lattice intertwiner block, such as the example given above, is useful input but must still be projected into channels and sewn with consistent Hermitian pairings to obtain the full diagonal-CFT four-point function. The corresponding numerical crossing tests, general defect-junction correlators, and RG endpoints are not claimed as outputs of this release. These are concrete follow-up calculations, with the spectrum and normalization data here providing their starting point.

Each exported number therefore has a mathematical location. Checking a rational recurrence is separate from proving that its coefficients are the characters of the constructed VOA; checking a lattice contraction is separate from proving full-CFT sewing. This separation is intended to make remaining gaps visible to both physicists and mathematicians.

A The explicit lattice construction

A.1 Lattice, cocycle, and currents

Use simple lattice vectors indexed $0, \dots, 7$. The Gram matrix has diagonal entries two and off-diagonal entries minus one on

$$(0,1), (1,2), (2,3), (3,4), (2,5), (6,7),$$

with all remaining off-diagonal entries zero. Fix the cocycle

$$\epsilon(a,b) = (-1)^{\sum_i a_i b_i + \sum_{i>j} a_i b_j G_{ij}}. \tag{79}$$

Write $L = E_6 \oplus A_2$ and $\mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}$. The underlying vector space is

$$B = S\big(\mathfrak{h} \otimes t^{-1}\mathbb{C}[t^{-1}]\big) \otimes \mathbb{C}_\epsilon[L].$$

A monomial $h_1(-n_1)\cdots h_r(-n_r)e^\alpha$ has weight $(\alpha,\alpha)/2 + \sum_i n_i$. In particular its weight is a nonnegative integer. The exponent in (79) is bilinear modulo two, and hence defines a normalized two-cocycle. Directly,

$$\frac{\epsilon(\alpha,\beta)}{\epsilon(\beta,\alpha)} = (-1)^{(\alpha,\beta)}, \qquad \epsilon(\alpha,-\alpha) = (-1)^{(\alpha,\alpha)/2}.$$

These identities fix the signs of all root brackets. For example, $e_0^\alpha e^{-\alpha} = -\alpha(-1)\mathbf{1}$ for a root α. The exponential fields obey

$$Y(e^\alpha, z)e^\beta = \epsilon(\alpha,\beta)z^{(\alpha,\beta)} \exp\left(\sum_{n>0} \frac{\alpha(-n)}{n} z^n\right) e^{\alpha+\beta}. \tag{80}$$

Together with the Heisenberg commutation relations, this formula specifies the exact mode engine used by the certificates.

Put $\alpha_i = (0,\dots,1,\dots,0)$ and $\alpha_L = (-1,-2,-3,-2,-1,-2,0,0)$. With

$$(c_0,\dots,c_7) = (5,8,9,8,5,0,2,2), \qquad h_R = \sum_i c_i\alpha_i,$$

the two current triples, in the order (H,E,F), are

$$(H_L, E_L, F_L) = (\alpha_L(-1)\mathbf{1}, e^{\alpha_L}, -e^{-\alpha_L}), \tag{81}$$

$$(H_R, E_R, F_R) = \left(h_R(-1)\mathbf{1}, \sum_{c_i\neq 0} e^{\alpha_i}, -\sum_i c_i e^{-\alpha_i}\right). \tag{82}$$

Current letters also denote their zero-mode actions when applied to a state. Thus, for example, $F_R E_R v$ means $(F_R)_0(E_R)_0 v$.

Lemma A.1 (The two affine levels)**.** *The triples* (81)–(82) *commute and generate* $A_{1,1}\otimes A_{1,39}$.

Proof. We have $(\alpha_L,\alpha_L) = 2$ and $(\alpha_L,\alpha_i) = 0$ whenever $c_i \neq 0$. Also $(h_R,\alpha_i) = 2$ for every $c_i \neq 0$, and $(h_R,h_R) = 78$. Formula (80) gives $[H_0,E_0] = 2E_0$, $[H_0,F_0] = -2F_0$ and $[E_0,F_0] = H_0$ for both triples. The double poles are

$$(H_L)_1H_L = 2\mathbf{1}, \quad (E_L)_1F_L = \mathbf{1}, \quad (H_R)_1H_R = 78\mathbf{1}, \quad (E_R)_1F_R = 39\mathbf{1}.$$

Orthogonality gives all mixed singular products equal to zero. These are the affine relations at levels one and thirty-nine. The positive-definite lattice representation is integrable; equivalently one may use the principal $\mathfrak{sl}_2$ embeddings in A_5 and A_2, whose levels 35 and 4 add in the diagonal. Thus the generated factors are the simple positive-integral-level affine VOAs. Their tensor product is simple and embeds in B. □

A.2 The defining field and the positive real form

The exact coefficient convention uses unnormalized simple-root Heisenberg oscillators and a torus gauge in which the principal raising current is the sum of the corresponding simple-root vertices. The lowering current is $-\sum_i c_i e^{-\alpha_i}$, with

$$(c_0,\dots,c_7) = (5,8,9,8,5,0,2,2).$$

The torus rescales simple-root vertices by $(1/\sqrt{5}, 1/\sqrt{8}, 1/3, 1/\sqrt{8}, 1/\sqrt{5}, 1, 1/\sqrt{2}, 1/\sqrt{2})$. In this convention define

$$\begin{aligned} u = {} & e^{(0,0,0,0,0,-1,1,1)} \\ & + \frac{\sqrt{13}+3\sqrt{-3}}{20} e^{(0,0,0,1,1,-1,0,0)} \\ & + \frac{\sqrt{13}-3\sqrt{-3}}{20} e^{(1,1,0,0,0,-1,0,0)}. \end{aligned} \tag{83}$$

The compact real form and PCT operation must be transported through this gauge. Ordinary coefficient conjugation in the displayed gauge is not a replacement for the ambient PCT map.

Here is the transport explicitly. Set $\widetilde{c} = (5, 8, 9, 8, 5, 1, 2, 2)$ and let g be the lattice torus automorphism which fixes the oscillators and multiplies e^{α} by $\prod_i \widetilde{c}_i^{-\alpha_i/2}$. If θ is the ordinary lattice PCT involution, the transported involution is $\theta_g = g\theta g^{-1}$. For real lattice oscillators,

$$\theta_g\big(h_1(-n_1)\cdots h_r(-n_r)e^{\alpha}\big) = (-1)^r\Big(\prod_i \widetilde{c}_i^{\alpha_i}\Big)h_1(-n_1)\cdots h_r(-n_r)e^{-\alpha}. \tag{84}$$

It acts antilinearly on coefficients, so fixes $\sqrt{13}$ and negates $\sqrt{-3}$. The transported Hermitian form is $\langle v, w\rangle_g = \langle g^{-1}v, g^{-1}w\rangle$ and is positive definite. Formula (84) squares to the identity. The same torus element acts on V_{E_8}: the positive real character on L extends by its real linear logarithm to the common Cartan space. We transport the form and PCT on H by that element as well, so the entire inclusion $U \subset B \subset H$ uses one unitary convention. It preserves the current algebra, including the root factor even though its two root vectors have different norms in this gauge.

All three momenta in (83) have squared norm four and Cartan weight $(1, 13)$. Substitution in the lattice formula verifies that u is a highest vector of current type $(2, 14)$, where the entries now denote representation dimensions. In the original positive unitary basis, $g^{-1}u$ is, up to a common scalar,

$$e^a + \zeta e^b + \overline{\zeta}e^c, \qquad \zeta = \frac{\sqrt{13} + 3i\sqrt{3}}{\sqrt{40}}, \qquad |\zeta| = 1,$$

with a, b, c the three displayed momenta. Thus the exact construction fixes a relative phase; there is no remaining continuous parameter.

The ambient and affine conformal vectors are fixed by these conventions:

$$\omega_B = \frac{1}{2}\sum_{i,j}(G^{-1})_{ij}\alpha_i(-1)\alpha_j(-1)\mathbf{1}, \tag{85}$$

$$\omega_{A_{1,k}} = \frac{1}{2(k+2)}\left(\tfrac{1}{2}H_{-1}H + E_{-1}F + F_{-1}E\right). \tag{86}$$

The conformal vector of U is ω_B, not the sum of the two affine conformal vectors. Their central charges sum to $1 + 117/41 < 8$.

A.3 The constructed spaces in weights two and three

Certificate input A.2 (Actual initial spaces)**.** The saved exact construction gives a PCT-stable 120-dimensional space $S_2 \subset U_2$, containing u, ω_B, the affine vacuum descendants, two neutral primitive fields, and eight charged current multiplets. The current multiplet dimensions are

$$(1, 5), (1, 9), (1, 9), (1, 13), (2, 2), (2, 6), (2, 8), (2, 14).$$

These contribute 96 vectors, in addition to the 22 affine vacuum vectors and two neutral fields. The exact generated grade-three space $S_3 \subset U_3$ has dimension 956. Its normal subspace $N_3 = K_{-1}S_2 + TS_2$ has dimension 634, and

$$S_3 = N_3 + \operatorname{span}\{a_0 b : a, b \in S_2\}, \qquad \dim(S_3/N_3) = 322.$$

The spaces are current-stable. These are dimensions of constructed finite spaces, not initial assertions that they exhaust U_2, U_3.

The source records are `hi_cft_20260923_c8_exact_fields.json`, `hi_cft_20260923_c8_exact_unitary_certificate.json`, and `hi_cft_20260923_c8_minimal_generators.json`, together with the exact grade-three checkpoint bound into the final ledger.

The minimal-generator record is logically prior to the closure records. It starts with the six specified currents, ω_B, and u, then closes their weight-two span under current zero modes and u_1. Every intermediate vector is therefore an actual U-vector. The final span has dimension 120 and contains each independently specified highest vector and neutral conformal vector. The equality with the saved space S_2 is checked only after this construction. This is the step that relates the large certificate presentation to the short definition (1); an ambient weight-two dimension count alone would not do so.

For reference the two neutral fields, denoted e, f, satisfy

$$e + f = \omega_B - \omega_{A_{1,1}} - \omega_{A_{1,39}}, \qquad c_e = \frac{2400}{1271}, \quad c_f = \frac{70}{31}. \tag{87}$$

They commute as Virasoro fields. The exact current and stress eigenvalues on the eight primitive highest vectors are as follows. The two occurrences of $(1, 9)$ are distinct vectors.

Current dimensions (n, m)	e_1 eigenvalue	f_1 eigenvalue
$(1, 5)$	798/1271	38/31
$(1, 9)$	200/1271	42/31
$(1, 9)$	1512/1271	10/31
$(1, 13)$	420/1271	20/31
$(2, 2)$	192/1271	49/31
$(2, 6)$	1830/1271	3/31
$(2, 8)$	752/1271	24/31
$(2, 14)$	180/1271	13/31

Each row obeys

$$\frac{n^2 - 1}{12} + \frac{m^2 - 1}{164} + h_e + h_f = 2.$$

The affine vacuum space contributes 22 dimensions in weight two: 4 from $A_{1,1}$, 9 from $A_{1,39}$, and $3 \cdot 3$ mixed current products. The level-one null multiplet removes five of the nine universal weight-two states of its affine factor. Together with the two singlets and the 96 charged states, this gives 120. The exact normal-space and rank calculations certify these spaces inside the lattice algebra, including their intersections.

B Finite closure and the all-weight criterion

Write $Y(a, z) = \sum_{n \in \mathbb{Z}} a_n z^{-n-1}$ and $T = L(-1)$. A singular product has mode index $n \geq 0$. The zero mode a_0 is a derivation and commutes with T.

Definition B.1 (Differential normal span)**.** For a homogeneous positive-weight subspace $S \subset V$, let $\mathcal{N}(S)$ be the linear span of the vacuum and every expression obtained from vectors of S by translation and arbitrary parenthesized (-1) products. Let $\mathcal{N}_d(S) = \mathcal{N}(S) \cap V_d$. This definition makes translation and normal multiplication closed operations on $\mathcal{N}(S)$; closure under singular products is a separate assertion. We use the usual convention that a set strongly generates when its differential normal span is the whole algebra.

Lemma B.2 (Filtered Wick reduction)**.** *Let V be a nonnegatively integer-graded vertex algebra with $V_0 = \mathbb{C}\mathbf{1}$, and let S have strictly positive weights. Fix w. Suppose every singular product*

of two homogeneous generators of total weight less than w belongs to $\mathcal{N}(S)$. Then the same is true for two normal expressions of total weight less than w. It is also true at total weight w if at least one expression is a derivative of a positive-weight expression or a normal product of two positive-weight expressions.

Proof. Use induction on total input weight. A vacuum factor has zero singular bracket. Sesquilinearity removes a derivative and lowers the total input weight; the resulting powers of the formal variable and translations preserve coefficientwise membership in $\mathcal{N}(S)$. Skew symmetry allows a composite input to be placed on the right. The noncommutative Wick identity is

$$[x_\lambda : yz :] =: [x_\lambda y]z : + : y[x_\lambda z] : + \int_0^\lambda [[x_\lambda y]_\mu z]\, d\mu. \tag{88}$$

The first two brackets have smaller total input weight, since an omitted factor has positive weight. Their coefficients therefore admit homogeneous expressions in $\mathcal{N}(S)$. A coefficient $x_j y$ in the integral has weight $\mathrm{wt}(x) + \mathrm{wt}(y) - j - 1$. Its pairing with z has total weight at most $\mathrm{wt}(x) + \mathrm{wt}(y) + \mathrm{wt}(z) - 1$, so induction applies again. If the coefficient is scalar, its bracket is zero. Normal multiplication preserves the first two terms by definition. This proves both assertions, including the case in which both inputs are composite. Neither a preferred parenthesization nor independence of normal words was used. □

Theorem B.3 (Recursive strong-generation criterion)**.** *Let V be a nonnegatively graded vertex algebra with $V_0 = \mathbb{C}\mathbf{1}$. Choose finite homogeneous subspaces $S_d \subset V_d$, $1 \le d \le h$, where $h \ge 2$, and let $\mathcal{N}$ be their differential normally ordered span, allowing every parenthesization. Put $K = S_1$ and $A = S_2$. Assume*

1. *$k_n s, a_n s \in \mathcal{N}$ for every $k \in K$, $a \in A$, $s \in S_d$, and $n \ge 0$;*
2. *for $d \ge 3$,*
$$S_d \subset A_0 S_{d-1} + \mathcal{N}_d(S_1 + \cdots + S_{d-1}).$$

Then $\mathcal{N}$ is a vertex subalgebra. In particular, the chosen spaces strongly generate the vertex algebra that they generate.

Proof. The first assumption, the derivation property and $[T, a_0] = 0$ give $a_0\mathcal{N} \subset \mathcal{N}$ for $a \in A$. Explicitly, $a_0(x_{-1}y) = (a_0x)_{-1}y + x_{-1}(a_0y)$, so induction on the construction of a normal expression proves this stability before any singular-product closure is known.

Now induct lexicographically on $(\mathrm{wt}(s) + \mathrm{wt}(t), \mathrm{wt}(s))$ for generator pairs s, t. Left weight one or two is the first assumption. Otherwise the second assumption reduces the left input to $s = a_0 b$, with $\mathrm{wt}(b) = \mathrm{wt}(s) - 1$, plus normal remainders. The latter are covered by Lemma B.2: a homogeneous remainder of weight $\mathrm{wt}(s)$ in lower-weight generators has no single undifferentiated generator of that weight. Use

$$(a_0 b)_n t = a_0(b_n t) - b_n(a_0 t). \tag{89}$$

The first term has smaller total pair weight before applying a_0, which preserves $\mathcal{N}$. The known normal expression for $a_0 t$ has weight $\mathrm{wt}(t) + 1$. Every derivative or composite term reduces total pair weight in the second term. The only remaining possibility is a single undifferentiated generator of weight $\mathrm{wt}(t) + 1$; its pairing with b has the original total weight and smaller left weight. It is therefore an earlier induction case. If $\mathrm{wt}(t) = h$, this single-generator possibility is absent.

All generator singular products close. Wick induction then gives closure for all normal expressions. Translation and normal multiplication preserve $\mathcal{N}$ by definition. For $r \geq 1$,

$$x_{-r}y = \frac{1}{(r-1)!}(T^{r-1}x)_{-1}y,$$

so all negative modes close too. □

For the Wick identities see [8, Section 2]. The full local derivation is `hi_cft_20260923_next_structure_recursive_strong_certificate.md`.

Remark B.4 (Finite scope of the input)**.** Every membership in the first hypothesis has output weight at most $h+1$. Thus the criterion avoids checking all generator pairs through weight $2h-1$. The second hypothesis is essential: it asserts actual zero-product identities, rather than agreement of a dimension or a character coefficient. The proof takes place in the given ambient vertex algebra and makes no assertion that its normal words are free. This resolves the all-weight implication formerly marked F2.

B.1 The finite spaces and prerequisite controls

Choose actual quotient lifts $P_3 \subset S_3$ and an actual triplet $P_4 \subset U_4$, with joint current highest weight $(0,2)$, and set

$$N_4 = K_{-1}S_3 + TS_3 + (S_2)_{-1}S_2, \qquad W_4 = N_4 + P_4, \tag{90}$$

$$N_5 = K_{-1}W_4 + TW_4 + (S_2)_{-1}S_3. \tag{91}$$

These denote full differential normal spaces, using the lower closure identities to reorder expressions. A computational solver may use a proper subset of actual words in these spaces.

The triplet is obtained from an actual a_0p by current-zero-mode projectors and lowering. It satisfies $P_4 \subset (S_2)_0S_3 + N_4$.

Lemma B.5 (Bounds for the current projectors)**.** *On ambient weight d, a Cartan element represented by $h \in L \otimes \mathbb{R}$ has eigenvalues m satisfying $m^2 \leq 2d(h,h)$. At weight four the two current highest weights are therefore at most 4 and 24.*

Proof. Cartan zero modes act on a Fock monomial of momentum α by (h,α), and commute with its oscillators. The grading gives $(\alpha,\alpha) \leq 2d$, so the estimate is Cauchy–Schwarz in the positive lattice form. Every eigenvalue is integral. Apply the estimate to $(h,h) = 2, 78$ and $d = 4$, and then to the highest vectors in the finite-dimensional semisimple current decomposition. □

Consequently finite polynomials in FE give exact projectors. For an input of weight $(0,2)$, one may use

$$\prod_{\ell=1}^{2}\left(1 - \frac{F_LE_L}{\ell(\ell+1)}\right), \qquad \prod_{\ell=2}^{12}\left(1 - \frac{F_RE_R}{\ell(\ell+1)-2}\right).$$

The raising operator acts first in every FE factor. Indeed, on the weight-m line in an $\mathfrak{sl}_2$ representation of highest weight j, FE acts by $(j-m)(j+m+2)/4$. The first polynomial retains left highest weight zero. At right weight two the second retains highest weight two and kills every possible larger highest weight. Thus all denominators are nonzero and the projector has a finite, justified range.

Certificate input B.6 (Lower identities and the actual triplet)**.** The lower-closure artifact contains 3,970 completed exact identities. The actual P_4 vectors satisfy 18 current-zero-mode identities and six independently checked inclusions

$$L(1)P_4 \subset S_3, \qquad L(2)P_4 \subset S_2.$$

These direct Virasoro checks are available before the weight-four gate.

The two gates left for the all-weight argument are

$$(S_2)_0 S_3 \subset W_4 \qquad \text{(Gate 4)}, \tag{92}$$
$$(S_2)_0 P_4 \subset N_5 \qquad \text{(Gate 5)}. \tag{93}$$

C Gate 4: exact seeds, raising, and singlet recovery

Put $P = S_3/N_3$, of independently known dimension 322. For a fixed joint-highest primitive field $a \in S_2$, the map

$$f_a : P \longrightarrow B_4/W_4, \qquad [p] \longmapsto [a_0 p]$$

is well defined because the lower relations give $a_0 N_3 \subset N_4$. Current covariance implies that its kernel is stable under both commuting raising operators E_L, E_R.

Lemma C.1 (Raising saturation)**.** *If $C_a \subset \ker f_a$ and $P = E_L P + E_R P + C_a$, then $f_a = 0$.*

Proof. On $M = P/\ker f_a$ the hypothesis says $M = E_L M + E_R M$. The commuting ideal (E_L, E_R) acts nilpotently because all current weights are bounded. Iterating gives $M = (E_L, E_R)^r M = 0$ for sufficiently large r. □

In this application nilpotence is entirely finite-dimensional. At weight three Lemma B.5 bounds the two Cartan eigenvalues by $|m_L| \le 3$ and $|m_R| \le 21$. Either raising operator increases $m_L + m_R$ by two. Every word of length 25 in the two raising operators therefore vanishes on P. The quotient by $\ker f_a$ inherits these operators because the kernel is stable. This gives an explicit termination bound for the argument; there is no limiting step.

The exact raising coordinates comprise 644 images. Their rank is 294. For each of nine independent highest left fields, the completed seed sets and these images have rank 322. A nonzero minor at the good split prime $p = 1{,}000{,}003$ proves a nonzero characteristic-zero minor. This proves spanning because the exact ambient quotient dimension is already 322. It does not give an upper rank for an unknown weight-four normal space.

The original exact run supplies 2,145 seed identities. The saturation calculation asks for 130 further inputs; 118 have direct exact residual certificates. The remaining twelve outputs have weight $(0,0)$. An exact replay of the known kernel and twelve additional identities at output weight $(0,2)$ shows that both raising currents kill each of these twelve remaining classes modulo W_4.

Lemma C.2 (Recovery from a joint-singlet projection)**.** *Let $t \in B_4$ have weight $(0,0)$, with $E_L t, E_R t \in W_4$. Define*

$$\Pi = \prod_{j \in \{2,4\}} \left(1 - \frac{4F_L E_L}{j(j+2)}\right) \prod_{j \in \{2,4,\ldots,24\}} \left(1 - \frac{4F_R E_R}{j(j+2)}\right). \tag{94}$$

If Πt is a linear combination of Πn_i, with $n_i \in N_4$, then $t \in W_4$.

Proof. Both W_4 and N_4 are stable under current zero modes. Every FE factor kills $[t] \in B_4/W_4$, so $[\Pi t] = [t]$. Each Πn_i remains in $N_4 \subset W_4$. Thus $[t] = 0$. Equivalently, the weight-zero highest class is a singlet in the finite semisimple current module. No dimension assertion about the selected normal columns is needed. □

The prime search selects 21 independent projected normal columns. All twelve projected identities have complete residual zero over $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$. The final verifier repeats those residuals by independent integer denominator clearing, authenticates the raising premise, and repeats all nine saturation ranks. By Lemmas C.1 and C.2, all $9 \cdot 322 = 2898$ required highest-left identities hold. Current covariance gives the charged descendants. The second neutral field follows from $\omega_B - \omega_{K_L} - \omega_{K_R}$ minus the retained neutral field. This proves (92) with the certified lower controls.

C.1 Positive modes available after Gate 4

The normal-product argument at the next gate uses positive modes on S_3. We establish them here, before their first use. This also explains why no additional unverified positive-mode gate is required.

Lemma C.3 (Virasoro and current bootstrap)**.** *The lower identities, the direct P_4 controls, and Gate 4 imply*

$$(S_2)_1 S_3 \subset S_3, \qquad (S_2)_2 S_3 \subset S_2, \qquad (S_2)_3 S_3 \subset K, \tag{95}$$

$$K_1 W_4 \subset S_3, \qquad K_2 W_4 \subset S_2, \qquad K_3 W_4 \subset K, \tag{96}$$

$$L(1) N_5 \subset W_4, \qquad L(2) N_5 \subset S_3. \tag{97}$$

The next mode in each of the first two lines is scalar, and all further modes vanish.

Proof. The exact lower inputs give all singular products of K and S_2 with S_2 in the expected lower spaces. The provenance $S_3 = N_3 + (S_2)_0 S_2$ and the current commutator give

$$K_0 S_3 \subset S_3, \qquad K_1 S_3 \subset S_2, \qquad K_2 S_3 \subset K.$$

For instance,

$$k_1(a_0 b) = (k_0 a)_1 b + (k_1 a)_0 b + a_0 (k_1 b). \tag{98}$$

The Virasoro commutators for a weight-two vector are

$$[L(1), a_n] = (2 - n) a_{n+1} + (L(1)a)_n, \tag{99}$$

$$[L(2), a_n] = (3 - n) a_{n+2} + 3(L(1)a)_{n+1} + (L(2)a)_n. \tag{100}$$

They show $L(1)S_3 \subset S_2$ and $L(2)S_3 \subset K$. Applied to the three summands of N_4, they give $L(1)N_4 \subset S_3$ and $L(2)N_4 \subset S_2$. For a normal input $a_{-1}b$ the latter formula includes the scalar term $(L(2)a)_{-1}b$; it must not be omitted. It lies in S_2. The six direct controls of Certificate B.6 extend these inclusions to W_4 without using Gate 5.

Each of the eight charged primitive multiplets and the two neutral singlets in S_2 is quasiprimary. Indeed, $L(1)$ is equivariant for current zero modes, and none of their current types is $(3,1)$ or $(1,3)$, the two types in K. For such a vector a and $p \in S_3$, Gate 4 and the preceding controls give

$$\begin{aligned} 2a_1 p &= L(1)(a_0 p) - a_0 L(1) p \in S_3, \\ a_2 p &= L(1)(a_1 p) - a_1 L(1) p \in S_2, \\ 2a_3 p &= L(2)(a_1 p) - a_1 L(2) p \in K. \end{aligned}$$

In the last line the possible $(L(2)a)_1$ term is zero since $L(2)a$ is scalar. The affine normal and derivative part of S_2 is handled by the current iterate identities. This proves (95) on all of S_2.

The actual construction gives $P_4 \subset (S_2)_0 S_3 + N_4$. Equation (98), now with $b \in S_3$, gives $K_1 P_4 \subset S_3$; the same inclusion for N_4 follows on its normal generators. Since currents are primary of weight one,

$$[L(1), k_1] = -k_2, \qquad [L(2), k_1] = -k_3.$$

The controls on $L(1)W_4, L(2)W_4$ consequently give (96). The remaining modes follow from grading.

Finally apply $L(1), L(2)$ to $N_5 = K_{-1}W_4 + TW_4 + (S_2)_{-1}S_3$. The current and translation summands follow from the established inclusions and the Virasoro relations. For the third summand, the required formulas are

$$\begin{aligned} L(1)(a_{-1}p) &= 3a_0 p + (L(1)a)_{-1}p + a_{-1}L(1)p, \\ L(2)(a_{-1}p) &= 4a_1 p + 3(L(1)a)_0 p + (L(2)a)_{-1}p + a_{-1}L(2)p. \end{aligned}$$

Every term belongs respectively to W_4 or S_3 by Gate 4, (95), and the lower identities. Reordering the last normal term uses only already established lower products. This proves (97). □

D Gate 5 from Jacobi identities

The proof uses the full normal spaces in (90)–(91), not only the selected solver columns. To see that the displayed formula for N_5 includes every lower normal word of weight five, remove a leading derivative, or split a normal product into weights $1+4$ or $2+3$. Reversed splits are reordered by skew symmetry for normal products. Its correction terms are derivatives of singular products, already controlled by Lemma C.3 and the lower identities. Quasi-associativity changes the parenthesization with correction terms of the same type. Thus the resulting words lie in $K_{-1}W_4 + TW_4 + (S_2)_{-1}S_3$.

Lemma D.1 (Normal-space controls)**.** *Before using Gate 5, the lower controls and Gate 4 give*

$$(S_2)_0 N_4 \subset N_5, \qquad (N_3)_0 S_3 \subset N_5. \tag{101}$$

Proof. The first follows by zero-mode derivation and skew reordering. For the second, $(Ta)_0 = 0$ and, for a current x,

$$(x_{-1}a)_0 p = \sum_{i \geq 0} x_{-1-i}(a_i p) + \sum_{i \geq 0} a_{-1-i}(x_i p). \tag{102}$$

All inner products belong to the required lower spaces. To make the translation reduction explicit, repeatedly use

$$x_{-r}v = \frac{T(x_{-(r-1)}v) - x_{-(r-1)}Tv}{r-1}, \qquad r \geq 2.$$

The base terms are precisely those in (91); the same identity applies to $a_{-r}v$ with $a \in S_2$. Thus no unproved P_3-by-P_3 closure enters (101). All positive inner modes in (102) are supplied by (95) and the current controls established before this section. More explicitly the first sum has inner vectors in $W_4, S_3, S_2, K, \mathbb{C}\mathbf{1}$, in that order; the second has inner vectors in $S_3, S_2, K, \mathbb{C}\mathbf{1}$. Both sums are finite by grading. □

If $P_4 \subset N_4$, Gate 5 already follows. Otherwise $Q = W_4/N_4$ is the irreducible current triplet of highest weight $(0,2)$, since the actual P_4 is irreducible. Let q_-, q_0, q_+ be its weight basis, and set

$$\beta(a,p) = [a_0 p] \in Q, \qquad F(a,q) = [a_0 \widetilde{q}] \in B_5/N_5.$$

The second map does not depend on the representative $\widetilde{q} \in W_4$, by (101).

Lemma D.2 (Jacobi reduction)**.** *If $a, b \in S_2$ satisfy $a_0 b \in N_3$, then for $p \in S_3$,*

$$F(a, \beta(b,p)) = F(b, \beta(a,p)). \tag{103}$$

Proof. The difference is the class of $(a_0 b)_0 p$, which vanishes by (101) and $[a_0, b_0] = (a_0 b)_0$. □

D.1 The charged families

Let $a_0, \dots, a_7$ denote the eight charged highest fields, with current weights

$$(0,4), (0,8), (0,8), (0,12), (1,1), (1,5), (1,7), (1,13).$$

In a product such as $(a_i)_0 p$, the parenthesized subscript denotes the field label and the next subscript denotes its mode. The saved exact relations give

$$\beta(a_3, p_3) = c_3 q_-, \qquad \beta(a_6, p_6) = c_6 q_-, \qquad c_3 c_6 \neq 0,$$

where $p_3 =$ `r3_000910` has weight $(0,-14)$, and $p_6 =$ `r3_000899` has weight $(-1,-9)$. The coefficients are authenticated in the final Jacobi certificate. Exact combinations of the lower relations establish $(a_i)_0 a_j \in N_3$ for the following pairs:

i	j	Input	Weight of $(a_i)_0 p_j$
0	3	p_3	$(0,-10)$
1,2	3	p_3	$(0,-6)$
3	6	p_6	$(-1,3)$
4	6	p_6	$(0,-8)$
5	3	p_3	$(1,-9)$
6	3	p_3	$(1,-7)$
7	3	p_3	$(1,-1)$

Every listed output weight lies outside Q, so $\beta(a_i, p_j) = 0$. Equation (103) and $c_j \neq 0$ give $F(a_i, q_-) = 0$. Raise this resulting zero with a_i held highest to obtain the other two right inputs. Current covariance gives all left descendants.

Remark D.3. The weight exclusion is used only at the displayed bottom inputs. It need not hold after raising p_j; using it there would be an incorrect stronger assertion. We raise the proved zero identity instead.

D.2 The neutral family

Let C be the full current span, inside $P = S_3/N_3$, of brackets of charged weight-two fields. The charged-span certificate reconstructs 28 actual charged seeds from the exact lower relations. Together with the 644 lowering images, of rank 294, these have rank 322. The independent verifier also checks both $\mathfrak{sl}_2$ commutators and all cross-axis commutators against the saved exact raising matrices. Since C is a current submodule, the nilpotent-ideal argument gives

$$P = F_L P + F_R P + C \quad \Longrightarrow \quad P/C = 0.$$

Once again the finite-prime full rank is a positive spanning witness inside an independently exact 322-dimensional space.

For the retained neutral singlet e, write

$$e_0 a_3 = \sum_i t_i (x_i)_0 y_i + n, \qquad n \in N_3,$$

with charged x_i, y_i. Apply zero modes to p_3. By Jacobi, the charged conclusion and (101),

$$(e_0 a_3)_0 p_3 = \sum_i t_i [(x_i)_0, (y_i)_0] p_3 + n_0 p_3 \in N_5.$$

Indeed Gate 4 puts each inner product in W_4, and the next charged product is controlled on both N_4 and P_4. A second Jacobi identity gives

$$e_0((a_3)_0 p_3) = (a_3)_0 (e_0 p_3) + (e_0 a_3)_0 p_3 \in N_5.$$

The nonzero coefficient of q_- proves the neutral bottom identity; covariance gives the triplet. The other neutral field is obtained from the total stress minus the two affine Sugawara stresses and e. This proves Gate 5. Its 24 independent highest channels comprise 23 charged channels and one neutral channel; they are consequences of the displayed identities, not 24 separately recomputed grade-five lattice products.

E From the finite gates to properness

Lemma C.3 supplies the positive-current products and $L(1)N_5 \subset W_4$, $L(2)N_5 \subset S_3$ before Gate 5. For a quasiprimary primitive $a \in S_2$, Gate 5 now yields, for $p \in P_4$,

$$\begin{aligned} 2a_1 p &= L(1)(a_0 p) - a_0 L(1) p \in W_4, \\ a_2 p &= L(1)(a_1 p) - a_1 L(1) p \in S_3, \\ 2a_3 p &= L(2)(a_1 p) - a_1 L(2) p \in S_2, \\ a_4 p &= L(2)(a_2 p) - a_2 L(2) p \in K. \end{aligned}$$

The next mode is scalar. The 30 earlier prime-only $a_1 P_4$ tests are therefore derived exactly here; they are not an omitted gate.

All hypotheses of Theorem B.3 hold for K, S_2, S_3, P_4. Their normal span is a vertex subalgebra of U containing K, u, ω_B, hence equals U. A normal word in positive-weight generators cannot acquire weight one unless it is a weight-one generator. Thus $U_1 = K$, proving Proposition 1.1.

Proof of Corollary 1.2. A differential normal word of weight two uses only K and S_2; its current descendants are already in S_2. A word of weight three uses only K, S_2, S_3, and the lower normal words lie in N_3. Thus the strong-generation result proves $U_2 = S_2$ and $U_3 = S_3$. The exact PCT certificate gives $\theta_g S_2 = S_2$, while $\theta_g K = K$. Since $S_2 \subset U$ contains the original generators, $\theta_g U = U$. The inherited positive form makes U a unitary subalgebra of B, hence simple by [10, Proposition 5.29]. It contains ω_B, so has central charge eight, the same nonnegative grading, and $U_0 = \mathbb{C}\mathbf{1}$. The character assertion is now the definition of the graded trace. □

REVIEW — F1: what the finite certificate establishes

This proof of properness has the strongest computational support in the manuscript. It still relies on correctness of the lattice mode engine, the saved actual-vector recipes, and the stated algebraic lemmas.

Hashes authenticate the records; hashes alone do not prove those lemmas or independently reconstruct every earlier mode calculation. The Lean checks in Appendix N cover selected abstract implications, not the VOA construction or the complete strong-generation argument. No second implementation of the entire construction is claimed. In particular, $\dim U_4 = 6010$ is not used as an input to this proof.

F Regularity from a positive conformal-vector identity

This section concerns the actual PCT-stable subalgebra $U \subset B$ defined in the construction. Its conclusions do not require properness. We use the convention $Y(a,z) = \sum_{n\in\mathbb{Z}} a_n z^{-n-1}$ and write $L^e(n) = e_{n+1}$ for the Virasoro operators of a conformal vector e. The subalgebras used below need not contain the ambient conformal vector. We call an inclusion $A \subset V$ *grading compatible* if the A-weight of each homogeneous vector of A equals its V-weight. Thus a mode a_n changes both gradings by $\mathrm{wt}(a) - n - 1$; the equality of the two energy operators is not required on all of V. All subalgebras in the positive identity below are grading compatible and stable under the ambient PCT involution.

Certificate input F.1 (Embedded structural data)**.** The following are assertions about actual fields and their exact modes in the lattice construction, in addition to the finite closure ledger.

1. The Potts fields lie in U, are PCT stable, and generate the regular unitary algebra $K(\mathfrak{sl}_2, 3)$, with its translation and intrinsic grading inherited from the ambient VOA. The exact records are `hi_cft_20260923_proof_potts.json` and `hi_cft_20260923_proof_potts_exact.json`.

2. The actual fields z, a of weights two and three lie in U, are PCT stable, have central charge $8/5$, and satisfy the stress, primary, and translation identities in `hi_cft_20260923_proof_pf13_exact.json`, `hi_cft_20260923_proof_pf13_membership.json`, and `hi_cft_20260923_proof_remaining_coset.json`. Their full singular first rows agree with the parafermion rows at $c = 8/5$, $\lambda = 14/473$, as polynomials in $z, a, a_1 a, a_1(a_1 a)$. The polynomial records are `hi_cft_20260923_proof_pf13_upper_poles.json`, `hi_cft_20260923_proof_pf13_nonlinear.json`, `hi_cft_20260923_proof_pf13_truncation.json`, `hi_cft_20260923_proof_pf13_simple_pole_algebra.json`, and `hi_cft_20260923_proof_pf13_dly.json`. The membership record also gives the positive identity (104).

3. The lattice–Potts core T below is embedded with its stated charge realization. The six joint spectral vectors used to construct its order-thirteen extension are actual vectors of U_2, with the eigenvalues printed below. Their exact source is `hi_cft_20260923_next_structure_core_extension.json`; the construction of T is described in `hi_cft_20260923_proof_report.md`, Section 4.

4. The zero-mode action on the complementary eighty currents is irreducible, and the four mixed lattice tops of dimension eighty-one are irreducible as U-tops. These are exact field-action assertions, as recorded in `hi_cft_20260923_proof_current_module_v2.json` and `hi_cft_20260923_next_81top_all.json`, respectively.

All polynomial equalities here mean exact vector equalities in the fixed lattice realization, not projected or floating-point matches.

The input from the local construction is the identity

$$\omega_U = \omega_{A_{1,1}} + \frac{3403}{4563}\omega_{A_{1,39}} + \frac{25}{6}\overline{p} + \frac{25}{26}\overline{z}. \tag{104}$$

Here p and z are the conformal vectors of actual embedded copies of the three-state Potts algebra $K(\mathfrak{sl}_2, 3)$ and the parafermion algebra $K(\mathfrak{sl}_2, 13)$; a bar denotes normalized averaging over the compact current group. Each subalgebra is regular and unitary, and its intrinsic grading agrees with the ambient grading on its states. The averages take place in the finite-dimensional space U_2. More concretely, p, z lie in the already constructed current-stable space S_2, whose invariant subspace is spanned by the four mutually orthogonal stresses $\omega_{A_{1,1}}, \omega_{A_{1,39}}, e, f$. For either vector v, its Haar average is therefore

$$\overline{v} = \sum_{w \in \{\omega_{A_{1,1}}, \omega_{A_{1,39}}, e, f\}} \frac{\langle v, w \rangle_g}{\langle w, w \rangle_g} w.$$

Thus the exact coefficients in (104) can be verified inside S_2 without assuming $S_2 = U_2$ or properness. The compact orbit of a conformal vector has compact convex hull in the real space of PCT-fixed weight-two vectors. Its Haar average belongs to that hull. Carathéodory's theorem therefore replaces each average by a finite convex combination of orbit points. In particular,

$$\omega_U = \sum_{i=1}^{N} c_i e_i, \qquad c_i > 0. \tag{105}$$

Zero coefficients, if any, are omitted. All simple modules of each regular subalgebra A_i have nonnegative lowest weights, with zero occurring only for its vacuum module.

REVIEW — R1: the embedded regular factors are local inputs

Certificate F.1 states the model-specific field inputs explicitly. Proposition F.2 below proves the transport from the complete checked parafermion rows to the actual factor identification. It does not replace verification of the listed lattice identities, Potts embedding, or positive stress identity. These inputs are separate from the finite closure theorem.

Proposition F.2 (Finite reconstruction of the parafermion factor)**.** *Let V, V' be simple vertex algebras of CFT type generated by a conformal vector L and a weight-three primary a. Set $b = a_1 a$ and $d = a_1 b$, of weights four and five. Suppose the full singular products $a_n a, a_n b, a_n d$, $n \geq 0$, agree as homogeneous differential normal polynomials in L, a, b, d, with the same normal ordering. Suppose also that their self-row is*

$$a_5 a = \frac{c}{3}\mathbf{1}, \quad a_4 a = 0, \quad a_3 a = 2L, \quad a_2 a = TL, \quad a_1 a = b, \quad a_0 a = \frac{1}{2}Tb - \frac{1}{12}T^3 L,$$

and that $a_n b = 0$ for $n \geq 4$, $a_3 b \in \mathbb{C}a$, and $a_2 b \in \mathbb{C}Ta$. Then there is an isomorphism $V \cong V'$ identifying L, a, b, d. In particular $a_1 d$ and both simple poles must be specified polynomials; they cannot be left as independent new generators.

Proof. We first recover the remaining generator OPEs without a circular normal-product reduction. The Virasoro rows follow by commuting Virasoro modes through $b = a_1 a$ and $d = a_1 b$:

$$L(m)b = (2m+1)a_{m+1}a, \qquad L(m)d = (2m+1)a_{m+1}b + a_1 L(m)b \quad (m \geq 1).$$

These are linear in generators and derivatives under the displayed hypotheses. Thus Virasoro creation operators can be moved to the outside of any normal word using already known

commutators. Sesquilinearity removes derivatives. Since three non-Virasoro factors have weight at least nine, the possible two-factor cores through weight eight are exactly

weight bound	cores besides individual fields
6	$:aa:$
7	$:aa:,\ :ab:$
8	$:aa:,\ :ab:,\ :bb:,\ :ad:$.

Reversing the order of a core uses skew symmetry and its already known pair bracket. Commuting an external a_n or b_n through outer Virasoro creation operators reduces their number; any resulting negative mode is itself a normal product.

Write A_r for the evaluation of a_nP, $n \geq 0$, on every normal polynomial P of weight at most r, and B_r for the corresponding assertion with b_n. We use the noncommutative Wick identity [8, Section 2]

$$[x_\lambda : yz :] =: [x_\lambda y]z : + : y[x_\lambda z] : + \int_0^\lambda [[x_\lambda y]_\mu z]\, d\mu.$$

The following order is acyclic. First A_6 is known: the only two-factor core is $:aa:$, and $[a_\lambda a]$ is linear in the vacuum and derivatives of L, b, whose brackets with a are given. The iterate identity

$$b_n s = a_1(a_n s) - a_0(a_{n+1}s) + a_{n+1}(a_0 s) - a_n(a_1 s) \tag{106}$$

with $s = b$ now determines $b_n b$, because each inner $a_j b$ has weight at most six. Next A_7 follows: its new core $:ab:$ uses only the just-computed b, b row in the Wick integral. Equation (106) with $s = d$ then determines $b_n d$, since every inner $a_j d$ has weight at most seven.

At this point B_6 is known. For its core $:aa:$ the Wick integral brackets a coefficient of $[b_\lambda a]$, of weight at most six, with a, which is A_6 after skew symmetry. Single-field inputs use the known b, b and b, d rows. We obtain A_8 next: its core $:bb:$ uses B_6 on coefficients of $[a_\lambda b]$, and its core $:ad:$ uses the known L, d and b, d rows. Finally B_7 follows from B_6 on the coefficients of $[b_\lambda a]$ in the new core $:ab:$. With $s = d$, the second iterate identity

$$d_n s = a_1(b_n s) - a_0(b_{n+1}s) + b_{n+1}(a_0 s) - b_n(a_1 s)$$

computes $d_n d$: the first two terms use A_8, and the last two use B_7. All generator singular products have therefore been expressed by the same polynomials in the two algebras.

For the uniqueness step, take the vertex subalgebra $D \subset V \times V'$ generated by the diagonal pairs of the four named fields. Common polynomial OPE closure and filtered Wick induction show that D is their differential normal span. Every generator has positive weight, so $D_0 = \mathbb{C}(\mathbf{1}, \mathbf{1})$ and $D_n = 0$ for $n < 0$. It follows that D has a unique maximal proper graded ideal: every proper graded ideal has zero degree-zero part, and so does their sum. The coordinate projections are surjective because L, a generate each target. They preserve the conformal vector and hence the grading. Their kernels are maximal proper graded ideals because V, V' are simple. The kernels coincide, which gives the asserted isomorphism. No freeness or independence of normal monomials has been used. □

The exact parafermion records instantiate this proposition at $c = 8/5$ and $\lambda = 14/473$. The upper-pole record supplies the self-row and the remaining positive poles, the nonlinear record supplies $a_0 b$, and the truncation record supplies $a_1 d$. The algebraic simple-pole derivation supplies $a_0 d$ from the independently checked upper poles and skew symmetry. The Dong–Lam–Yamada change-of-generators record matches the seven weight-six decoupling coefficients with those of $K(\mathfrak{sl}_2, 13)$, using [48, Appendix B]. The common first rows are the recursive formulas

(5.4), (5.7), and (A.2) of Linshaw [47]. Simplicity of the actual generated subalgebra follows from its PCT stability; its translation operator agrees with the ambient one by the complementary positive energy argument. Thus the proposition transports the checked polynomial identities to an identification with the parafermion algebra. The exact membership, PCT, translation, and polynomial identities remain the explicit finite inputs to this identification.

Lemma F.3 (Orthogonal energy bound)**.** *Let V be a unitary VOA of CFT type admitting an identity of the form (105), with the properties just stated. Then V is C_2-cofinite.*

Proof. For an ordinary simple A_i-module M, put

$$C_{2,A_i}(M) = \operatorname{span}\{a_{-2}m : a \in A_i,\ m \in M\}.$$

These quotients are finite-dimensional by [13, Proposition 5.2]. They are graded for the intrinsic operator $L^{e_i}(0)$. There are finitely many simple A_i-modules. Hence there is a finite number D_i bounding the intrinsic energy in all their C_{2,A_i} quotients; this includes their lowest weights. Regularity decomposes the weak A_i-module V into ordinary simples. Consequently every intrinsic eigenvector of energy above D_i lies in $C_{2,A_i}(V)$. The intrinsic energy preserves each finite-dimensional ambient grade V_n and is self-adjoint for its positive form. We may therefore diagonalize it on V_n. Since

$$C_{2,A_i}(V) \subset C_2(V),$$

every vector in $V_n \cap C_2(V)^{\perp}$ has zero projection to the intrinsic eigenvectors of energy above D_i. It follows that

$$\langle L^{e_i}(0)v, v\rangle \le D_i\|v\|^2 \qquad (v \in V_n \cap C_2(V)^{\perp}).$$

For $v \in V_n \cap C_2(V)^{\perp}$, the conformal-vector identity gives

$$n\|v\|^2 = \sum_i c_i\langle L^{e_i}(0)v, v\rangle \le \Big(\sum_i c_i D_i\Big)\|v\|^2.$$

The subspace $C_2(V)$ is ambient graded. Its orthogonal complement in each finite-dimensional V_n represents $(V/C_2(V))_n$, so the last inequality makes the quotient vanish for $n > \sum_i c_i D_i$. Finiteness of each V_n proves the claim. No simultaneous diagonalization of the different energy operators is required. The same proof applies directly to averages, since every conjugate factor has the same bound D_i. □

Lemma F.4 (Highest spaces for a compatible subalgebra)**.** *Let $A \subset V$ be grading compatible and regular, and let M be a weak V-module. Suppose a finite-dimensional subspace $E \subset M$ is stable under every ambient degree-zero mode from A and is killed by every mode from A of negative ambient degree. Then E is a direct sum of irreducible Zhu-algebra modules belonging to the tops of simple ordinary A-modules. In particular $L^A(0)|_E$ is diagonalizable, with eigenvalues among their lowest weights.*

Proof. Write

$$\Omega_A(M) = \{v \in M : a_n v = 0 \text{ for all homogeneous } a \in A,\ n \ge \mathrm{wt}_A(a)\}.$$

Compatibility puts E inside this space. Regularity decomposes $M\downarrow_A$ as an algebraic direct sum of ordinary simples, and Ω_A commutes with that sum. The highest space of an ordinary simple is its top. Zero modes endow $\Omega_A(M)$ with the corresponding semisimple Zhu-algebra action; its invariant finite-dimensional subspace E is therefore a sum of simple tops. This also explains why one must impose the lowering condition for all states of A, rather than only for a chosen generating set. □

Lemma F.5 (Vacuum-like vectors)**.** *Let M be a weak module over a VOA V. If $v \in M$ satisfies $L_M(-1)v = 0$, then $a_n v = 0$ for all $a \in V$ and $n \geq 0$, and $a \mapsto a_{-1}v$ is a V-module homomorphism $V \to M$ sending $\mathbf{1}$ to v.*

Proof. Translation covariance gives $L_M(-1)a_{n+1}v = -(n+1)a_n v$. Lower truncation and downward induction give $a_n v = 0$ for every $n \geq 0$. The module Jacobi identity then gives $b_m(a_{-1}v) = (b_m a)_{-1}v$; equivalently this follows by taking the constant coefficient in weak associativity, since the fields applied to v have no negative powers. This is the asserted homomorphism. □

Lemma F.6 (Lowest weights and projectivity of the vacuum)**.** *Let V satisfy Lemma F.3, and suppose every component algebra has nonnegative simple lowest weights and its vacuum is the unique simple of weight zero. Every simple ordinary V-module has nonnegative lowest conformal weight, the only weight-zero simple is V, and V is projective among grading-restricted generalized modules.*

Proof. Let $E = M_h$ be the lowest space of a simple ordinary module. For a homogeneous state a, $[L_V(0), a_n] = (\mathrm{wt}_V(a) - n - 1)a_n$ on every module. Thus degree-zero modes preserve E, and lowering modes kill E. Lemma F.4 implies that every $L^{e_i}(0)|_E$ is diagonalizable with nonnegative real spectrum. Taking traces, rather than assuming the operators commute, gives

$$h \dim E = \sum_i c_i \operatorname{Tr}_E L^{e_i}(0) \geq 0. \tag{107}$$

In particular h is real and nonnegative. If $h = 0$, every summand vanishes. Hence every partial energy is zero on E, which lies in sums of vacuum tops for every A_i. Consequently $L^{e_i}(-1)E = 0$, and (105) gives $L_V(-1)E = 0$. By Lemma F.5, any nonzero $v \in E$ gives a nonzero module map $V \to M$, hence an isomorphism.

Now let $\pi : N \twoheadrightarrow V$ be an epimorphism in the generalized module category. C_2-cofiniteness gives finite length, and the preceding argument excludes negative weights in all its composition factors. The finite generalized weight-zero space $N_{[0]}$ therefore has no lower-weight target for a lowering mode. It is again in every $\Omega_{A_i}(N)$ and is invariant under degree-zero modes. On this space $L_V(0)$ is nilpotent, so

$$0 = \operatorname{Tr}_{N_{[0]}} L_V(0) = \sum_i c_i \operatorname{Tr}_{N_{[0]}} L^{e_i}(0).$$

Diagonalizability and nonnegativity of each partial energy imply it is zero. The vacuum-top argument gives $L_V(-1)N_{[0]} = 0$. Generalized weight decompositions are preserved by π, so one can choose $v \in N_{[0]}$ with $\pi(v) = \mathbf{1}$. Then

$$s : V \longrightarrow N, \qquad s(a) = a_{-1}v, \qquad \pi s = \mathrm{id}_V,$$

is a section. Every epimorphism onto the vacuum splits. □

Lemma F.7 (A projective unit in a finite braided r-category)**.** *Let $\mathcal{C}$ be a finite complex linear abelian braided monoidal category with simple unit, right-exact tensor product, and Grothendieck–Verdier duality with dualizing object $\mathbf{1}$. If $\mathbf{1}$ is projective, then $\mathcal{C}$ is rigid and semisimple.*

Proof. Realize $\mathcal{C}$ as finite-dimensional modules for a finite-dimensional algebra, denote underlying vector-space dimension by d_0, and let P correspond to its regular module. Every Z is a quotient of $P^{\oplus d_0(Z)}$. For fixed X, right exactness gives

$$d_0(Z \otimes X) \leq C_X d_0(Z), \qquad \dim \mathrm{End}(X^{\otimes n}) \leq d_0(\mathbf{1})^2 C_X^{2n} < n!$$

for large n, where $C_X = \max\{1, d_0(P \otimes X)\}$. This uses only right exactness. For nonzero indecomposable X, Grothendieck–Verdier evaluation $DX \otimes X \to \mathbf{1}$ is nonzero, hence surjective onto the simple unit, and splits by projectivity. A finite abelian category is idempotent complete, and its Hom spaces are finite-dimensional. Thus X is non-negligible and has moderate growth in the precise sense of [14, Theorem 1.1]; that theorem proves X rigid. Finite direct sums give rigidity of all objects. Tensoring is now exact, and

$$\mathrm{Hom}(X, -) \cong \mathrm{Hom}(\mathbf{1}, - \otimes X^*)$$

is exact. Every object is projective, so $\mathcal{C}$ is semisimple. □

Theorem F.8 (Positive stresses imply strong rationality)**.** *A simple unitary VOA of CFT type satisfying the hypotheses of Lemma F.6 is strongly rational. In particular, the actual generated algebra U is strongly rational, independently of its properness in B.*

Proof. Unitarity gives self-contragredience. Lemmas F.3 and F.6 give C_2-cofiniteness and a projective vacuum. The category of grading-restricted generalized modules is finite abelian and braided, its tensor product is right exact, and contragredient duality is Grothendieck–Verdier duality with dualizing object $V' = V$; see [15, Sections 2.3 and 3]. Lemma F.7 proves semisimplicity, hence rationality in the C_2-cofinite setting. This argument does not assume that the nonvacuum modules are unitarizable. The local argument is recorded in `hi_cft_20260923_next_structure_rationality.md`. □

G Branching and the boundary fusion ring

From this point we use the properness conclusion and $\dim U_1 = 6$ of the finite closure argument. Before that conclusion is supplied, every statement in this section is conditional on the proper branch.

G.1 A universal top bound and actual lattice branching

The sharper bound uses an actual regular subalgebra $A_L \subset U$. In the notation of the local construction, a rational lattice–Potts core $\mathcal{T}$ of central charge $14/5$ and $K(\mathfrak{sl}_2, 13)$ of central charge $8/5$ have pointed charges $(j, k) \in (\mathbb{Z}/13)^2$ with quadratic form $-(j^2 + k^2)/13$. Put $A = \mathcal{T} \otimes K(\mathfrak{sl}_2, 13)$. The field certificate supplies nonzero weight-two vectors v with

$$(\omega_U - \omega_A)_1 v = 0, \qquad z_1 v = \frac{12}{13} v, \qquad (\omega_{\mathcal{T}})_1 v = \frac{14}{13} v, \qquad p_1 v = \frac{2}{3} v,$$

at Cartan weights $(\pm 1, \pm 5)$ and $(0, \pm 8)$. These are the top characters of the simple-current charges $(5, 1)$ and its inverse, up to reversing the second charge. The label is forced by actual, unreduced lowest weights: the τ-labelled $\mathcal{T}$ module of charge 5 starts at $31/65$, whereas $14/13 - 31/65$ is not integral, and the only parafermion modules starting at $12/13$ are the two inverse simple currents.

Here is why the witness gives an embedded extension. The kernel of $(\omega_U - \omega_A)_1$ in U is a connected unitary extension of A with conformal vector ω_A. Regularity of A and finite ambient homogeneous spaces make this extension finite. In a connected unitary extension, invertible constituents have multiplicity at most one and their support is a subgroup: the invariant pairing gives a nonzero product of a constituent with its inverse into the vacuum line; associativity then makes multiplication by it invertible on each supported invertible line. Integer grading makes that support isotropic. The only isotropic lines for the displayed quadratic form are

$k = 5j$ and $k = -5j$, since $5^2 = -1 \pmod{13}$. The witnessed nonzero charge generates one of these lines, and both cannot occur because together they generate nonintegral-spin charges. Consequently, after the harmless choice of parafermion charge orientation,

$$A_L = \bigoplus_{k=-5j} \mathcal{T}_{1,j} \otimes \mathrm{PF}_{0,k} \subset U, \qquad \mathrm{Rep}(A_L) \simeq \mathrm{Fib} \boxtimes \mathrm{SO}(3)_{13}.$$

The field witness and extension argument are in `hi_cft_20260923_next_structure_order13_core.md` and `hi_cft_20260923_next_structure_core_extension.json`. There are fourteen simple modules, labelled $x_a = (1, a)$ and $y_a = (\tau, a)$ for $0 \le a \le 6$. Indeed the free order-thirteen extension removes the hyperbolic pointed factor; no orbit has a stabilizer, so there is no fixed-point resolution. The two remaining labels are the Fibonacci label and the even affine label $2a$.

We give enough charge data to reconstruct their tops. For $\epsilon \in \{1, \tau\}$ put

$$p_1(m) = \begin{cases} 0, & 3 \mid m, \\ 2/3, & 3 \nmid m, \end{cases} \qquad p_\tau(m) = \begin{cases} 2/5, & 3 \mid m, \\ 1/15, & 3 \nmid m. \end{cases}$$

The actual lowest weights of the lattice–Potts modules are

$$h_{\mathcal{T}}(\epsilon, j) = \min_{m \equiv j \ (13)} \left(\frac{m^2}{156} + \frac{\mathbf{1}_{m \text{ odd}}}{4} + p_\epsilon(m) \right). \tag{108}$$

An even minimizing m contributes a left affine singlet and an odd one a left doublet; their right Cartan charge is m. Choosing $|m| \le 6$ bounds the minimum by $14/13$, while $|m| \ge 13$ gives $m^2/156 > 14/13$. Thus $-12 \le m \le 12$ contains every minimizer. For the representative $-6 \le k \le 6$, the parafermion formula is

$$h_{\mathrm{PF}}(a, k) = \frac{a(a+1)}{15} - \frac{k^2}{13} + \max(0, |k| - a). \tag{109}$$

The last term is the least affine grade needed for charge $2k$ outside the finite $\mathfrak{sl}_2$ top. This is equivalently the actual-top formula in [16, Section 2.4]; the parafermion top is one-dimensional. Finally minimize $h_{\mathcal{T}}(\epsilon, j) + h_{\mathrm{PF}}(a, -5j)$ over $j \in \mathbb{Z}/13$, using the representative in (109).

Let E be any simple U-module top and write x_a, y_a also for the nonnegative integer multiplicities of these A_L tops in E. Lemma F.4 justifies taking complete A_L tops here: the finite space E is invariant under all their Zhu zero modes, not merely contained in the sum of their highest spaces. Set $D = \dim E$. For each top, let L be the trace of the left affine energy, $Q = \mathrm{Tr}\, H_R(0)^2$, and P, Z the Potts and parafermion energy traces. Its contribution to (107) is

$$W = L + \frac{83}{6084} Q + \frac{25}{6} P + \frac{25}{26} Z. \tag{110}$$

This is a trace formula on a complete right $\mathfrak{sl}_2$ representation, not an assertion about a single partial-energy eigenvector. In fact, for an irreducible right $\mathfrak{sl}_2$ representation of highest weight r,

$$\frac{\mathrm{Tr}\, H_R(0)^2}{\dim} = \frac{r(r+2)}{3}, \qquad L^{A_{1,39}}(0) = \frac{r(r+2)}{164}.$$

Thus the Sugawara trace is $3Q/164$, and multiplying by $3403/4563$ gives $83Q/6084$. Haar

conjugation preserves traces on E. The exact dimensions and trace contributions are

top	d	$4680W$	top	d	$4680W$
x_0	1	0	y_0	1	7800
x_1	1	600	y_1	7	36960
x_2	5	10440	y_2	15	110880
x_3	9	56640	y_3	2	4080
x_4	2	5520	y_4	14	81480
x_5	14	148560	y_5	6	18240
x_6	6	54240	y_6	2	4920

For clarity, the lowest core weights and all minimizing absolute charges used in this table are

	h_{A_L}	$\lvert j\rvert$		h_{A_L}	$\lvert j\rvert$
x_0	0	0	y_0	2/5	0
x_1	2/15	0	y_1	8/15	0, 5
x_2	2/5	0, 3	y_2	4/5	0, 2, 3, 5
x_3	4/5	0, 2, 3, 6	y_3	1/5	2
x_4	1/3	6	y_4	11/15	1, 2, 4, 6
x_5	1	1, 4, 6	y_5	2/5	1, 4
x_6	4/5	4	y_6	1/5	4

For each positive j both signs occur. A convenient independent check on every row is

$$Z = h_{A_L} d - L - Q/156 - P, \qquad W = \frac{L}{26} + \frac{7Q}{936} + \frac{125P}{39} + \frac{25}{26} h_{A_L} d.$$

Complete right $\mathfrak{sl}_2$ characters imply $y_2 \geq x_5$ and $y_4 \geq x_5 + x_6$: these compare charges $10, 12$ and charges $7, 9$, respectively, with the common left doublet removed in the second comparison. Direct comparison of the displayed table gives

$$\begin{aligned} 4680(\tfrac{17}{10} D - W_{\text{total}}) = {} & 8460(y_2 - x_5) + 28716(y_4 - x_5 - x_6) \\ & + 7956x_0 + 7356x_1 + 29340x_2 + 14964x_3 + 10392x_4 \\ & + 22212x_6 + 156y_0 + 18732y_1 + 11832y_3 \\ & + 1188y_4 + 29496y_5 + 10992y_6 \geq 0. \end{aligned} \tag{111}$$

Since $W_{\text{total}} = hD$, this proves $0 \leq h \leq 17/10 < 2$ for *every* simple U-module, without assuming it is unitarizable.

REVIEW — R2: scope of the core and finite-top certificates

The charge minimizations and the nonnegative identity (111) are specified above; their finite exact check is `hi_cft_20260923_next_structure_order13_top_check.json`. The embedded lattice–Potts core and the spectral field witness remain construction inputs, supplied by the exact core-extension certificate. The finite-top irreducibility below is likewise imported from exact field actions, particularly `hi_cft_20260923_next_81top_all.json`; it should not be replaced by numerical cyclicity alone.

The strongly rational restriction $B \downarrow_U$ is semisimple. Its simple summands have integer lowest weights, hence only weights $0, 1$ are possible. The vacuum occurs once, while the complementary current space has dimension $86 - 6 = 80$. Its certified irreducible zero-mode action therefore implies

$$B \downarrow_U = U \oplus X, \qquad h_X = 1, \qquad \dim X_{\text{top}} = 80. \tag{112}$$

The four mixed nontrivial $E_6 \oplus A_2$ lattice sectors start at weight 1, with 81-dimensional tops. Their next possible top weight would be 2, excluded by (111). Their exact top-action certificates use a one-dimensional Cartan eigenline, positive-metric adjoint identities, and a cyclic span of dimension 81. The adjoint identities make the generated operator algebra a star algebra; orthogonal complements are therefore invariant, and the unique eigenline plus cyclicity proves irreducibility.

Write the actual lattice extension as $H = B \oplus J \oplus J^2$. Its restriction is consequently

$$H \downarrow_U = \mathbf{1} \oplus X \oplus Y \oplus Y^*, \qquad h_Y = 1, \quad \dim Y_{\mathrm{top}} = 81. \tag{113}$$

At this stage $Y \cong Y^*$ has not been assumed. The three types $\mathbf{1}, X, Y$ are distinguished by their lowest spaces.

G.2 The rank-four or rank-six alternative

Put $\mathcal{C} = \mathrm{Rep}(U)$ and $\mathcal{F} = \mathcal{C}_H$, the category of *all* right modules over the extension algebra H. The strict positivity of nonvacuum lowest weights gives the positive spherical structure through character asymptotics. Since H is holomorphic, it is Lagrangian and the center correspondence [39, Section 4] gives $\mathcal{C} \simeq Z(\mathcal{F})$. Write $s = \mathrm{FPdim}_{\mathcal{C}} B$. Extension dimension identities give

$$d_X = s - 1, \qquad d_Y = d_{Y^*} = s, \qquad \mathrm{Dim}\,\mathcal{C} = 9s^2, \quad \mathrm{Dim}\,\mathcal{F} = 3s.$$

Pseudo-unitarity of $\mathcal{F}$ follows from that of its center and the squared global-dimension and Frobenius–Perron-dimension identities. No C^* structure is needed in this section. All categorical dimensions below use the positive spherical structure and therefore equal Frobenius–Perron dimensions.

For center induction I and forgetful functor F, one has $H = I(\mathbf{1})$ and $FI(\mathbf{1}) = \bigoplus_{a \in \mathrm{Irr}\,\mathcal{F}} a \otimes a^*$. Adjunction therefore gives

$$\mathrm{rank}\,\mathcal{F} = \dim \mathrm{End}_{\mathcal{C}}(H) = \begin{cases} 4, & Y \ncong Y^*, \\ 6, & Y \cong Y^*. \end{cases} \tag{114}$$

Lemma G.1 (Automorphisms give invertible boundary objects)**.** *For a pseudo-unitary fusion category $\mathcal{F}$, algebra automorphisms of its canonical Lagrangian algebra $I(\mathbf{1})$ inject into* $\mathrm{Inv}(\mathcal{F})$.

Proof. Normalize the special Frobenius structure so that $m\Delta = \mathrm{id}$ and set $f * g = m(f \otimes g)\Delta$. If f is an algebra automorphism, then $f * f = f$, and the same holds for its composition inverse. Under the two-product description [17, Lemma 3.1 and Corollary 3.4], composition is fusion-ring multiplication and the primitive convolution idempotents are $d_a[a]$. Thus both f and f^{-1} have expansions with coefficients 0 or $d_a > 0$. Their fusion product is $[\mathbf{1}]$. Nonnegative fusion coefficients rule out cancellation. Every pair of objects in the two supports must therefore multiply to copies of the unit; rigidity then forces each pair to consist of inverse invertible objects. The two supports are singletons and their dimensions are one. Composition preserves this assignment, proving the injection. □

The actual order-three automorphisms of H/B consequently inject into $\mathrm{Inv}(\mathcal{F})$. This injection is already spherical: the proof used positivity of dimensions and fusion coefficients, without using adjoints of morphisms.

Lemma G.2 (No sign extension)**.** *The nonvacuum summand X of $B \downarrow_U = U \oplus X$ is not invertible.*

Proof. If X were invertible, self-duality of B and connectedness would give $X^2 = \mathbf{1}$. The extension would have its $\mathbb{Z}/2$ grading, and its sign automorphism g would satisfy $B^g = U$. On currents this would give $(E_6 \oplus A_2)^g = U_1 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, of rank two.

Every finite-order automorphism of a complex simple Lie algebra is conjugate to a toral automorphism times a diagram automorphism; see [18, Section 5.2]. It therefore fixes a toral subalgebra whose dimension is the number of orbits of the diagram automorphism on simple roots. In type E_6 this number is at least four, and in type A_2 at least one. Since the nonisomorphic ideals E_6, A_2 are preserved separately, $(E_6 \oplus A_2)^g$ contains a toral subalgebra of dimension at least five. Such a subalgebra cannot lie in the rank-two Lie algebra U_1, a contradiction. □

If rank six had six invertibles, then $3s = 6$ and $d_X = s - 1 = 1$, contradicting Lemma G.2. Thus there are exactly three invertibles in either rank. Their associator is also determined: the local B-module category has metric group

$$(\mathbb{Z}/3)^2, \qquad q(a,b) = \frac{2a^2 + b^2}{3} \pmod 1.$$

The actual extension H/B selects the diagonal isotropic line $L_+ = \{(a,a)\}$. Its complement $L_- = \{(a,-a)\}$ is also isotropic, and $(\mathbb{Z}/3)^2 = L_+ \oplus L_-$. The pointed subcategory on L_- is symmetric with trivial twists, hence Tannakian; as a tensor category it is the untwisted $\mathrm{Vec}_{\mathbb{Z}/3}$. Free H/B-module induction is tensor on this subcategory. It is fully faithful since

$$\mathrm{Hom}_{H/B}(x \otimes (H/B), y \otimes (H/B)) \cong \mathrm{Hom}_{\mathrm{Rep}(B)}(x, y \otimes (H/B))$$

has dimension $\delta_{x,y}$ for $x, y \in L_-$: distinct elements of L_- represent distinct cosets of L_+. Every free module is a direct sum of these three simple modules, and separability makes every module a summand of a free module. Thus induction identifies the entire all-module category for H/B with this untwisted pointed category. Viewing these modules in $\mathrm{Rep}(U)$ forgets only the locality condition for the intermediate algebra B; it preserves H-module morphisms and relative tensor products. This gives the required full tensor embedding into $\mathcal{F}$. Merely knowing the group of invertibles would not determine this associator.

G.3 Excluding rank four

At rank four the fusion ring is near-group:

$$a^3 = 1, \qquad a\rho = \rho a = \rho, \qquad \rho^2 = 1 + a + a^2 + m\rho.$$

There are exactly two self-dual simples, $\mathbf{1}$ and ρ. Consequently [19, Theorem 1.1], which assumes pseudo-unitarity, restricts m to $0, 2, 3, 6$. For $m = 0$, $s = 2$ contradicts Lemma G.2. For $m = 2$, the categorical reduction [20, Theorem 7.1 and Corollary 7.4] supplies the twisted A_4 double. Its electric regular algebra gives a holomorphic $c = 8$ extension H' with $U = (H')^{A_4}$ by the extension correspondence [40]. The nonvacuum lowest weights are positive, so H' is of CFT type, and [41, Theorem 1] identifies it with V_{E_8}. For completeness, the nontrivial order-two fixed Lie algebras are D_8 and $E_7 + A_1$, with dimensions $120, 136$ and adjoint traces $-8, 24$. At order three the possibilities are A_8, $E_6 + A_2$, $D_7 + \mathbb{C}$, and $E_7 + \mathbb{C}$, with dimensions $80, 86, 92, 134$ and traces $-4, 5, 14, 77$. These lists follow by selecting Kac coordinates of total mark two or three in the affine E_8 diagram, whose marks are $1, 2, 2, 3, 3, 4, 4, 5, 6$. For order three the trace is $(3 \dim \mathfrak{g}^g - 248)/2$, since the two nontrivial eigenspaces are dual. Thus the two relevant traces are bounded below by -8 and -4. Averaging over A_4 gives

$$\dim(E_8^{A_4}) \geq \frac{248 + 3(-8) + 8(-4)}{12} = 16,$$

contradicting $\dim U_1 = 6$.

The remaining cases require a spherical induction argument before any comparison with affine characters. We give this reduction explicitly.

Lemma G.3 (Near-group induction)**.** *Suppose $m = 3k$, and put $d^2 = 3 + 3kd$, $s = 2 + kd$. There are integers $p, q \geq 0$, $p + q = k$, and a primitive cube root ζ such that the known center objects have forgetful images*

object	$F(\text{object})$	*twist*
X	$1 + k\rho$	1
Y, Y'	$1 + a + k\rho,\ 1 + a^2 + k\rho$	$1, 1$
C	$a + a^2 + k\rho$	ζ
D, D'	$a + p\rho,\ a^2 + p\rho$	ζ^2, ζ^2
E, E'	$a + q\rho,\ a^2 + q\rho$	ζ^2, ζ^2

Every further simple Z_j forgets to $n_j\rho$, $n_j > 0$. Its twist θ_j satisfies

$$\sum_j n_j^2 = 6 + 3k^2 + 4pq, \tag{115}$$

$$S_1 := \sum_j n_j^2\theta_j = k^2 - kd - (p-q)^2\zeta^2,$$

$$\sum_j n_j^2\theta_j^2 = \overline{S_1} + \varepsilon\sqrt{9k^2 + 12}, \qquad \varepsilon = \nu_2(\rho) \in \{1, -1\}. \tag{116}$$

Proof. The four one-dimensional representations of the fusion algebra have formal codegrees $3s, 9s/d^2, 3, 3$. Thus induction of the unit is $1 + X + Y + Y'$, with dimensions $1, d^2/3, s, s$ and trivial twists. Since $FI(1) = 4 \cdot 1 + a + a^2 + 3k\rho$, adjunction and comparison of rational and irrational dimension coefficients give the first two rows. Also

$$FI(a) = 1 + 4a + a^2 + 3k\rho, \qquad FI(a^2) = 1 + a + 4a^2 + 3k\rho.$$

The two inductions have norm four, share one constituent, and contain Y, Y' respectively. Their remaining norm-three parts are therefore multiplicity-free, with a single common constituent C. Write its ρ multiplicity as c and the other two as p, q. Then $c + p + q = 2k$.

The untwisted pointed subcategory gives $\nu_3(a) = 1$, so positivity in the twist-trace formula forces every constituent of $I(a)$ to have cube-root twist. Its first indicator, separated over the disjoint fields $\mathbb{Q}(d)$ and $\mathbb{Q}(\zeta)$, gives

$$2 + 2\theta_C + \theta_D + \theta_E = 0, \qquad k + c\theta_C + p\theta_D + q\theta_E = 0.$$

The first equality forces $\theta_C = \zeta$ and $\theta_D = \theta_E = \zeta^2$, up to simultaneous conjugation; the second gives $c = k$ and $p + q = k$. Ribbon duality preserves the primed twists. Subtracting the nine known squared dimensions from $9s^2$ gives (115). The first two twist traces of $I(\rho)$ give (116), using $3/d = d - 3k$ and $3s/d = \sqrt{9k^2 + 12}$. □

Proposition G.4 (The multiplicity-three modular representation)**.** *For $k = 1$, the center has the modular data of $G_{2,3} \otimes A_{2,1}$ or their simultaneous reversal.*

Proof. One of p, q is zero. The corresponding D, D' are invertible and generate a nondegenerate pointed category $\mathcal{P}$ with twists $1, \zeta^2, \zeta^2$. Write $Z(\mathcal{F}) = \mathcal{M} \boxtimes \mathcal{P}$ using its modular centralizer. The known objects form three $\mathcal{P}$-orbits; representatives $1, X, C$ in $\mathcal{M}$ have dimensions $1, 1+d, s$ and twists $1, 1, \zeta$. Balancing verifies their trivial monodromy with D from $DX = E$ and

$DC \in \{Y, Y'\}$. The remaining free orbits, together with (115), leave three simples in $\mathcal{M}$, all of dimension d.

Take $\zeta = e^{2\pi i/3}$, the other choice conjugating the answer. For their twists z_1, z_2, z_3, the two moments become

$$\sum_i z_i = \alpha := \frac{-1 - i\sqrt{7}}{2}, \qquad \sum_i z_i^2 = -\frac{1}{2} - i(\varepsilon - \tfrac{1}{2})\sqrt{7}.$$

The second sum would have absolute value four for $\varepsilon = -1$, so $\varepsilon = 1$. Since $\alpha^2 + \alpha + 2 = 0$, Newton's identity and the unit modulus of all z_i give elementary symmetric functions $e_2 = \overline{\alpha}$, $e_3 = 1$. Thus the twists are the roots of $t^3 - \alpha t^2 + \overline{\alpha} t - 1$, namely $\zeta_7^3, \zeta_7^5, \zeta_7^6$.

The actual extension algebra $1 + X$ lies in $\mathcal{M}$. Its local module category has dimension three and is the pointed category with nontrivial twist ζ, fixed by its multiplicative central charge. Its two nonvacuum simples must both restrict to C: each underlying object has dimension s, cannot contain the unit by induction reciprocity, and no other combination of the six known dimensions has these properties. Hence the branching columns are $1 + X, C, C$. Modular intertwining for this extension [21, Theorems 4.1 and 4.5] fixes the first three rows of $\widehat{S} = \sqrt{3} s S_{\mathcal{M}}$:

$$\begin{pmatrix} 1 & 1+d & s & d & d & d \\ 1+d & 1 & s & -d & -d & -d \\ s & s & -s & 0 & 0 & 0 \end{pmatrix}.$$

All six objects are self-dual, the last three by their distinct twists. Their fusion matrices are real symmetric, so Verlinde diagonalization and the positive vacuum row make $S_{\mathcal{M}}$ real. For the remaining block K, orthogonality and the modular relation yield

$$\sum_j K_{ij} = d, \qquad \sum_j K_{ij} z_j = d + i\sqrt{3} s\, z_i^{-1}.$$

These determine each real row: three distinct points on a circle are not collinear. The affine $G_{2,3}$ data [22, Table 4 and equation (3.8)] realize these rows and twists in the reversed orientation; conjugating gives the other possibility. Thus the entire representation, rather than merely its twist list, is forced. □

Lemma G.5 (The balanced multiplicity-six pattern is impossible)**.** *In Lemma G.3 with $k = 2$, one has $\{p, q\} = \{0, 2\}$.*

Proof. Suppose $p = q = 1$, and put $d = 3 + 2\sqrt{3}$, $r = 1 + d$. The known dimensions are 1, $1 + 2d$, three copies of $2r$, and four copies of r. The remaining dimensions are $n_j d$, with $\sum_j n_j^2 = 22$. Write $u = (2 - \sqrt{3})/12$, $v = (2 + \sqrt{3})/12$. The normalized vacuum row is

$$(u, v, 1/3, 1/3, 1/3, 1/6, 1/6, 1/6, 1/6, n_j\sqrt{3}/12).$$

Signed Galois symmetry [23, Theorem II(iii)], for an automorphism reversing $\sqrt{3}$, sends this row to the X row: its first entry becomes $v > 0$, uniquely identifying X and a positive sign. The forced row is

$$(v, u, 1/3, 1/3, 1/3, 1/6, 1/6, 1/6, 1/6, -n_j\sqrt{3}/12).$$

The object X is self-dual because its dimension is unique. Verlinde's formula consequently gives

$$N_{XX}^X = \frac{v^3}{u} + \frac{u^3}{v} + \frac{3}{9} + \frac{4}{36} - \frac{22}{48} = \frac{97}{72} + \frac{32}{72} - \frac{33}{72} = \frac{4}{3},$$

contradicting integrality. Here $u + v = 1/3$, $uv = 1/144$ verify the first two terms directly. □

Lemma G.6 (A universal multiplicity-six character projection)**.** *The unbalanced case has a nondegenerate pointed factor $\mathcal{P}$ with twists $1, \zeta^2, \zeta^2$. In its modular centralizer there is a character projection with matrices*

$$S_v = -\frac{1}{2}\begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & -1 \\ 1 & -2 & 1 \end{pmatrix}, \qquad T_v = \operatorname{diag}(1, -i, -1), \tag{117}$$

before the common central-charge scalar, or the conjugate matrices.

Proof. The zero multiplicity gives the pointed factor just as for $k = 1$. In the centralizer the known simples have dimensions $1, 2d+1, 2d+2$ and twists $1, 1, \zeta$; all others have dimensions $n_j d$, with $\sum n_j^2 = 6$. The Gauss sum and the second indicator give

$$\sum n_j^2 z_j = -2 - 4i, \qquad \sum n_j^2 z_j^2 = -2 + 4i(1-\varepsilon).$$

The second equality excludes $\varepsilon = -1$ by absolute values. Since $(1+\Re z)(1+\Im z) \geq 0$ on the unit circle, summing gives $6 - 2 - 4 + \frac{1}{2}\Im(-2) = 0$. Thus each z_j is -1 or $-i$, with total squared multiplicity two or four, respectively.

Put

$$v_0 = \chi_1 - \chi_X, \qquad v_A = -\frac{1}{2}\sum_{z_j=-i} n_j \chi_j, \qquad v_B = -\sum_{z_j=-1} n_j \chi_j.$$

The vacuum row is $(u, v, 1/\sqrt{3}, n_j/4)$, with $u = (2\sqrt{3}-3)/12$, $v = (2\sqrt{3}+3)/12$; Galois symmetry gives the X row $(v, u, 1/\sqrt{3}, -n_j/4)$. Subtracting these rows yields $Sv_0 = -v_0/2 - v_A - v_B/2$. All three combinations are fixed by duality. Applying $S^2 = 1$ and $STS = iT^{-1}ST^{-1}$ to v_0 then determines

$$Sv_A = -v_0/2 + v_B/2, \qquad Sv_B = -v_0/2 + v_A - v_B/2,$$

which is (117). This proof permits either one dimension-$2d$ object or four dimension-d objects at twist $-i$; no unproved rank-nine classification has been used. □

Lemma G.7 (Pole cancellation)**.** *Let σ be a finite-image representation of $\mathrm{SL}_2(\mathbb{Z})$ with no invariant vector. Two weight-zero vector-valued modular functions, holomorphic on the upper half-plane, with representation σ and equal principal parts at infinity are equal.*

Proof. Their difference has no pole at infinity. Covariance makes every component holomorphic at every cusp of the finite-index kernel of σ. It descends to a holomorphic function on the compactified modular curve and is constant. The resulting constant vector must be invariant under σ, hence zero. □

Proposition G.8 (Current obstructions for the remaining near groups)**.** *Under the hypotheses above, $m = 3$ forces*

$$\dim U_1 = 15 + A + 10B \quad \text{or} \quad 14 + 42A, \qquad A, B \in \mathbb{Z}_{\geq 0},$$

whereas $m = 6$ forces

$$\dim U_1 = 8 + A \quad \text{or} \quad 5 + 12P, \qquad A, P \in \mathbb{Z}_{\geq 0}.$$

In particular none allows $\dim U_1 = 6$.

Proof. We give the theta normalizations, since a factor of two in the vacuum coordinate would change these obstructions. For $m = 3$, use the affine $G_{2,3}$ labels $1, 7, 27, 77, 14, 64$, and for each $A_{2,1}$ label put

$$v = (\chi_1 - \chi_{64}, -\chi_{77}, -\chi_{14}, -\chi_7).$$

The complement of the extension character is the even Fourier block on $\mathbb{Z}/7$, with entries

$$S_{00} = -1/\sqrt{7}, \quad S_{0r} = -2/\sqrt{7}, \quad S_{r0} = -1/\sqrt{7}, \quad S_{rs} = -2\cos(4\pi rs/7)/\sqrt{7},$$

and twist exponents $r^2/7$, $0 \le r \le 3$. Together with the even A_2 block, multiplication by η^8 identifies it with the Weil representation of a positive rank-four lattice with discriminant form $\mathbb{Z}_7(r^2/7) \oplus \mathbb{Z}_3(s^2/3)$. The actual extension gives $\chi_1 + \chi_{64} = \chi_B$. Thus a difference of two such character vectors has vacuum projection $\delta v_0 = 2\delta\chi_U$.

Take $L = L_7 \oplus A_2$, where L_7 has Gram matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 4 \end{smallmatrix}\right)$. For a basis u, v of L_7, write $w = v - u/2$ and $x = pu + tw$. Set $a = p^2$, $b = 7t^2/4$, and $c = (y, y)/2$ on the A_2 factor. The quadratics $H_1 = a - b$ and $H_2 = a + b - c$ are harmonic. Poisson summation gives vector-valued theta series of weight four. Their required coefficients are

(r, s)	q-power	Θ_{H_1}	Θ_{H_2}
$(1, 0)$	$1/7$	$-1/7$	$1/7$
$(3, 0)$	$2/7$	$3/7$	$4/7$
$(0, 0)$	1	2	-4

The first row comes from $2w/7$, the second from the two shortest vectors with $a = 1/4, b = 1/28$, and the last from two L_7 roots and six A_2 roots. The only nonvacuum polar top dimensions are A at $2/7$ and B at $1/7$, with zero allowed for an integer-shifted top. Subtract the affine reference, which has 22 currents and $A = 7, B = 0$. Pole matching gives

$$\eta^8 \delta v = \lambda\Theta_{H_1} + \mu\Theta_{H_2}, \qquad \lambda = 7 - A + 4B, \quad \mu = 7 - A - 3B.$$

This identity follows from Lemma G.7: after division by η^8, the remainder has no poles; its only possible T-fixed line is $(r, s) = (0, 1)$, which the Fourier S matrix does not preserve. The vacuum coefficient therefore gives $2(\dim U_1 - 22) = 2\lambda - 4\mu$, the first formula.

For the reverse orientation, realize the negative discriminant form by the positive lattice with Gram matrix

$$G_7^- = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 6 \end{pmatrix}.$$

Its dual generator $\ell = (16, 11, 6, 1)/21$ has energy $8/21$; $(r, s) \mapsto (9r + 7s)\ell$ identifies the discriminant form. Project the theta series to functions even separately in r, s. Let $H = \|x_{A_3}\|^2 - 3\|x_{A_3^\perp}\|^2$ and $F = -6D_2\Theta$, where $D_2 = q\,d/dq - E_2/6$ is the modular derivative. The necessary coefficients after projection are

(r, s)	q-power	F	Θ_H
$(0, 0)$	0	1	0
$(0, 0)$	1	-84	24
$(2, 1)$	$2/21$	$3/14$	$-2/7$

There are twelve roots, all in A_3. The polar coset has shortest vector $(1, 2, 3, 4)/21$, perpendicular to A_3; the separately even projection halves its coefficient, since the paired coset has no vector of the same norm. If A is its actual polar top dimension, matching the coefficient $-A$ gives $\eta^8 v = F + (3 + 14A)\Theta_H/4$. Here T has no fixed vector, so pole cancellation again

proves equality. Since $\eta^8\chi_B = 1 + 78q + \cdots$, its vacuum coefficient is $2\dim U_1 - 94$. Thus $2\dim U_1 - 94 = -84 + 6(3 + 14A)$, giving $14 + 42A$.

For $m = 6$, tensor the projection in Lemma G.6 with its pointed factor and first take the orientation with twists $1, i, -1$. After multiplication by η^8 this is the Weil representation for $L = A_1 \oplus A_1 \oplus A_2$, restricted by interchange of the two A_1 factors and conjugation of the A_2 charge. The harmonic quadratic $H = \|x_{A_1^2}\|^2/2 - \|y_{A_2}\|^2/2$ has coefficients

$$[q^{1/4}]\Theta_{H,(10,0)} = \frac{1}{2}, \qquad [q]\Theta_{H,(00,0)} = -2.$$

The first counts the two half-roots in a nontrivial A_1 coset; the second counts four A_1 roots and six A_2 roots. Let $A = \sum n_j \dim(M_j)_{\mathrm{top}}$ over the quarter-twist polar modules. This is a nonnegative integer, and its projected polar coefficient is $-A/2$. Use as reference the order-three simple-current extension of $A_{2,9}$ tensored with $A_{2,1}$: it has 16 currents and $A = 8$. Its projected representation agrees with (117); its matrices are also given in [24, Lemma 3.4 and Proposition 3.5]. Thus cancellation gives

$$\eta^8\delta v = (8 - A)\Theta_H, \qquad 2(\dim U_1 - 16) = -2(8 - A).$$

As in the first orientation above, the sole T-fixed coordinate is not S-invariant. This gives $\dim U_1 = 8 + A$.

For the reverse orientation use the positive lattice with Gram matrix

$$G_6^- = \begin{pmatrix} 2 & -1 & -1 & -1 \\ -1 & 2 & 0 & 0 \\ -1 & 0 & 2 & 0 \\ -1 & 0 & 0 & 4 \end{pmatrix}.$$

Its first three basis vectors span A_3; if the basis is a, b, c, e, then $w = e + a + (b + c)/2$ is perpendicular to A_3 and has norm three. The discriminant generators $(b + e)/2, (c + e)/2, 2w/3$ have quadratic values $3/4, 3/4, 2/3$ and mutually integral pairings. The polar coset is represented by $-w/3$, with energy $1/6$. This shortest vector is unique, and the projection does not halve its coefficient. With $H = \|x_{A_3}\|^2 - 3\|x_{A_3^\perp}\|^2$ and $F = -6D_2\Theta$,

coordinate	q-power	F	Θ_H
vacuum	0	1	0
vacuum	1	-84	24
polar	1/6	0	-1

Let P be the weighted top dimension in the half-twist coordinate with nonvacuum pointed charge; the weighting is by the integers n_j. Contragredience equates the two pointed-charge sums, so averaging does not change this integer. Pole cancellation gives $\eta^8 v = F + P\Theta_H$: there is no T-fixed coordinate in this orientation. Comparing the vacuum coefficient yields $2\dim U_1 - 94 = -84 + 24P$, hence $\dim U_1 = 5 + 12P$. □

The representations used in the last proof have finite image because they are Weil representations, with the finite-order eta multiplier removed. Thus the proof does not assume a dimension formula for a space of vector-valued modular forms. The coefficients A, B, P are weighted dimensions of actual polar tops, and distinct modules need not have linearly independent characters.

Remark G.9 (Scope of the rank-four exclusion)**.** The universal induction, balanced-pattern obstruction, weighted projection, and theta coefficient normalizations are printed above. Their

source calculations are `hi_cft_20260923_next_m3_universal_data.md`, `hi_cft_20260923_next_structure_m6_universal_projection.md`, `hi_cft_20260923_next_structure_m3_character_obstruction.md`, and `hi_cft_20260923_next_structure_m6_character_obstruction.md`. Lemma G.2 removes the formerly used pointwise stabilizer calculation $\mathrm{Aut}(B/U) = 1$ from this proof. No classification restricted to C^* categories is used here.

G.4 The rank-six ring and exclusion of higher multiplicity

Rank four being excluded, (114) gives $H = \mathbf{1} \oplus X \oplus 2Y$. The composition algebra of $\mathrm{End}_{\mathcal{C}}(H)$, identified with the complex fusion algebra by [17, Corollary 3.4], is

$$K_0(\mathcal{F}) \otimes \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}),$$

In particular it is noncommutative. Let a generate the three invertibles. A permutation of order three on three noninvertibles is either transitive or trivial. If the left action were trivial, duality would make the right action trivial as well. The two nontrivial character idempotents of $\mathbb{C}[\mathbb{Z}/3]$ would then be one-dimensional direct ideals. Their four-dimensional complementary ideal would have to be $M_2(\mathbb{C})$, but the dimension character is nonzero on that ideal, a contradiction.

Choose a noninvertible ρ. The right action commutes with the free left cyclic action, so it is either the same cycle or its inverse. The same cycle would make the fusion ring generated by two commuting elements a, ρ, which is impossible. Thus $\rho a = a^2\rho$. If $\rho^* = a^k\rho$, then $(a^i\rho)^* = a^{i+k}\rho$; involutivity gives $2k = 0$ in $\mathbb{Z}/3$, hence $k = 0$. All three noninvertibles are self-dual. Frobenius reciprocity gives one copy of the unit and no other invertible in ρ^2. Write its remaining coefficients as m_0, m_1, m_2. The identity $\rho^2 a = a\rho^2$ cyclically equates these coefficients, so

$$a^3 = 1, \qquad \rho a = a^2\rho, \qquad \rho^2 = 1 + m(\rho + a\rho + a^2\rho), \qquad m \geq 1. \tag{118}$$

The following induction calculation determines the multiplicity. Put

$$d = \frac{3m + \sqrt{9m^2+4}}{2}, \quad s = 1 + d^2, \quad \Delta = 9m^2 + 4, \quad R = \rho + a\rho + a^2\rho.$$

Lemma G.10 (Center induction and two moments)**.** *For a pseudo-unitary categorification of* (118) *whose pointed subcategory is untwisted, center induction satisfies*

$$I(1) = 1 + X + 2Y, \quad d_X = d^2, \quad d_Y = s, \qquad I(a) = I(a^2) = C_0 + C_1 + C_2,$$

where $\theta_X = \theta_Y = 1$, $d_{C_j} = s$, and the C_j twists are $1, \zeta_3, \zeta_3^2$. Every other central simple Z_j satisfies $F(Z_j) = n_j R$, with $n_j \in \mathbb{Z}_{>0}$, and

$$\sum_j n_j^2 = 4m^2 + 2,$$

$$\sum_j n_j^2 \theta_j = -\frac{m^2 + m\sqrt{\Delta}}{2}, \tag{119}$$

$$\sum_j n_j^2 \theta_j^2 = \frac{-m^2 + (2\varepsilon - m)\sqrt{\Delta}}{2}, \qquad \varepsilon = \nu_2(\rho) \in \{1, -1\}. \tag{120}$$

Proof. The complex fusion algebra has two characters $a \mapsto 1$, $\rho \mapsto d, -d^{-1}$, and the two-dimensional representation

$$a \longmapsto \begin{pmatrix} \zeta_3 & 0 \\ 0 & \zeta_3^2 \end{pmatrix}, \qquad \rho \longmapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Their formal codegrees are respectively $3s, 3s/d^2, 3$. The induction–formal-codegree correspondence [25, Theorems 2.5 and 2.13] gives $I(1) = 1 + X + 2Y$, with the stated dimensions and trivial twists. Conjugation by a cycles the three noninvertibles, so their multiplicities in the forgetful image of a central object coincide. Since

$$FI(1) = 6 \cdot 1 + 3mR,$$

adjunction and dimensions give $F(X) = 1 + mR$ and $F(Y) = 2 \cdot 1 + mR$.

Similarly,

$$FI(a) = FI(a^2) = 3a + 3a^2 + 3mR.$$

Thus the two induced objects have self-inner-products and mutual inner-product equal to three. They are equal, multiplicity-free sums of three simples C_j, disjoint from $I(1)$. Adjunction gives $F(C_j) = a + a^2 + b_j R$, where $b_j \geq 0$ and $\sum_j b_j = 3m$. Untwistedness gives $\nu_3(a) = 1$. The center-twist formula [26, Theorem 4.1] therefore reads

$$\sum_j d_{C_j} \theta_{C_j}^3 = \operatorname{Dim} \mathcal{F} = \sum_j d_{C_j}.$$

Positive dimensions and equality in the triangle inequality imply $\theta_{C_j}^3 = 1$. The first indicator of a is zero, so

$$\sum_j (2 + 3b_j d)\theta_{C_j} = 0.$$

The real quadratic field $\mathbb{Q}(d)$ is disjoint from $\mathbb{Q}(\zeta_3)$. Equating its two coefficients shows that the three twists are $1, \zeta_3, \zeta_3^2$ once each and $b_j = m$ for all j. All invertible constituents have now been exhausted by induction of $1, a, a^2$; every remaining forgetful image is $n_j R$.

The center dimension identity is

$$9s^2 = 1 + d^4 + 4s^2 + 9d^2 \sum_j n_j^2,$$

which simplifies to the first asserted equation. Finally adjunction gives

$$I(\rho) = mX + mY + m(C_0 + C_1 + C_2) + \sum_j n_j Z_j.$$

Its first and second twist traces are $\operatorname{Dim} \mathcal{F}\, \nu_1(\rho) = 0$ and $\operatorname{Dim} \mathcal{F}\, \nu_2(\rho) = 3s\varepsilon$. Using $d^{-1} = d - 3m$ gives (119) and (120). □

The roots in these equations form a multiset of exactly $4m^2 + 2$ elements, with each θ_j repeated n_j^2 times. No distinctness of the twists has been assumed.

Write $\Delta = t^2 D$ with D squarefree. Since $(3m)^2 < \Delta < (3m + 1)^2$, we have $D > 1$. Reduction modulo three and seven gives $D \equiv 1 \pmod 3$ and $7 \nmid D$: the latter uses that $m^2 \equiv 5 \pmod 7$ has no solution. Except when $D = 10$,

$$\frac{\varphi(2D)}{\sqrt{D}} \geq \frac{12}{\sqrt{13}}. \tag{121}$$

Indeed for odd D the left side is the product of $(p-1)/\sqrt{p}$ over its prime factors. A single prime must be at least 13, and two distinct primes contribute at least $40/\sqrt{55} > 12/\sqrt{13}$; all further factors exceed one. For even D there is an additional factor $\sqrt{2}$. A single odd prime is 5 (the exceptional $D = 10$) or at least 11, and two odd primes again give a larger product.

Apply [19, Theorem 1.3] to two copies of the root multiset in (119), to remove any half-integral coefficients. Together with (121) this gives

$$4m^2+2 \geq \frac{mt}{2}\varphi(2D) \geq \frac{6m\sqrt{9m^2+4}}{\sqrt{13}}.$$

For $m>1$ this is impossible, since

$$36m^2(9m^2+4)-13(4m^2+2)^2 = 4(m^2-1)(29m^2+13) > 0.$$

The exceptional squareclass requires the following equality statement.

Lemma G.11 (The conductor-forty bound)**.** *If a multiset of N roots of unity has sum $-a-b\sqrt{10}$, where $a,b\in\mathbb{Z}_{\geq 0}$, then $N\geq a+8b$. If equality holds and the sum of the squared roots lies in $\mathbb{Q}(\sqrt{10})$, that second sum is a.*

Proof. Choose a cyclotomic field containing all roots and $\mathbb{Q}(\sqrt{10})$. Take the union of the multiset over its Galois group G fixing $\sqrt{10}$, and put $M=|G|$. This gives a multiset of MN roots, with sum $-Ma-Mb\sqrt{10}$, whose multiplicities are constant on G-orbits. Multiplicities remain integral throughout.

If an orbit consists of primitive nth roots and $40\nmid n$, its sum is the Ramanujan sum $\mu(n)\in\{0,1,-1\}$ and its size is $\varphi(n)$. Its rational contribution therefore costs at least one root per unit. A negative contribution realizes equality only for the singleton orbit $\{-1\}$.

If $40\mid n$, the primitive roots split into two equal orbits, distinguished by the primitive quadratic character of conductor 40. Their sums are

$$\frac{\mu(n)\pm\tau_n(\chi_{40})}{2}.$$

Here $\mu(n)=0$, and the induced Gauss sum vanishes unless $n=40u$ with u squarefree and coprime to 40. In that case

$$\tau_n(\chi_{40}) = \mu(u)\chi_{40}(u)\sqrt{40}.$$

This formula follows by separating the factors coprime to 40 in the Chinese remainder theorem; an extra power of a prime dividing the conductor gives zero. Thus an orbit contributes zero or $\pm\sqrt{10}$, at cost at least eight roots. Equality for a nonzero contribution requires $n=40$.

The rational and irrational contributions come from disjoint types of orbit, so $MN\geq Ma+8Mb$. Equality leaves only Ma copies of $\{-1\}$ and Mb copies of the negative quadratic orbit of primitive fortieth roots. The latter orbit has squared sum zero: reduction of its exponents modulo 20 gives every unit once, since $\chi_{40}(x+20)=-\chi_{40}(x)$, and the resulting sum is $\mu(20)=0$. The averaged squared sum is therefore Ma. By the additional hypothesis, Galois averaging multiplied the original second sum by M. Dividing by M gives the assertion. □

In our application m,t are even and $a=m^2/2$, $b=mt/2$. Lemma G.11 gives

$$4m^2+2 \geq \frac{m^2}{2}+4mt$$

For $m\geq 4$ one has $t/m>\sqrt{9/10}>15/16$, making the right side strictly larger than $17m^2/4>4m^2+2$. The remaining $m=t=2$ is the equality case, so the squared sum must be 2, whereas (120) gives -2 or $-2-4\sqrt{10}$. Thus $m=1$. The same argument, including the indispensable second-moment hypothesis, is recorded in `hi_cft_20260923_next_structure_rank6_indicators.md` and `hi_cft_20260923_next_rank6_root_audit.md`.

Consequently

$$d=\frac{3+\sqrt{13}}{2},\qquad s=\frac{13+3\sqrt{13}}{2},\qquad d_X=\frac{11+3\sqrt{13}}{2}. \tag{122}$$

Lemma G.12 (The six residual twists)**.** *For $m = 1$, there are six residual central simples, each with $n_j = 1$. Their twists are the six primitive thirteenth roots with quadratic nonresidue exponents, and $\nu_2(\rho) = 1$.*

Proof. Equation (119) gives a multiset of six roots with sum $(-1 - \sqrt{13})/2$. Average it over a cyclotomic Galois group fixing $\sqrt{13}$, as in Lemma G.11. An orbit with nonzero $\sqrt{13}$ coefficient has coefficient of absolute value $1/2$ and at least six members. To see the equality condition, the induced quadratic Gauss-sum formula is nonzero only at primitive orders $13u$, with u squarefree and coprime to 13. The orbit size is $6\varphi(u)$, so equality permits only $u = 1, 2$.

The entire available length is already needed for the irrational coefficient. Hence no rational-only orbit or cancellation can occur. The two possible negative quadratic orbits, of orders 13 and 26, have sums $(-1 - \sqrt{13})/2$ and $(1 - \sqrt{13})/2$, respectively. The required rational coefficient forces every orbit to be the former. Consequently every original root is one of the six nonresidue thirteenth roots. These six roots are linearly independent over $\mathbb{Q}$: a rational relation would give a polynomial of degree at most twelve, with zero constant term, divisible by $\Phi_{13}(t) = 1 + t + \cdots + t^{12}$, and hence zero. Comparison with the quadratic period shows that each root occurs once. As each object was repeated n_j^2 times, all six objects have $n_j = 1$ and distinct twists.

Multiplication by two sends nonresidues modulo thirteen to residues, so the second sum is $(-1 + \sqrt{13})/2$. Equation (120) now gives $\varepsilon = 1$. □

The center thus has twelve simples. Its dimensions and twist exponents are

number	1	1	4	6
dimension	1	d^2	s	$3d$

$$0, 0, 0, 0, \tfrac{1}{3}, \tfrac{2}{3}, \tfrac{2}{13}, \tfrac{5}{13}, \tfrac{6}{13}, \tfrac{7}{13}, \tfrac{8}{13}, \tfrac{11}{13}.$$

The complete S, T data follow from [27, Theorem A(2), equivalently Theorem 5.1]: this statement assumes a complex spherical untwisted HI category. The auxiliary metric group has prime order 13 and its negative quadratic Gauss sum fixes its isometry class. This identifies the standard Haagerup modular data. It does not by itself identify a braided category; that stronger conclusion is treated separately below and in the analytic section.

G.5 An optional algebraic classification route

There is a second possible route from this fusion ring to an abstract ribbon identification. We isolate its extra input rather than folding an unreplayed classification into the preceding modular-data argument. Vercleyen [43, Section 4.4 and Table 9.1, p. 171] reports an exhaustive classification of complex pivotal multiplicity-free fusion categories of rank at most seven. Its list for $\mathrm{HI}(\mathbb{Z}_3)$ has four entries. Gauge equivalence there permits complex nonzero scalars and ring automorphisms; no unitarity assumption is imposed. The solution method uses exact polynomial and algebraic-number operations, with a separate numerical redundancy check. The complete notebooks and decision logs are available on request rather than included in the thesis [43, Sections 4.1.7–4.1.8 and 4.4.1]. We have not independently replayed that exhaustion calculation.

Proposition G.13 (Ribbon identification with an additional classification input)**.** *Assume, in addition to the hypotheses used above, that the four-entry list in [43, Table 9.1] exhausts complex pivotal fusion categories with the* $\mathrm{HI}(\mathbb{Z}_3)$ *fusion ring. Then $\mathcal{F}$ is tensor equivalent to $\mathcal{H}_2$ or $\mathcal{H}_3$, and there is an abstract ribbon equivalence*

$$\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3).$$

In particular this modular category is abstractly unitarizable.

Proof. Evans–Gannon [44, Theorem 3(ii)] construct four inequivalent pivotal categories with this ring. Two have noninvertible dimension $(3+\sqrt{13})/2$ and are the unitary categories $\mathcal{H}_2$ and $\mathcal{H}_3$; the other two have dimension $(3-\sqrt{13})/2$. Their theorem classifies the endomorphism realizations considered there; the additional exhaustion hypothesis is what makes this a complete list for the present abstract $\mathcal{F}$.

A pivotal structure on any categorification of this ring is unique when it exists. Indeed, the ratio of two such structures is a tensor automorphism of the identity. Write its scalar on a simple Z as t_Z. Since ρ^2 contains both $\mathbf{1}$ and ρ, $t_\rho^2 = 1 = t_\rho$. It also contains $a\rho$ and $a^2\rho$, so their scalars are one, and $a \otimes \rho = a\rho$ gives $t_a = 1$. Thus all six scalars are one. The negative-dimensional examples cannot acquire a positive spherical structure by repivotalization. The positive spherical category $\mathcal{F}$ is therefore one of the two positive examples.

The categories $\mathcal{H}_2$ and $\mathcal{H}_3$ are Morita equivalent by [36, Theorem 1.4], so their centers are braided equivalent. Combining this with $\mathrm{Rep}(U) \simeq Z(\mathcal{F})$ gives the claimed braided equivalence. Both sides have the positive spherical structures, which are unique; the induced pivotal and twist structures are consequently preserved. This makes the equivalence ribbon. Transporting the unitary structure from the Haagerup center proves abstract unitarizability. □

This proposition does not establish positivity of the canonical VOA fusion pairings or the analytic realization of the transported adjoint. Those are separate complete-unitarity assertions addressed later. Neither this classification input nor the proposition is needed for the character computation that follows.

H The complete character vector

Let $q = e^{2\pi i\tau}$. Character exponents are $h - c/24 = h - 1/3$. Using the preceding twist list and strict positivity of nonvacuum lowest weights, the only allowed nonvacuum pole is $\gamma q^{-7/39}$ in the twist-$2/13$ coordinate. Here $\gamma \in \mathbb{Z}_{\geq 0}$ is its top dimension if that module starts at $2/13$, and is zero if its lowest weight is shifted upward. The vacuum pole is $q^{-1/3}$ with coefficient one.

Lemma H.1 (Uniqueness from the principal part)**.** *Two character vectors for this modular representation with the same γ coincide in every weight.*

Proof. The modular representation has finite image by the congruence theorem [23, Theorem I]. A constant vector must be fixed by both S and the *character* T matrix, which includes $e^{-2\pi i/3}$. The only T-fixed coordinate is the object of categorical twist $e^{2\pi i/3}$. Its S image has a nonzero vacuum component, so it cannot span an invariant line. There are therefore no invariant constants, and Lemma G.7 applies. □

Evans–Gannon [9, Sections 1.4 and 2.4, equation (2.7)] provide a fundamental matrix Ξ whose vacuum and d_3 columns have precisely the two principal parts above. Thus, in their labeling,

$$\boldsymbol{\chi} = \Xi_{:,0} + \gamma \Xi_{:,d_3}, \qquad \dim U_1 = 6 + 13\gamma.$$

The actual six-current result forces $\gamma = 0$. Their recursive construction of Ξ and the uniqueness lemma give the entire vector, not just a finite coefficient match. In particular,

$$\mathrm{ch}\, U = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + 6010q^4 + \cdots). \tag{123}$$

The lowest weights and top dimensions are

label	h	$\dim M_{\mathrm{top}}$	label	h	$\dim M_{\mathrm{top}}$
0	0	1	d_1	6/13	7
b	1	80	d_2	11/13	42
a	1	81	d_3	15/13	119
c_0	1	81	d_4	5/13	5
c_1	1/3	3	d_5	7/13	13
c_2	2/3	27	d_6	8/13	14

In particular the final maximum is 15/13, improving the preliminary 17/10 bound. The proof record is `hi_cft_20260923_next_structure_conditional_characters.md`. Although this vector fixes every unflavoured homogeneous dimension, it does not determine the separate branching functions for the fourteen-module core A_L, and it was not used to prove properness.

Corollary H.2 (The four unmixed lattice restrictions)**.** *The two lattice modules labelled* $(0,\pm1)\in(\mathbb{Z}/3)^2$ *restrict to the simple* U*-module* c_1*, and the two labelled* $(\pm1,0)$ *restrict to* c_2*.*

Proof. The first pair has lowest weight 1/3, top dimension three, and weights in $1/3+\mathbb{Z}_{\geq0}$. The next possible summand top, 4/3, exceeds the established maximum 15/13. The character list contains a unique simple of lowest weight 1/3, with top dimension three, so semisimplicity gives precisely one copy of c_1. The second pair has lowest weight 2/3, top dimension 27, and next possible top 5/3; the same argument gives c_2. No additional finite-top irreducibility calculation is required. □

I From vacuum unitarity to complete unitarity

The exact strong-generation certificates concern the actual subVOA $U\subset B\subset H=V_{E_8}$. In particular, they establish $\dim U_1=6$ and $U\neq B$; they do not establish positivity on all modules or on fusion products. We now separate those questions. *Vacuum unitarity* is the positive invariant form inherited by the PCT-stable subVOA U. *Strong unitarity* means that all simple ordinary U-modules admit compatible positive invariant forms. *Complete unitarity* additionally includes positivity of the canonical Hermitian fusion forms, giving the VOA representation category its unitary modular tensor structure. An abstract ribbon equivalence to a unitary category does not, by itself, prove this last positivity statement.

We use the preceding algebraic analysis as input: regularity of U, the actual restrictions

$$B{\downarrow_U}=U\oplus X,\qquad H{\downarrow_U}=U\oplus X\oplus Y^{\oplus2},\qquad h_X=h_Y=1, \tag{124}$$

the twelve simple algebraic U-modules, and the $\mathrm{HI}(\mathbb{Z}/3)$ fusion ring of the all-module category for the original extension. Every use of these inputs inherits the qualifications attached to their preceding proofs. Agreement of modular data alone is never used as a classification of tensor categories.

REVIEW — Analytic comparison

The independent sector count and linear reconstruction resolve the earlier obligations A2 and A3. A separate pairwise comparison of bounded annulus insertions resolves A1 and supplies the positivity hypothesis. It uses a common localizer for each pair; a global choice of lifts for all individual annuli is unnecessary. Complete unitarity and the specified VOA extensions then follow from the same construction inputs. The net category also has a proof independent of A1. These arguments are mathematical deductions for review, not an assertion of independent expert validation.

I.1 The subnet and its sector count

There are two net constructions in use. Write $\mathcal{N}_V$ for the net generated by the closed smeared fields of a strongly local VOA V, and temporarily write $\mathcal{A}_V^{\mathrm{bl}}$ for the net generated by Tener's bounded localized insertions. For $V = U, B, H$ both exist: strong locality passes to unitary subalgebras, as do bounded localized vertex operators. The conformal vectors are the ambient ones, so the rotations and central charge agree in each construction.

First work with the bounded-insertion inclusion. Regularity of U and Tener [29, Corollary 4.24] give

$$[\mathcal{A}_H^{\mathrm{bl}} : \mathcal{A}_U^{\mathrm{bl}}] < \infty.$$

The ambient net is the holomorphic lattice net, by Tener [28, Proposition 7.5]. Hence Longo [32, Theorem 24] makes $\mathcal{A}_U^{\mathrm{bl}}$ completely rational. Tener's Lemma 7.6 gives a vacuum net isomorphism $\mathcal{A}_U^{\mathrm{bl}} \cong \mathcal{N}_U$, using the ambient lattice comparison. It follows that $\mathcal{N}_U$ is completely rational as well. Only this isomorphism-invariant property is transferred here: we do not infer equality of the two concrete net constructions or an identification of their embedded extensions. In particular, this step does not assume strong unitarity of U.

The actual smeared-field inclusion $\mathcal{N}_U \subset \mathcal{N}_H$ will be used for the sector calculation. Its branching and index are established below, after the elementary domain lemma needed for reconstruction.

Proposition I.1 (The actual net branching and sector count; A2)**.** *Assume the algebraic branching and rank-six ring deductions of Section G, including their certificate hypotheses. The restriction of the holomorphic H net has precisely the three inequivalent irreducible $\mathcal{N}_U$ summands corresponding to (124), with multiplicities $1, 1, 2$. Its boundary fusion category $\mathcal{F}_{\mathrm{net}}$ has rank six, untwisted invertible subcategory $\mathrm{Vec}_{\mathbb{Z}/3}$, and the $\mathrm{HI}(\mathbb{Z}/3)$ fusion ring. Consequently*

$$\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{F}_{\mathrm{net}}), \qquad \#\,\mathrm{Irr}\,\mathrm{Rep}(\mathcal{N}_U) = 12.$$

Proof. Every field $v \in U$ satisfies polynomial energy bounds on H, since it is a field of the strongly local lattice VOA and the conformal Hamiltonian is unchanged. Its restriction to an algebraic unitary summand satisfies the same bound. Consequently the finite-energy spaces are cores for the smeared fields in the respective Hilbert completions. Apply Lemma I.2 below to the orthogonal summand projections. This proves that the algebraic decomposition of H is a Hilbert direct-sum decomposition for the actual represented subnet. Conversely, a subnet intertwiner commutes with its local diffeomorphisms and hence with L_0. The equality of closed fields in reconstruction, or Fourier coefficients of their matrix elements, then makes its finite-energy restriction a U-module map. Thus simple summands stay irreducible and inequivalent, and the multiplicities remain $1, 1, 2$.

The inclusion is irreducible. To see the local assertion explicitly, let $x \in \mathcal{N}_H(I) \cap \mathcal{N}_U(I)'$. Locality makes x commute with $\mathcal{N}_U(I')$ as well. Strong additivity of $\mathcal{N}_U$ shows that the algebras of I and its complementary interval generate the global represented subnet: remove the two endpoints from any test interval and apply strong additivity twice. Thus x commutes with the global subnet and, in particular, its local diffeomorphisms and common rotations. Its action on the vacuum lies in the one-dimensional zero-energy space of H, so the separating property of the vacuum makes it scalar. The precise finite-index input is now Kawahigashi–Longo [33, Proposition 2.3]: an irreducible local extension of a net with finitely many irreducible DHR sectors, and in which every sector is a direct sum of finite-statistical-dimension sectors, has finite index. These hypotheses follow from the complete rationality already established for $\mathcal{N}_U$. Thus the actual inclusion has finite index. Its dual canonical sector is the restriction of the H vacuum computed above, and its dimension is the inclusion index. Its algebra A_{net}

is Lagrangian because $\mathcal{N}_H$ is holomorphic. The preceding comparison also gives the algebra identity

$$\mathrm{End}(A_{\mathrm{net}}) \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}). \tag{125}$$

Here the multiplication is composition, rather than a dimension count alone. Center induction identifies its dimension with $\mathrm{rank}(\mathcal{F}_{\mathrm{net}})$, and identifies the algebra with the complex fusion algebra of $\mathcal{F}_{\mathrm{net}}$. Thus the latter has rank six and is noncommutative.

We next identify the invertible objects using only the actual $\mathbb{Z}/3$ lattice-extension automorphisms of H/B. They act on the net inclusion and hence on its canonical algebra. The automorphism–invertible-object injection of Lemma G.1 embeds this group in $\mathrm{Inv}(\mathcal{F}_{\mathrm{net}})$. Since the category has rank six, the latter group has order three or six. If its order were six, the category would be pointed of global dimension six. The intermediate lattice inclusion has $[\mathcal{N}_H : \mathcal{N}_B] = 3$, so multiplicativity of index would give $[\mathcal{N}_B : \mathcal{N}_U] = 2$. An irreducible local index-two extension is the fixed-point extension of its sign involution: its canonical algebra is $\mathbf{1} \oplus \beta$, where $\beta^2 = \mathbf{1}$, and $\mathbf{1} \oplus (-\mathrm{id}_\beta)$ is its nontrivial algebra automorphism. Theorem 6.9 of [10] lifts this vacuum-preserving net automorphism to a unitary automorphism g of the actual B. Their Proposition 7.6 gives $\mathcal{N}_{B^g} = \mathcal{N}_B^g = \mathcal{N}_U$, and injectivity of their unitary-subalgebra/subnet correspondence (Theorem 7.5) gives $B^g = U$. This contradicts Lemma G.2. Consequently

$$\mathrm{Inv}(\mathcal{F}_{\mathrm{net}}) \cong \mathbb{Z}/3.$$

This argument needs no assertion about the full pointwise stabilizer $\mathrm{Aut}(B/U)$.

The lattice net $\mathcal{N}_B$ has discriminant metric group $\mathbb{Z}/3 \times \mathbb{Z}/3$ with $q(a,b) = (2a^2 + b^2)/3 \pmod{\mathbb{Z}}$. The extension H/B is its diagonal isotropic line $L_+ = \{(a,a)\}$, and $L_- = \{(a,-a)\}$ is a transverse isotropic line. The subcategory $\mathcal{T}_-$ supported on L_- is Tannakian and is tensor equivalent to the untwisted $\mathrm{Vec}_{\mathbb{Z}/3}$. Let Q_+ be the extension algebra supported on L_+. Free-module induction $t \mapsto t \otimes Q_+$ is a tensor functor from $\mathcal{T}_-$ to the category of all Q_+-modules. For simple $t, t' \in \mathcal{T}_-$, its adjunction gives

$$\mathrm{Hom}_{Q_+}(t \otimes Q_+, t' \otimes Q_+) \cong \mathrm{Hom}(t, t' \otimes Q_+) = \begin{cases} \mathbb{C}, & t = t', \\ 0, & t \neq t'. \end{cases}$$

Here transversality $L_+ \cap L_- = \{0\}$ gives the last equality. The target has dimension $9/3 = 3$, so these three distinct invertible images exhaust its simple objects. Induction is therefore a tensor equivalence. Forgetting the intermediate B-action embeds this category fully faithfully and tensorially in $\mathcal{F}_{\mathrm{net}}$; relative tensor product over the same H algebra is unchanged. Thus the entire pointed subcategory is untwisted.

We may now apply the rank-six ring and indicator argument of Section G to $\mathcal{F}_{\mathrm{net}}$ itself. Its hypotheses are the fusion-algebra decomposition above, the three invertibles, and their untwisted associator. They have just been proved for this category, and its spherical structure is positive because it is a C^* fusion category. The argument gives (118) with multiplicity one. In particular, for $d = (3 + \sqrt{13})/2$ and $s = 1 + d^2$, the center calculation gives dimensions

$$1,\ d^2,\ \underbrace{s, s, s, s}_{4},\ \underbrace{3d, 3d, 3d, 3d, 3d, 3d}_{6}. \tag{126}$$

The same center-induction and indicator calculation gives the last six twists with exponents $\{2, 5, 6, 7, 8, 11\}/13$. This is the source of the net count twelve. Finally the Lagrangian-algebra theorem gives $\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{F}_{\mathrm{net}})$, and the displayed center dimensions give its rank twelve. No VOA character transformation matrix was assigned to the net category. □

For comparison, the first paragraph also follows for the bounded-insertion inclusion directly from Tener [28, Proposition 5.5 and Theorem 5.6]. The argument above has the additional

advantage that the later automorphism comparison refers throughout to the original lattice fields. The purely categorical deductions used in the proof are recorded in `hi_cft_20260923_next_structure_rank46.md`, `hi_cft_20260923_next_structure_rank6_indicators.md`, and `hi_cft_20260923_next_structure_rank6_center.md`.

I.2 Reverse reconstruction and exhaustion

Henriques–Tener [30, Theorem 9.5 and Corollary 9.7] reconstruct unitary modules and identify their closed smeared fields. Remark 9.8 extends the construction beyond the finite-multiplicity hypothesis. This extension matters: finite-dimensional energy eigenspaces must not be silently assumed for the net representation one is trying to reconstruct. Their Theorem 8.9 identifies the reconstructed vacuum VOA of $\mathcal{N}_U$ with the original U, since U is strongly local.

The closure step can be isolated from the VOA theory. Its formulation also explains why a formal mode identity alone is insufficient.

Lemma I.2 (Reduction on a common core)**.** *Let $T_0 : D \to \mathcal{K}$ be a densely defined closable operator and P an orthogonal projection. Suppose $PD \subset D$ and $T_0 P\xi = PT_0\xi$ for $\xi \in D$. Then P reduces $T = \overline{T_0}$ and T^*, and it commutes with the von Neumann algebra generated by T. In particular, this holds simultaneously for a family of fields satisfying these hypotheses on a common core.*

Proof. If $\xi_j \in D$ converges to $\xi \in D(T)$ in graph norm, then $P\xi_j \to P\xi$ and $T_0 P\xi_j = PT_0\xi_j \to PT\xi$. Thus $P\xi \in D(T)$ and $TP\xi = PT\xi$. The same assertion holds for $1 - P$. It follows that T is the direct sum of its closed restrictions to $P\mathcal{K}$ and $(1 - P)\mathcal{K}$; taking adjoints gives the corresponding direct sum for T^*. Its polar partial isometry and the spectral projections of $|T|$ are therefore block diagonal. These generate the von Neumann algebra of T, proving the last assertion. Notice that no separate assertion that D is a core for T^* is required. □

Proposition I.3 (Reverse module comparison; A3)**.** *For this regular U, reconstruction sends each irreducible $\mathcal{N}_U$ sector to an irreducible ordinary unitary U-module. It preserves L_0 and induces a fully faithful linear $*$-functor on finite direct sums,*

$$\mathcal{R} : \mathrm{Rep}(\mathcal{N}_U) \longrightarrow \mathrm{Rep}^u(U).$$

The functor preserves conformal Hamiltonians. No tensorator or compatibility with braiding is asserted.

Proof. Let $\mathcal{K}$ be an irreducible net representation and put

$$M = \bigcup_{R>0} 1_{[0,R]}(L_0)\mathcal{K}$$

as in Henriques–Tener, Remark 9.8. This is a dense admissible weak U-module with a positive invariant Hermitian form. Theorem 9.5(ii) and its extension in that remark identify the closed reconstructed smeared field with $\pi_I(Y_U(v, f))$ for $\mathrm{supp}(f) \subset I$. By Definition 9.4, M is a core for this closed operator. The proof of Corollary 9.6 applies unchanged on bounded-energy vectors, so the VOA zero mode of the conformal vector is precisely the self-adjoint rotation generator on this domain.

Regularity decomposes the weak module M algebraically into ordinary simple modules. Choose a nonzero simple summand S. Its inherited form is positive and invariant. Since its L_0 is the restriction of the net Hamiltonian, its energy decomposition is

$$S = \bigoplus_{n\geq 0} S_{h+n}, \qquad \dim S_{h+n} < \infty, \quad h \geq 0.$$

The Hilbert closure $\overline{S}$ is the orthogonal Hilbert sum of these finite-dimensional spaces. In particular it reduces L_0, and the projection P onto $\overline{S}$ satisfies

$$P1_{[0,R]}(L_0)\mathcal{K} \subset \bigoplus_{h+n\le R} S_{h+n} \subset S, \qquad PM \subset S \subset M. \tag{127}$$

This establishes the domain invariance before any commutation with fields is used. It requires finite energy multiplicities only for the chosen ordinary summand, not for $\mathcal{K}$.

For a homogeneous $v \in U$, the unitary invariance identity expresses the adjoint of a mode $v_{(n)}$, on M, as a finite linear combination of modes of the vectors $L_1^j\theta v$, where θ is PCT. Every such mode preserves S. Testing against $s \in S$ therefore gives

$$\langle Pv_{(n)}\xi, s\rangle = \langle \xi, v^\dagger_{(n)}s\rangle = \langle P\xi, v^\dagger_{(n)}s\rangle = \langle v_{(n)}P\xi, s\rangle.$$

Both outer vectors lie in S, by (127), whence $Pv_{(n)}\xi = v_{(n)}P\xi$ for $\xi \in M$. The smeared fields on M are Hilbert-space vectors, as in the paragraph preceding Henriques–Tener Definition 9.4. Taking their energy components, or equivalently their convergent Fourier sums, now yields

$$PY^M(v,f)\xi = Y^M(v,f)P\xi, \qquad \xi \in M.$$

There is no assertion here that a smeared field maps M into itself. Only its value in $\mathcal{K}$ and the boundedness of P are needed. Lemma I.2 extends this identity to the closures and their adjoints.

The local algebras are generated by the closed smeared fields. Normal representations preserve this generation statement (equivalently, apply them to the polar decompositions and spectral projections). Thus P is a net intertwiner. Irreducibility and $S \neq 0$ imply $P = 1$. Equation (127) then implies $M = S$. This proves ordinaryness and simplicity, including the removal of the infinite multiplicities permitted in Remark 9.8.

Finally consider morphisms. A net intertwiner commutes with the canonical diffeomorphism action, because that action is generated by the local diffeomorphism groups. It therefore preserves bounded-energy vectors. Commutation with the reconstructed fields gives commutation with their Fourier coefficients, hence a module map. Conversely, an isomorphism of simple ordinary unitary modules is a scalar multiple of a unitary. To verify this without presupposing boundedness, compare the two invariant forms on each finite-dimensional energy space. The resulting grade-preserving positive operator is an algebraic U-endomorphism, by invariance of the forms. An eigenvector in the finite-dimensional lowest space has an eigenspace that is a nonzero U-submodule. Simplicity makes that eigenspace the whole module, so the comparison operator is a single positive scalar on all grades. The isomorphism therefore has a bounded unitary rescaling. Its unitary extension intertwines the smeared fields on their common cores and hence their closures, so it intertwines the nets. Between inequivalent simples both morphism spaces vanish. Matrices of these maps handle finite direct sums and respect adjoints. This proves full faithfulness and the asserted linear $*$-functor. □

The preliminary version of this deduction is recorded in `hi_cft_20260923_next_m3_unitarity_scope.md`.

Proposition I.4 (Strong unitarity from sector exhaustion)**.** *Under the preceding algebraic inputs, including the count of twelve simple U-modules, every simple U-module is unitarizable, and its character is the character of a net sector.*

Proof. Full faithfulness makes the twelve net simples pairwise inequivalent simple ordinary U-modules. There are exactly twelve algebraic simple modules, so this injection is a bijection. Each image carries its reconstructed positive form, and L_0 is preserved. Finite direct sums give the corresponding statement for ordinary modules. □

This counting step supplies essential surjectivity only as a linear comparison. Neither fusion compatibility, equality of categorical S matrices, nor positivity of VOA fusion forms follows from it.

I.3 Pairwise annulus compatibility

The next step concerns the specific geometric annulus system of Tener [28, Definition 3.18 and Corollaries 3.20–3.22]. Let (B, A) be a localized pair for an interval I, with incoming factor A and outgoing factor B. Subscripts distinguish their actions on the vacuum and on a reconstructed module. For an insertion point $z \in \mathrm{int}(B, A)$, put

$$X_v(B, z, A) = \overline{B_U Y_U(v, z) A_U} \in \mathcal{A}_U^{\mathrm{bl}}(I).$$

Here the bar denotes the bounded operator represented by the sesquilinear insertion form. At a point outside the incoming annulus one must not silently treat the unsmeared formal field as an everywhere defined Hilbert-space operator. The earlier abstract isomorphism $\mathcal{A}_U^{\mathrm{bl}} \cong \mathcal{N}_U$ does not identify these operators. The following proposition includes the specific vacuum identification needed to compare them, and then its module version.

Proposition I.5 (Pairwise annulus compatibility; A1)**.** *For Tener's geometric system above, the bounded-insertion and smeared-field nets agree on the original vacuum Hilbert space: $\mathcal{A}_U^{\mathrm{bl}}(I) = \mathcal{N}_U(I)$ for every interval. The Henriques–Tener geometric point insertion for the corresponding localized annulus pair is $X_v(B, z, A)$. If M is an ordinary unitary module reconstructed from a sector π_M, then*

$$\pi_M\big(X_v(B, z, A)\big) = \overline{B_M Y_M(v, z) A_M}, \qquad v \in U, \quad z \in \mathrm{int}(B, A). \tag{128}$$

The right side exists as a bounded extension of the insertion form. The assertion includes the generators $B_U A_U$ when the insertion interior is empty. It requires no global choice of central lifts for individual incoming annuli.

We first record the domain and continuation arguments used in the proof.

Lemma I.6 (Dense inverse-energy domains)**.** *Let M be an ordinary unitary module and let A_M, C_M be the actions of geometric annuli on its Hilbert completion. Suppose*

$$A_M C_M = \lambda r^{L_0}, \qquad 0 < r < 1, \quad \lambda \neq 0. \tag{129}$$

Then $M^{\mathrm{fin}} \subset \mathrm{Ran}\, A_M$ and $A_M^{-1} M^{\mathrm{fin}} = C_M M^{\mathrm{fin}}$ is dense. The same conclusion holds for the outgoing adjoint factor.

Proof. The geometric annulus actions are injective and have dense range by Henriques–Tener Lemma 4.15. Equation (129) gives, for $v \in M^{\mathrm{fin}}$, $v = A_M C_M(\lambda^{-1} r^{-L_0} v)$, proving the range assertion. It also gives $C_M M^{\mathrm{fin}} \subset A_M^{-1} M^{\mathrm{fin}}$. Conversely, if $A_M \xi \in M^{\mathrm{fin}}$, then

$$A_M \xi = A_M C_M \lambda^{-1} r^{-L_0} A_M \xi, \qquad \xi = C_M \lambda^{-1} r^{-L_0} A_M \xi.$$

The last equality uses injectivity of A_M, and proves the reverse inclusion. Finally M^{fin} is dense and C_M is bounded with dense range; hence $C_M M^{\mathrm{fin}}$ is dense. This is the argument of Henriques–Tener Lemma 7.9 with a central scalar retained. □

For the geometric annuli under consideration the factorization itself is elementary. The incoming annulus is embedded in the unit disc, its smooth inner Jordan domain contains zero, and its outer parametrization is the identity. Choose a sufficiently small round disc $r\mathbb{D}$

inside that inner domain. The intervening annulus C has its outer parametrization chosen to sew to the inner boundary of A, and its inner parametrization is the standard round one. Then AC is the round annulus. Adjusting the lift of C by the central $\mathbb{C}^\times \times \mathbb{Z}$ fiber makes the product a lift of the standard round annulus, giving (129). In particular the range and density requirements in Tener's definition of incoming annulus [28, Definition 4.2] hold on each reconstructed module.

Define

$$\mathcal{D}_A = \{\xi \in \mathcal{K}_M : A_M\xi \in M^{\mathrm{fin}}\}, \qquad \mathcal{D}_{B^*} = \{\eta \in \mathcal{K}_M : B_M^*\eta \in M^{\mathrm{fin}}\}. \tag{130}$$

Both domains are dense by Lemma I.6.

Lemma I.7 (Continuation of the insertion form)**.** *Let O_{in} and O_{out} be the incoming and outgoing insertion interiors, and put*

$$\Omega = \mathrm{int}(\overline{O_{\mathrm{in}}} \cup \overline{O_{\mathrm{out}}}).$$

Suppose a bounded geometric insertion $G_v(z)$ is holomorphic on Ω. If its matrix elements equal the algebraic insertion form on $O_{\mathrm{in}} \cup O_{\mathrm{out}}$, for vectors in the two domains (130), the equality holds throughout Ω. It uniquely defines a bounded extension of that form.

Proof. For $\xi \in \mathcal{D}_A$ and $\eta \in \mathcal{D}_{B^*}$, consider

$$\langle Y_M(v,z)A_M\xi, B_M^*\eta\rangle \tag{131}$$

as a formal matrix coefficient. It is a finite Laurent polynomial: if v has weight k, and the input and output energies are p, q, only the mode with $q = p + k - n - 1$ can contribute. Both vectors have finite energy support. Thus the form and the geometric matrix coefficient are holomorphic on $\Omega \subset \mathbb{C}^\times$.

Every component of Ω meets $O_{\mathrm{in}} \cup O_{\mathrm{out}}$. Indeed, its complement inside Ω is contained in the union of the two boundaries; boundaries of open sets are closed with empty interior, and a finite union of such sets contains no nonempty open component. The identity theorem therefore extends the equality across every component, including any part of the sewing circle lying in Ω. For fixed z, the form is now the restriction of the bounded operator $G_v(z)$ to two dense test domains. This proves existence and uniqueness of its bounded extension. A graph-core assertion is neither used nor needed. □

Proof of Proposition I.5. Write $X = (\psi, \rho, t, \gamma)$ for the incoming tuple and $\widetilde{X}$ for the tuple whose adjoint gives the outgoing factor. If $(\varphi_s)_{s\geq 0}$ is the semigroup generated by $-z\rho(z)\partial_z$, put $F_X = \psi \circ \varphi_t$. The corresponding embedded annulus is

$$A_X = \overline{\mathbb{D}} \setminus F_X(\mathbb{D}), \qquad \varphi_{\mathrm{out}} = \mathrm{id}, \qquad \varphi_{\mathrm{in}} = F_X \circ \gamma^{-1}.$$

Its inner Jordan domain contains zero by Tener's Definition 3.16.

First identify the annulus operator. For the semigroup part, the smooth framing $h(\theta, s) = \varphi_{t-s}(e^{i\theta})$, $0 \leq s \leq t$, runs from the inner to the outer boundary. The flow identity

$$\varphi_u(z)\rho(\varphi_u(z)) = z\rho(z)\varphi_u'(z)$$

gives $(-h_s/h_\theta)\partial_\theta = i\rho(e^{i\theta})\partial_\theta$. In the convention of [31, Remark 4.2], $\pi(f\partial_\theta) = T(if)$, so the time-ordered exponential [31, formula (5.2) and Theorem 5.5] is exactly $e^{-tL(\rho)}$. Since ρ has only nonnegative Fourier modes, every bounded-energy subspace is invariant under $L(\rho)$; uniqueness of the evolution identifies this exponential with Tener's finite-energy definition.

Thin factors implement boundary reparametrization. Thus the annulus A_X has a central lift whose vacuum action is exactly $A_U = U(\psi)e^{-tL(\rho)}U(\gamma)^*$, with the prescribed scalar. The case $t = 0$ consists only of thin factors.

We choose lifts for the pair as follows. By [28, Lemma 3.19 and Corollary 3.20], there is a common localizer δ such that $\delta F_X \gamma^{-1} = \mathrm{id}$ on I', and likewise for $\widetilde{X}$. Hence δA_X and $\delta A_{\widetilde{X}}$ are localized annuli. Their localized central extensions have $\mathbb{C}^\times$ fibers and embed canonically in $\widetilde{\mathrm{Ann}}_c$, as in [31, Section 7]. Choose their lifts $D_A, D_{\widetilde{A}}$ with vacuum operators $U(\delta)A_U, U(\delta)B_U^*$, respectively. Such normalization is possible by the preceding operator identification and is unique in the scalar fiber, since annulus operators are nonzero. Choose a common universal lift $\widetilde{\delta}$ projecting to the fixed central lift used in $U(\delta)$, and set

$$\underline{A} = \widetilde{\delta}^{-1} D_A, \qquad \underline{B} = D_{\widetilde{A}}^\dagger \widetilde{\delta}.$$

The product $\underline{BA} = D_{\widetilde{A}}^\dagger D_A$ is localized in I. The local transport theorem [31, Theorem 7.4], equivalently [30, Proposition 4.17], identifies their actions on a sector:

$$A_M = U^M(\widetilde{\delta})^* \pi_{M,I}(U(\delta)A_U), \qquad B_M = \pi_{M,I}(B_U U(\delta)^*) U^M(\widetilde{\delta}).$$

These are precisely the paired actions of [28, Section 5.1, formula (5.2)]. Changing $\widetilde{\delta}$ by the deck transformation Z^m, where $Z = e^{2\pi i L_0}$, replaces A_M, B_M by $Z^{-m}A_M, B_M Z^m$. The operator Z commutes with every U-mode, since these change energy by integers. The insertion form is therefore unchanged, even at fractional lowest conformal weights. No scalar comparison in the vacuum is being used to erase a deck transformation in a nonvacuum sector.

Next reflect $A_{\widetilde{X}}$ by $j(z) = \bar{z}^{-1}$ and sew it to A_X along their standard unit circles. Since j fixes the circle pointwise, the pair condition in Tener's formula (3.2) makes the resulting annulus BA localized in I. Its insertion interior for this geometric representative is the union of the incoming and outgoing interiors with their interior seam points, namely the set Ω in Lemma I.7. For $z \in \Omega$, the geometric operator

$$G_v(z) = (\underline{BA})[v(z)]$$

is bounded, lies in $\mathcal{N}_U(I)$, and acts in any sector as $\pi_{M,I}(G_v(z))$, by [30, Lemma 6.8]. It is holomorphic in z by their Corollary 6.15.

For z in the incoming interior, sewing [30, Section 6.2, formula (76)] and Theorem 9.5(i) give

$$G_v^M(z)\xi = B_M Y_M(v, z)A_M\xi, \qquad \xi \in \mathcal{D}_A.$$

The theorem applies because A_X is a divisor of a round annulus, as shown above; it permits every central lift. The outgoing equality of matrix coefficients follows by taking adjoints, using [30, Proposition 6.19 and formula (81)] and the invariant unitary form of M. The reflected state is $e^{\bar{z}L_1}(-\bar{z}^{-2})^{L_0}\Theta v$, the usual antiholomorphic change of local coordinate, a finite expression on U. Lemma I.7 now identifies the matrix coefficients on all of Ω and gives their unique bounded extension. In the vacuum the reconstructed VOA is the original U by [30, Theorem 8.9]; hence this extension is precisely $X_v(B, z, A)$. In the sector the same calculation proves (128). Repeat the construction for every geometric pair representing the same generalized operator pair. The union of these sewn interiors is the prescribed $\mathrm{int}(B, A)$, as explained in [28, proof of Theorem 4.28]. The operators agree on overlaps by the uniqueness of the bounded extension of the insertion form, so the identity holds on that entire prescribed interior.

For a pair with no insertion points, local transport directly gives $\pi_{M,I}(B_U A_U) = B_M A_M$. This also covers empty insertion interiors and supplies all no-insertion generators. We have

thus proved $\mathcal{A}_U^{\mathrm{bl}}(I) \subset \mathcal{N}_U(I)$. Both nets act on the same vacuum Hilbert space; the bounded-insertion net has cyclic vacuum and the same conformal covariance by [28, Proposition 4.13]. Haag duality for this vacuum conformal net, together with locality of $\mathcal{N}_U$, gives

$$\mathcal{A}_U^{\mathrm{bl}}(I) \subset \mathcal{N}_U(I) \subset \mathcal{A}_U^{\mathrm{bl}}(I')' = \mathcal{A}_U^{\mathrm{bl}}(I).$$

This proves equality of the concrete nets and completes the proof. □

Theorem I.8 (Complete unitarity)**.** *Under the preceding algebraic construction inputs, U is completely unitary.*

Proof. Proposition I.4 unitarizes every simple module. Proposition I.5 supplies the representation required by Tener [29, Definition 2.18], for each such module; finite direct sums preserve the property. His Corollary 4.6 then supplies positivity of the canonical fusion forms. Strong unitarity must be established first: the unitarizable-fusion-product hypothesis in Theorem 4.4 cannot be used to bypass the exhaustion argument. □

The pairwise proof above does not require Gui's Condition II. His extension comparison assumes that condition, including complete unitarity and energy bounds and strong intertwining for a generating family of charged fields; his inheritance theorem concerns regular cosets [49, Definition 4.15 and Theorems 4.20, 6.11]. Those results alone do not supply the present subalgebra comparison. Henriques–Tener also construct bounded vacuum multidisc operators and prove the vacuum trace-class condition [50, Theorems 3.6 and 4.1]. Our comparison instead uses the local transport and module point-insertion theorems explicitly, with a common localizer cancelling the pair's lift ambiguity. A linear equivalence of unitary module categories alone would not give the positivity of the canonical VOA fusion forms.

J Unitary categorical identifications

Lemma J.1 (All modules over a commutative Q-system)**.** *Let $\mathcal{C}$ be a unitary braided fusion category and A a connected commutative Q-system in $\mathcal{C}$. The category $\mathcal{C}_A$ of all right A-modules, with relative tensor product, is a unitarizable fusion category. This assertion does not require the modules to be local.*

Proof. If $r : M \otimes A \to M$ is a right action, the braiding gives a left action $l = r \circ c_{A,M}$. Commutativity of A and the hexagon identity make these commuting actions, and hence an A-bimodule. Naturality of the braiding makes every right-module morphism a bimodule morphism for these choices. This realizes $\mathcal{C}_A$ as the full subcategory of bimodules with $l = r \circ c_{A,M}$. The relative tensor product is obtained by splitting the separability idempotent on $M \otimes N$. Its induced left and right actions again satisfy this relation, by the same hexagon identity and the balancing relation. Thus the subcategory is closed under relative tensor products, sums and summands, and its unit is A.

Every such bimodule is unitarizable by Gui [34, Theorem 2.28]; an isomorphism transporting the actions preserves the displayed relation. The unitary relative tensor product and its associativity and unit constraints are supplied by Gui, Theorem 3.13. The above full subcategory therefore inherits a C^* tensor structure. Separability gives semisimplicity, connectedness makes its unit simple, and the usual module-category construction over a connected separable algebra gives rigidity and finiteness. It is consequently a unitary fusion category up to tensor equivalence.

For clarity, locality is the additional monodromy condition $r \circ c_{A,M} \circ c_{M,A} = r$. We have not imposed it. Only the local subcategory inherits the ambient braiding by descent. □

Theorem J.2 (Unitary Haagerup-center identification)**.** *Under the hypotheses of Theorem I.8, and the preceding determination of the all-module fusion ring and its untwisted pointed subcategory, there is a unitary ribbon equivalence*

$$\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3).$$

For the original embedded extension H/U, its all-module category is tensor equivalent to one of $\mathcal{H}_2$ and $\mathcal{H}_3$.

Proof. Put $\mathcal{C} = \mathrm{Rep}(U)$ and let A be the algebra of the actual unitary extension H. Complete unitarity allows application of Gui [34, Theorems 2.21 and 2.28]: A is a standard commutative Q-system and its modules are unitarizable. Lemma J.1 makes $\mathcal{F} = \mathcal{C}_A$ a C^* fusion category. Here $\mathcal{C}_A$ means *all* modules. The local modules form $\mathcal{C}_A^0 \simeq \mathrm{Vec}$ because the extension is holomorphic, so substituting local modules would lose the Haagerup category.

Its already determined fusion ring is $\mathrm{HI}(\mathbb{Z}/3)$. The actual untwisted pointed subcategory makes the obstruction $c^{(0,3)}$ vanish, while the other obstruction is in

$$H^1\Big(\mathbb{Z}/2, H^2(\mathbb{Z}/3, \mathbb{C}^\times)\Big) = 0.$$

Indeed, the cyclic-group calculation gives $H^2(\mathbb{Z}/3, \mathbb{C}^\times) = \mathbb{C}^\times/(\mathbb{C}^\times)^3 = 0$; every nonzero complex number has a cube root. Also $(\mathbb{Z}/3)/2(\mathbb{Z}/3) = 0$, so no choice of distinguished noninvertible object introduces an additional classification orbit. These are the hypotheses in Izumi [35, Definition 2.7 and Theorems 5.9 and 9.1], giving the two unitary tensor classes $\mathcal{H}_2, \mathcal{H}_3$. No algebra structure on $1+\rho$ is inserted as an extra assumption. Grossman–Snyder [36, Theorem 1.4] place these categories in the same Morita class. Their centers are therefore braided equivalent. The positive spherical structures make this a ribbon equivalence, and the Lagrangian-center theorem gives

$$\mathcal{C} \simeq Z(\mathcal{F}) \simeq Z(\mathcal{H}_3).$$

Reutter [37, Theorem 5] provides the unitary braided realization of the equivalence, preserving the associated positive ribbon structures. □

This uses the published unitary classification as an external input; its classification equations have not been replayed here. It avoids the broader pivotal census, but it does not prove that the natural reconstruction functor $\mathcal{R}$ preserves Connes fusion, the VOA tensor product, or braiding. That particular comparison is a distinct statement and is not needed for the theorem just proved.

Corollary J.3 (Haagerup-center identification for the conformal net)**.** *Under the preceding algebraic inputs, the completely rational central-charge-eight net $\mathcal{N}_U$ satisfies*

$$\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{H}_3)$$

as a unitary ribbon category. Its actual extension to the E_8 lattice net has boundary category equivalent to $\mathcal{H}_2$ or $\mathcal{H}_3$. Neither conclusion uses Proposition I.5.

Proof. Proposition I.1 constructs $\mathcal{F}_{\mathrm{net}}$ as a unitary fusion category with the $\mathrm{HI}(\mathbb{Z}/3)$ fusion ring and an untwisted pointed subcategory. The Izumi classification and the vanishing obstruction groups used in the preceding proof therefore apply directly to this category. Thus $\mathcal{F}_{\mathrm{net}}$ is tensor equivalent to $\mathcal{H}_2$ or $\mathcal{H}_3$. Their Morita equivalence, the unitary Lagrangian-center theorem, and uniqueness of the unitary braided structure give $\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{F}_{\mathrm{net}}) \simeq Z(\mathcal{H}_3)$. No positivity statement about VOA intertwining-operator spaces is used in this argument. □

K Full net theory and a chosen E_8 VOA extension

Corollary K.1 (A chosen holomorphic net extension)**.** *Under the preceding algebraic inputs, there is an irreducible local conformal-net extension $\mathcal{N}_U \subset \mathcal{E}_3$ of central charge eight. It is holomorphic and has index*

$$[\mathcal{E}_3 : \mathcal{N}_U] = \mathrm{Dim}(\mathcal{H}_3) = 3(1+d^2), \qquad d = (3+\sqrt{13})/2.$$

Its all-module boundary category is $\mathcal{H}_3$, and its algebra-bimodule category is $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$, with 36 *simple objects. This construction does not require A1.*

Proof. Transport the canonical connected commutative Q-system $I_{\mathcal{H}_3}(\mathbf{1}) \in Z(\mathcal{H}_3)$ along Corollary J.3; call the result $Q_3 \in \mathrm{Rep}(\mathcal{N}_U)$. The local-net reconstruction theorem of Longo–Rehren [45, Theorem 4.9] produces the extension. Connectedness is precisely the multiplicity-one hypothesis for the identity sector, hence gives irreducibility. The Q-system has finite statistical dimension, which equals the displayed index; its commutativity is precisely the locality condition in that theorem. The conformal-net form of the correspondence is recalled in [46, Section 2.6]. The represented local diffeomorphism unitaries of the subnet give the conformal action of the extension, so the central charge remains eight.

The finite-index theorem [32, Theorem 24] gives complete rationality. Since Q_3 is Lagrangian, the index relation gives

$$\mu(\mathcal{E}_3) = \frac{\mu(\mathcal{N}_U)}{\dim(Q_3)^2} = 1,$$

and the extension is holomorphic. The canonical-algebra equivalences of [39, Corollary 4.1 and Proposition 4.4] identify the all-module category with $\mathcal{H}_3$. The corresponding bimodule category is $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$ by the double-centralizer description [42, Sections 4 and 8]; its simple objects are the $6 \cdot 6$ ordered pairs. As below, these algebra-bimodule defects preserve the original $\mathcal{N}_U$ subnet. □

No identification of $\mathcal{E}_3$ with the E_8 lattice net is made here. Holomorphicity and central charge eight are the conclusions of the cited net theorems; the VOA classification used later does not by itself classify arbitrary holomorphic conformal nets. Nor does the present construction identify a reconstructed VOA extension with the selected tensor category of VOA modules.

Proposition K.2 (Diagonal full theory)**.** *Under the preceding algebraic inputs, the completely rational net $\mathcal{N}_U$ admits its canonical unitary local diagonal full theory. With the twelve-module character determination, its partition function is*

$$Z_{\mathrm{diag}}(\tau,\bar{\tau}) = \sum_{i=0}^{11} \chi_i(\tau)\overline{\chi_i(\tau)}, \qquad c_L = c_R = 8. \tag{132}$$

Proof. Apply the diagonal Longo–Rehren construction, in the identity-coupling case of Rehren [38, Theorem 1.4, Proposition 1.5, and Corollary 1.6]. Propositions I.1 and I.3 identify all sector Hilbert spaces with the twelve ordinary modules and preserve their Hamiltonians. Hence their traces give (132). The annulus compatibility claim is not needed for this net construction or this linear character matching. □

Theorem K.3 (Chosen $\mathcal{H}_3$ extension and defects)**.** *Assume the hypotheses of Theorem J.2 and the preceding strict positivity of all nonvacuum lowest conformal weights. There exists a*

unitary holomorphic conformal extension $H' \supset U$ isomorphic to V_{E_8} whose all-module category is $\mathcal{H}_3$. For its extension algebra $A_3 \in \mathrm{Rep}(U)$,

$$\mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3, \qquad {}_{A_3}\mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}. \tag{133}$$

The second category describes algebra-bimodule defects preserving the U chiral subalgebra in the corresponding full extension theory and contains an $\mathcal{H}_3$ fusion subcategory. It has 36 simple objects, labelled by ordered pairs of simple objects of $\mathcal{H}_3$.

Proof. Let $I_{\mathcal{H}_3}$ be the right adjoint of the forgetful functor $Z(\mathcal{H}_3) \to \mathcal{H}_3$. Transport the canonical algebra $I_{\mathcal{H}_3}(\mathbf{1})$ along the unitary ribbon equivalence. By the canonical Lagrangian-algebra construction, as in Davydov–Müger–Nikshych–Ostrik [39, Corollary 4.1 and Proposition 4.4], the resulting A_3 is connected, commutative, separable, and twist-free, with

$$\mathrm{FPdim}(A_3)^2 = \mathrm{FPdim}(\mathrm{Rep}(U)), \qquad \mathrm{Rep}(U)^0_{A_3} \simeq \mathrm{Vec}, \qquad \mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3.$$

Trivial twist makes its constituent conformal weights integral. Connectedness and strict positivity away from the vacuum give a one-dimensional weight-zero space. The extension correspondence of Huang–Kirillov–Lepowsky [40, Theorems 3.2–3.6], together with Gui's unitary Q-system theorem, produces a unitary conformal VOA extension H'. Here the transported algebra is a Q-system: the canonical algebra in the unitary center has its standard positive Frobenius structure, and the equivalence is unitary. Thus Gui's theorem is applied to a specified positive algebra structure, rather than to an arbitrary separable algebra without a chosen $*$-structure.

For completeness, the finiteness condition on the extension can be checked at the level of C_2 spaces. Write $H' = \bigoplus_i M_i^{\oplus m_i}$ as a finite sum of ordinary U-modules and set

$$C_2^U(H') = \mathrm{span}\{u_{(-2)}w : u \in U,\ w \in H'\}.$$

The module C_2-cofiniteness theorem of Abe–Buhl–Dong [13, Proposition 5.2] implies $\dim H'/C_2^U(H') < \infty$. Since $C_2^U(H') \subset C_2(H')$, the extension is C_2-cofinite. Complete reducibility for the extension follows from the separable algebra/module correspondence in Huang–Kirillov–Lepowsky, Theorems 3.5–3.6. The unitary invariant form makes the simple extension self-contragredient. Together with the already verified CFT-type grading, these are the strong-rationality hypotheses. The Lagrangian condition gives $\mathrm{Rep}(H') \simeq \mathrm{Rep}(U)^0_{A_3} \simeq \mathrm{Vec}$, so it is holomorphic. Its central charge is eight, so Dong–Mason [41, Theorem 1] identifies H' with V_{E_8}.

For the canonical algebra, the algebra-bimodule duality gives the second equivalence in (133); see Müger [42, Sections 4 and 8]. One way to keep track of the opposite is to view $\mathcal{H}_3$ as a module category over its center. The two commuting actions on it are

$$(x \boxtimes y) \cdot m = x \otimes m \otimes y.$$

Composing two such functors reverses the order of the right factors, which accounts for "op" in the formula. The double-centralizer description identifies these endofunctors with the bimodule category of the canonical algebra. The equivalence transports with A_3. The first-factor embedding $x \mapsto x \boxtimes \mathbf{1}$ is fully faithful and tensor, and supplies the stated $\mathcal{H}_3$ fusion subcategory. There are $6 \cdot 6 = 36$ simple pairs, with dimensions $\dim(x)\dim(y)$. As a consistency check, their total squared dimension is $\mathrm{Dim}(\mathcal{H}_3)^2 = \mathrm{Dim}(\mathrm{Rep}(U))$, as required by Morita duality. □

Remark K.4 (Extension choice and preserved symmetry)**.** The selection of the canonical $\mathcal{H}_3$ algebra proves existence of H' with the specified all-module category. It does not identify A_3 with the original embedded H/U algebra; the original boundary category may be $\mathcal{H}_2$ or

$\mathcal{H}_3$. Both holomorphic VOAs being isomorphic to V_{E_8} does not identify their embeddings of U. Moreover, the defects in (133) preserve U and need not preserve the enlarged E_8 chiral algebra. The local chiral H'-module category is Vec.

The finite certificates and algebraic arguments supply the stated representation-theory inputs. Comparisons A2–A3 give sector exhaustion, strong unitarity, and the unitary Haagerup-center identification for the conformal net. The pairwise annulus comparison A1 then proves complete unitarity of the VOA, its specified unitary ribbon structure, and the chosen $\mathcal{H}_3$ VOA extension. The diagonal net theory and its character interpretation also have a proof without this last comparison. The earlier synthesis sources are `hi_cft_20260923_c8two_final_theorem.md` and `hi_cft_20260923_c8two_certificate_audit.md`; the explicit domain, lift, and insertion arguments above supply the analytic intermediate steps needed for those conclusions.

L Relation to the proposed Haagerup theories

Albert–Honda–Kaidi–Zheng explicitly assume the existence of a $c = 8$ Haagerup-center theory in Section II, footnote 4 of [11]. Their equation (12) displays the γ-dependent characters, and their extension discussion relates the center theory to the E_8 theory with Haagerup defects.

Under the stated construction inputs, Theorem 1.3 supplies a $\gamma = 0$ realization as a conformal net and a diagonal full net theory. Theorem 1.4 supplies the completely unitary VOA realization and a compatible E_8 VOA extension. The distinction between U-preserving and full-E_8-preserving defects is maintained. Neither this construction nor that finite gauging identifies the continuum limit of the ρ-projector chain. Section III of the cited paper proposes further asymmetric quotients towards charges two and six; those require additional conformal and defect data. No uniqueness theorem for all central-charge-eight CFTs is asserted here.

The ambient lattice is also compatible with the conditional characterization in [12, Section 4.3]: a rational chiral algebra with the Haagerup center at central charge eight must arise from a rank-two hypergroup action on $E_{6,1} \otimes A_{2,1}$. That characterization constrains a realization; it does not supply its generating fields. Conversely, the concrete inclusion constructed here does not identify the original extension's fusion category as $\mathcal{H}_3$ without the separate distinction between $\mathcal{H}_2$ and $\mathcal{H}_3$ discussed above. Keeping these two statements separate avoids making an existence argument depend on a desired choice of Morita representative.

M Certificate ledger and reproducibility

The following frozen records supply the finite construction inputs. The revision of 27 September 2026 additionally reran the original provenance check, minimal generation, grade-two exact OPE and PCT checks, the exact grade-three construction, and the final Gate 4, charged-span, and Gate 5 verifiers. Fresh outputs were written under `c8_revision_20260927/`; the frozen inputs were not overwritten. All paths are relative to the project directory. The filenames preserve an unambiguous connection to the actual records.

Result	Record and scope
Minimal generation	`hi_cft_20260923_c8_minimal_generators.json`: the short generating set in (1) generates all 120 vectors of S_2. This necessary provenance step was absent from the original 640-file bundle and is explicitly included in the supplemented ledger.
Initial spaces	`hi_cft_20260923_c8_exact_unitary_certificate.json` and `hi_cft_20260923_c8_grade3_exact_v2.json`: PCT, stress and singular OPE identities, and exact dimension 956.
Lower identities	`hi_cft_20260923_next_structure_lower_closure.json`: 3,970 exact relations.
Actual triplet	`hi_cft_20260923_next_structure_p4_controls.json`: 18 current-zero and six direct Virasoro controls.
Raising saturation	`hi_cft_20260923_c8two_structure_saturation.json`: 644 exact images in a quotient of dimension 322.
Additional direct seeds	`hi_cft_20260923_c8two_targeted_h4_trimmed_v2/summary.json`: 118 completed exact identities. The unfinished final block in this historical directory is not counted as a completed identity.
Raised seeds	`hi_cft_20260923_c8two_structure_h4_raised_exact/summary.json`: twelve additional ordinary exact identities.
Singlet premise	`hi_cft_20260923_c8two_h4_singlet_premise_verified.json`: all 24 raising conditions replayed.
Projected residuals	`hi_cft_20260923_c8two_h4_singlet_exact.json`: twelve full exact residuals, using 21 selected columns.
Full Gate 4	`hi_cft_20260923_c8two_gate4_verified.json`: all 2,898 identities, using singlet recovery and raising saturation.
Charged spanning	`hi_cft_20260923_c8two_charged_span_verified.json`: 28 reconstructed seeds, rank 322, and current commutators.
Full Gate 5	`hi_cft_20260923_c8two_gate5_verified.json`: all 24 channels by the stated Jacobi argument.
Combined provenance	`hi_cft_20260923_c8two_proof_bundle.json`: 640 authenticated files, including earlier exact inputs and proof notes. It checks provenance and completion, not formal correctness of every mathematical argument.

The final Gate 4 verifier replays the twelve projected identities using integer-cleared coefficients, separately from the rational producer. A field element is represented in the basis $1, s, t, st$, with $s^2 = 13$, $t^2 = -3$; all residuals are exact four-component rational identities. Prime calculations are used to select candidate columns and to certify positive spanning against an independently known upper dimension. A vanishing prime residual alone is never treated as a characteristic-zero containment proof.

The provenance-ledger replay can be written to a fresh output without changing the original certificate:

```
python hi_cft_safe_run.py --limit-mib 1024 -- \
  python hi_cft_20260923_c8two_proof_bundle.py \
  --output /tmp/haagerup-c8-proof-review.json
```

These lines represent a single shell command. The output path must not already exist. This rechecks the frozen source hashes and completion flags; it does not rerun all original

mode calculations. The original and supplemented ledgers together bind the listed certificate producers and verifiers, including the additional dependencies described below.

M.1 Fresh replay coverage

The supplemented ledger `c8_revision_20260927/finite_input_ledger_complete.json` retains the original provenance graph and explicitly binds the minimal-generation and exact grade-three records. This repairs an omission in the original bundle's declared roots; it does not modify the old bundle or infer a mathematical identity from a hash. The principal fresh outputs are listed below. A "replay" means the specific operation in the right column, rather than a full independent reconstruction of every earlier lattice calculation.

Output under `c8_revision_20260927/`	Recomputed assertion
`provenance_replay.json`	All 640 original source hashes and required completion flags agree.
`minimal_generators_replay.json`	The specified short generating set gives the whole exact 120-dimensional S_2.
`unitary_replay.json`	The PCT, stress, primarity and grade-two singular-product tests, including all eight primary multiplets, pass in the exact coefficient field.
`grade3_replay.json`	The characteristic-zero generated span has dimension 956, with normal subspace of dimension 634.
`gate4_replay.json`	Integer-cleared projected residuals, raising premises and saturation establish all $2,898$ required Gate 4 inputs.
`charged_span_replay.json`	The reconstructed charged seeds and lowering images span dimension 322; the saved current matrices satisfy both axis and cross-axis commutators.
`gate5_replay.json`	Exact quotient relations, nonzero bottom coefficients and authenticated inputs support all 24 channels of the printed Jacobi deduction.
`top_bound_replay.json`	Independent rational enumeration reconstructs all fourteen core tops, trace coefficients, the two current-character inequalities and the 17/10 certificate.
`core_extension_replay.json`	The zero-complementary-energy space in S_2 has dimension 25 and contains all six required order-thirteen spectral witnesses with their exact eigenvalues.
`top81_replay.json`	In each of four mixed lattice sectors, exact zero-mode actions and weighted adjoints generate the entire 81-dimensional top from a unique Cartan eigenline.
`potts_replay.json`	The explicit Potts primary has the required stress action and lies in the actual generated S_3.
`pf13_membership_replay.json`	The parafermion candidate lies in the generated Cartan-zero S_3, its PCT line is stable, and the positive stress identity has the stated exact coefficients.

The last two producers originally had fixed output paths. The adapter `c8_revision_20260927/replay_structural_input.py` redirects only that output construction to a fresh file, requires exactly one write, and compares the complete mathematical output with the frozen record, excluding timing and memory statistics. It authenticates the unchanged producer and checks that the original record remains unchanged. The parafermion polynomial OPE and decoupling records are authenticated inputs, not claimed as newly reconstructed by this membership replay.

These calculations ran serially with at most two compute threads and CPU affinity enforced. The permanent shared-desktop guard imposed a 6 GiB address ceiling, an 8 GiB process-group RSS watchdog and a 32 GiB host-available-memory reserve; the exact lattice replays used smaller address caps of at most 2 GiB. The grade-three replay took about 493 seconds and peaked at 0.190 GiB sampled group RSS. No swap capacity was included in the allocation budget. Per-job estimates and logs are recorded in `c8_revision_20260927/resource_review.md`.

N Limited formalization

The construction's two Lean files isolate implications that are useful independently of the VOA implementation. They use the existing Lean 4.33.1 and Mathlib 4.33.1 environment and were checked without `sorry` or new axioms. The axiom reports contain only Lean's standard propositional extensionality, quotient soundness and, for some statements, classical choice.

1. `lean/HaagerupIzumi/CFT/FiniteClosure.lean` proves raising saturation for a stable submodule, under an explicit joint nilpotence hypothesis. It also proves recovery from a projected residual and the abstract linear Jacobi-bottom-witness implication. The hypotheses identify exactly where the current-representation bounds and finite identities enter the manuscript's use of these lemmas.

2. `lean/HaagerupIzumi/CFT/TopBound.lean` proves the displayed fourteen-variable deficit identity and derives the cleared-denominator 17/10 inequality from nonnegative multiplicities and the two stated $\mathfrak{sl}_2$ inequalities. The independent rational checker separately reconstructs the table to which this identity is applied.

The physics revision adds `lean/HaagerupIzumi/CFT/PhysicsBounds.lean`, with two checked real-number implications. The theorem `modular_heavy_tail_comparison` derives $H \le r(L - L_d)/(1 - r)$ from $L + H = L_d + H_d$, $H \le rH_d$, and $0 \le r < 1$. The theorem `selfdual_tail_first_moment_comparison` derives $H \le L'/E_{\mathrm{cut}}$ from $E_{\mathrm{cut}} > 0$ and $E_{\mathrm{cut}}H \le L'$. These are the algebraic steps in the thermal tail argument; they do not prove its infinite-sum, modularity, differentiability, or interval-arithmetic premises. The successful compiler and standard-axiom report is `c8_physics_20260927/physics_lean.log`.

The safe compiler commands, issued serially from the project root, are

```
python lean/safe_lean.py \
  lean/HaagerupIzumi/CFT/FiniteClosure.lean
python lean/safe_lean.py \
  lean/HaagerupIzumi/CFT/TopBound.lean
python lean/safe_lean.py \
  lean/HaagerupIzumi/CFT/PhysicsBounds.lean
```

The driver invokes the same resource guard. These formalizations do not define a VOA, validate the lattice engine, prove filtered Wick reduction, construct the embedded parafermion

factor, or formalize the analytic comparison. Their value is to check the stated algebraic implications with their premises exposed, rather than to certify the entire manuscript.

O Compilation and review scope

The main source is `hi_cft_c8_proof.tex`; its companion sources are `hi_cft_c8_manuscript_structure.tex` and `hi_cft_c8_manuscript_analytic.tex`, with references in `hi_cft_c8_bibliography.tex`. Run `pdflatex` twice on the main source for cross-references and the table of contents. The document uses standard TeX packages and requires no shell escape.

The frozen computational records remain unchanged. Fresh replays, the supplemented provenance ledger and the two Lean modules are additional checks with the scopes stated above. The original ledger does not authenticate this revised exposition. The analytic comparison proof and the exact embedded-field inputs should be examined directly when assessing the VOA conclusion; neither a successful PDF build nor a modular-character match verifies these arguments.

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